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THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY

ELEC 2100 Signals and Systems

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Outline Chapter 3 Frequency domain analysis of CT signals

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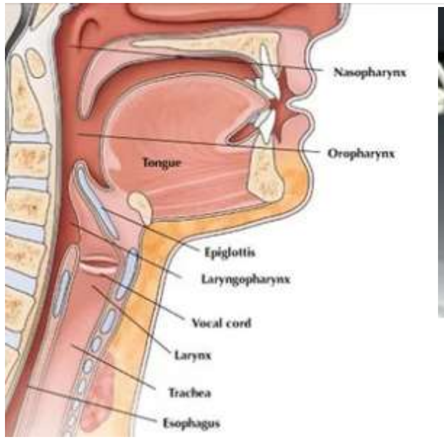
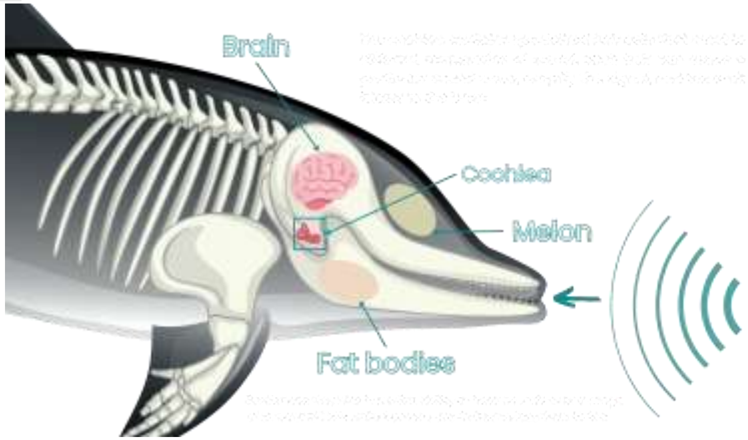
Main Content

1. Learning the concept of spectrum
2. Understanding frequency-domain analysis as a method of decomposing signals into frequency components, analyzing their responses, and reconstructing them

1. Frequency Characteristics and Spectrum

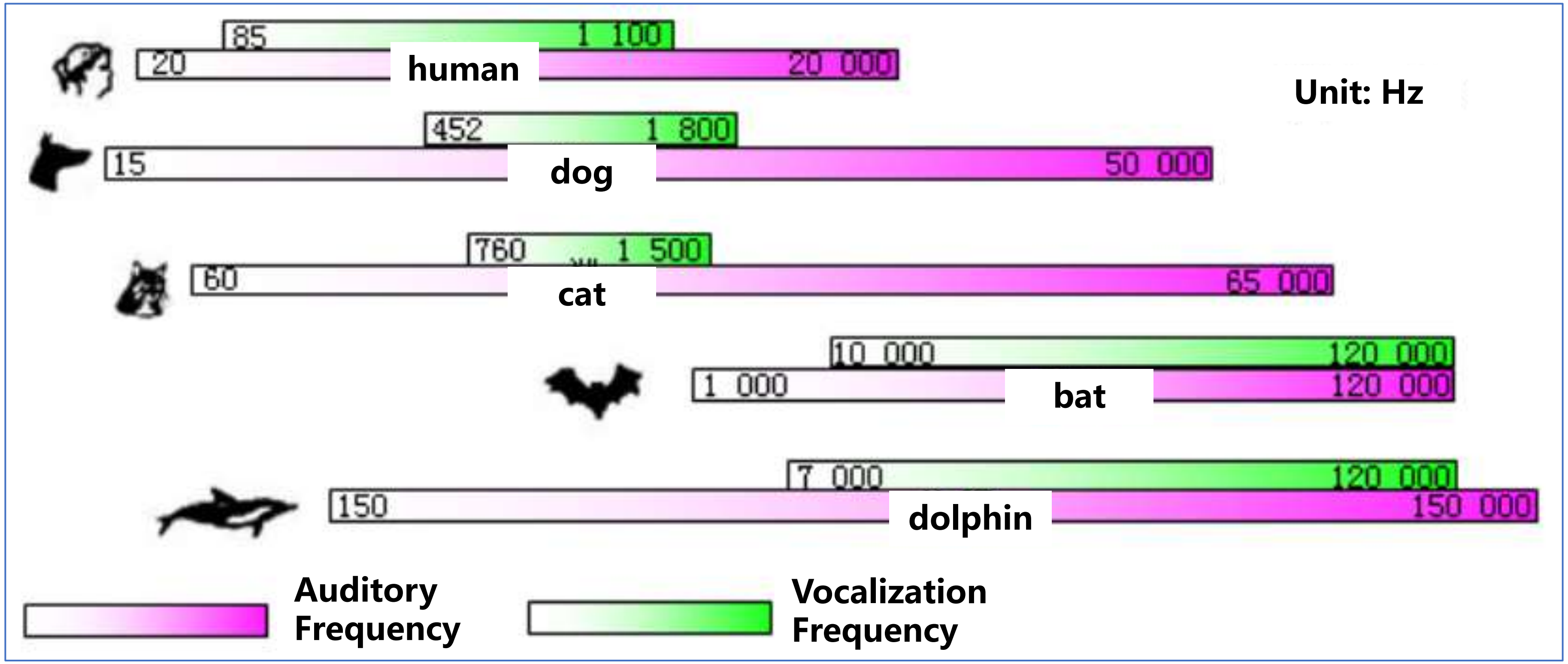
The vocal systems of humans, dolphins, and bats





The frequency characteristics of signals and systems are determined by their physical properties

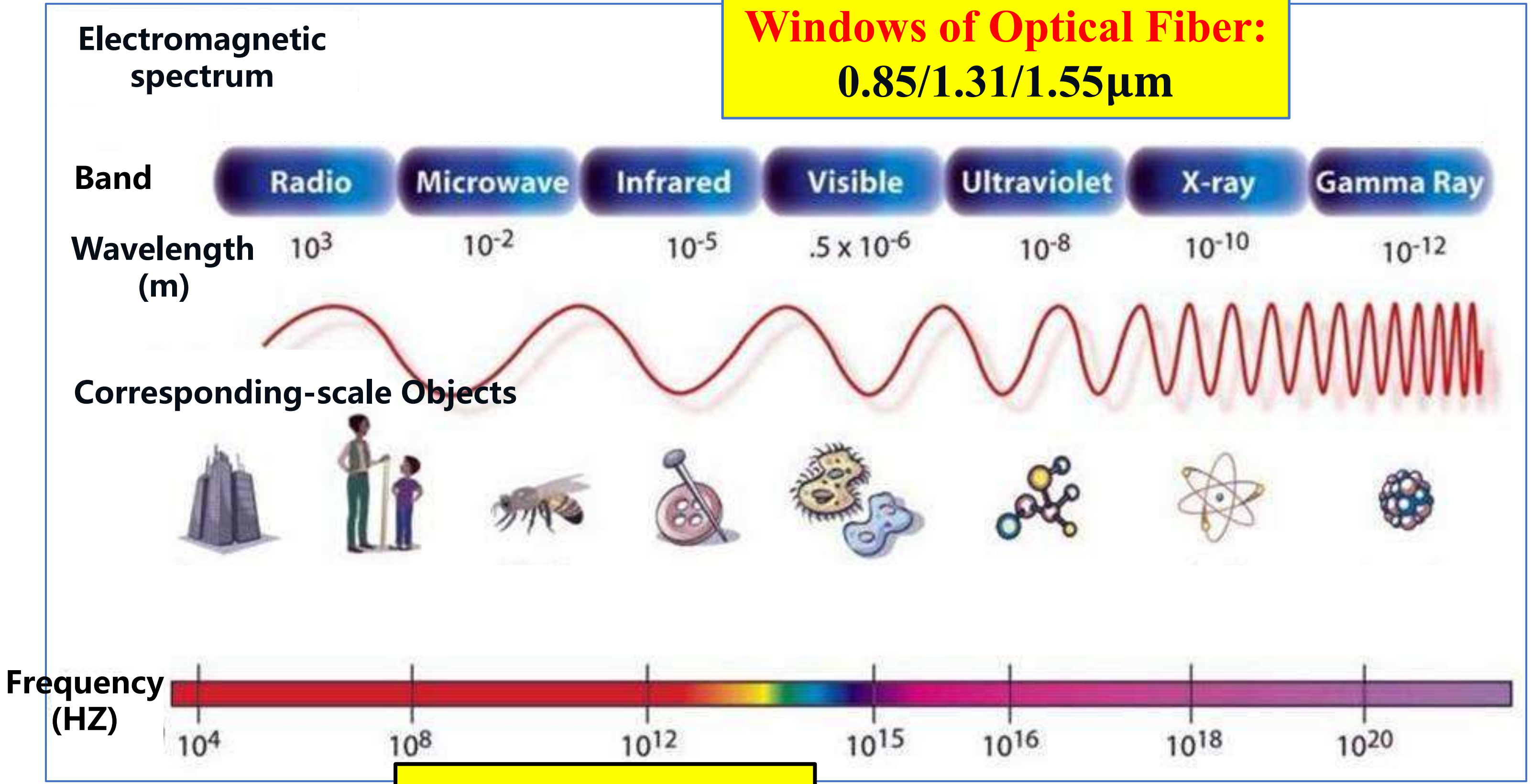
1. Frequency Characteristics and Spectrum



The **frequency characteristics** of signals and systems are determined by their **physical properties**

1. Frequency Characteristics and Spectrum

The Three Transmission Windows of Optical Fiber: 0.85/1.31/1.55µm

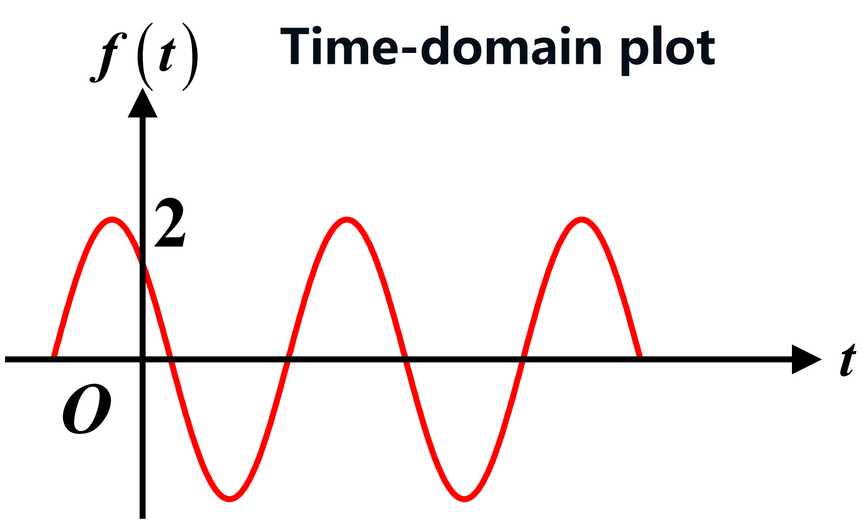


GSM 900/1800 MHz

1. Frequency Characteristics and Spectrum

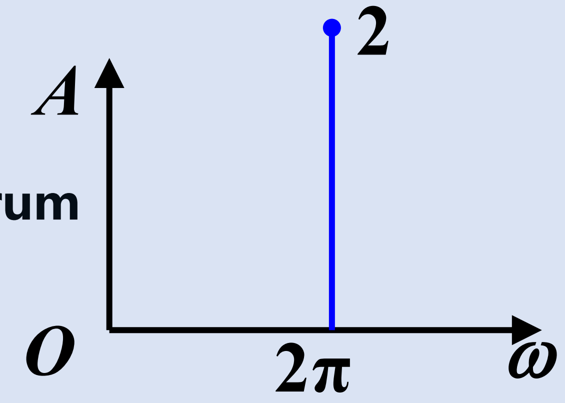
$$f(t) = 2\cos\left(2\pi t + \frac{\pi}{4}\right)$$

Time-domain plot

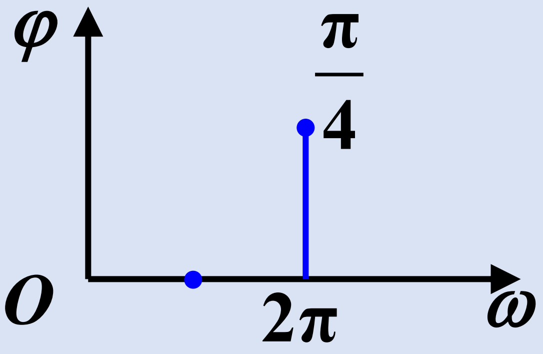


Amplitude	$A = 2$
Angular frequency	$\omega = 2\pi$
Initial phase	$\varphi = \frac{\pi}{4}$

Spectrum



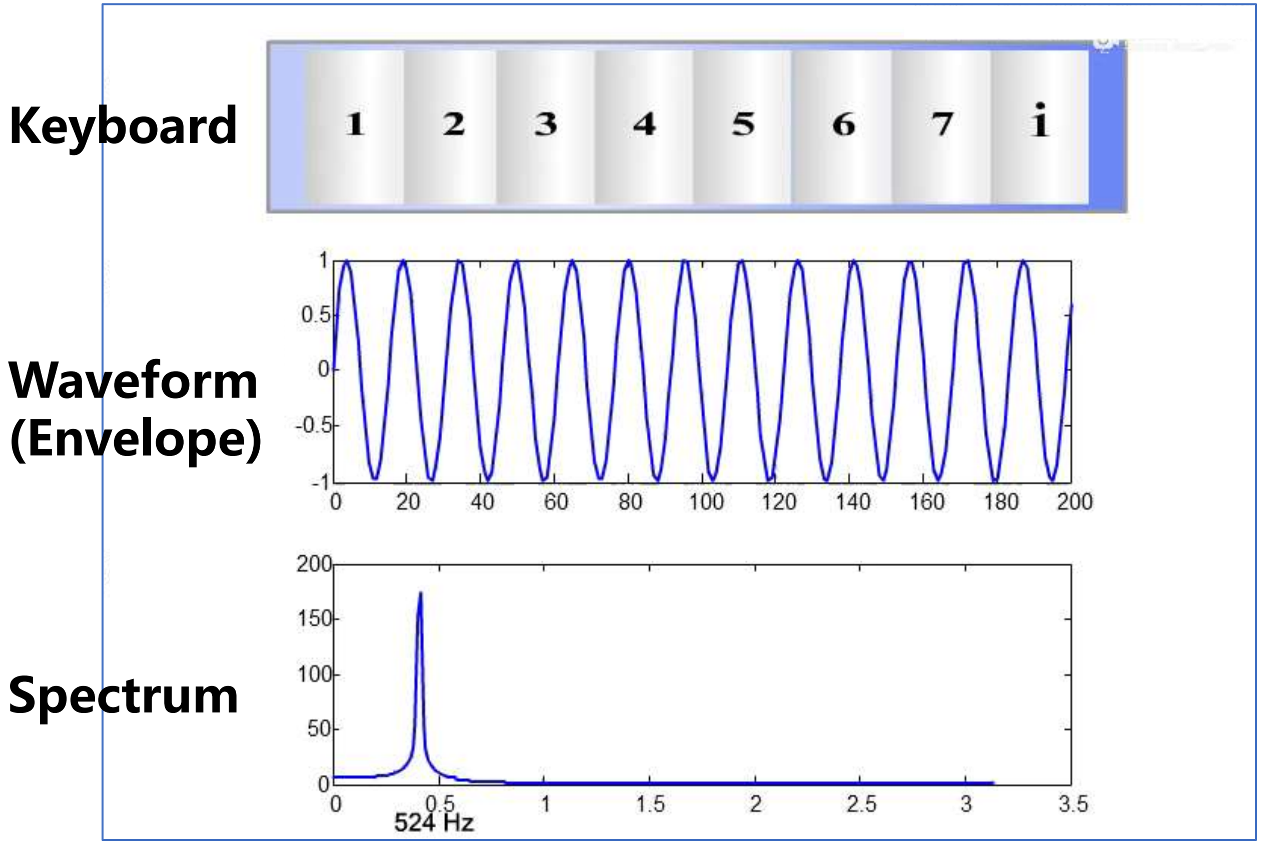
Amplitude spectrum



Phase spectrum

1. Frequency Characteristics and Spectrum

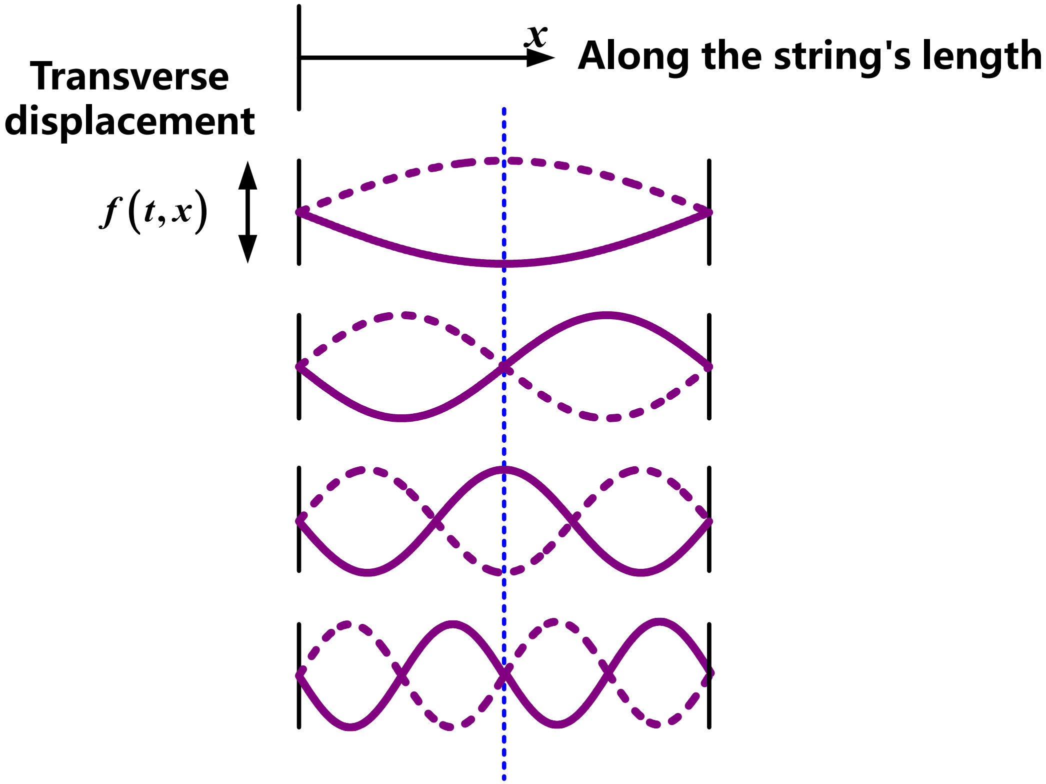
Musical Notation as Spectral Representation



1. Frequency Characteristics and Spectrum

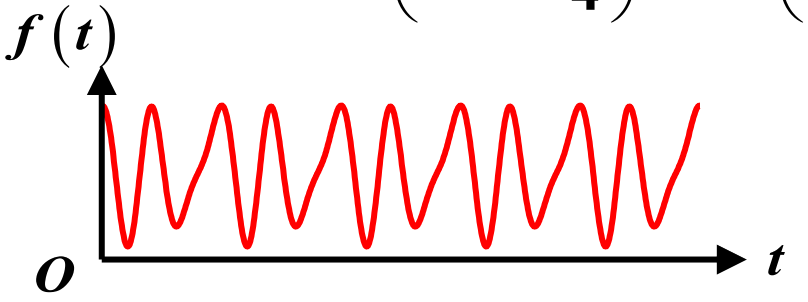
In 1748, Euler, in his study of vibrating strings, discovered that the oscillation modes of strings were all sinusoidal functions of x and exhibited harmonic relationships

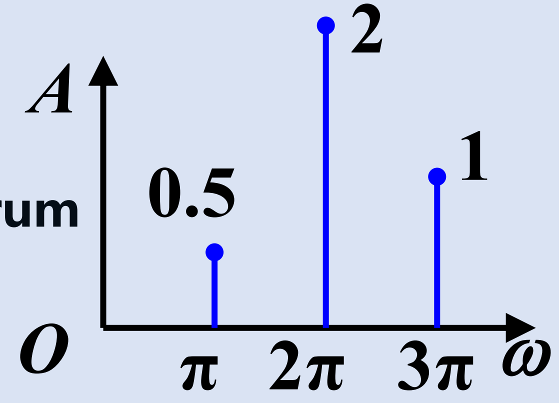
- **Fundamental (First Harmonic)** $\sin(\omega_0 t)$
- **Second Harmonic** $\sin(2\omega_0 t)$
- **Third Harmonic** $\sin(3\omega_0 t)$
- **Fourth Harmonic** $\sin(4\omega_0 t)$
- ……



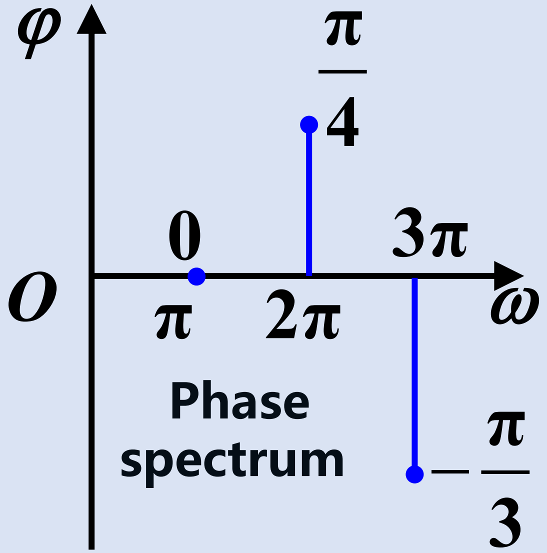
1. Frequency Characteristics and Spectrum

$$f(t) = 0.5\cos(\pi t) + 2\cos\left(2\pi t + \frac{\pi}{4}\right) + \cos\left(3\pi t - \frac{\pi}{3}\right)$$





Amplitude spectrum



Phase spectrum

2. Overview of Fourier Frequency Analysis

Periodic signals can all be expanded/decomposed into combinations of their fundamental frequency and harmonic trigonometric components (single-frequency components)

Jean Baptiste Joseph Fourier (1768–1830)
Baron, French mathematician and physicist

2. Overview of Fourier Frequency Analysis

Signal and System Frequency Domain Analysis Method

$e(t)$

$e(t)$
$h(t)$

Periodic Signal
Fourier Series

Aperiodic/Periodic Signal
Unit Impulse Response
Fourier Transform

$\cos(n\omega_0 t)$
$\sin(n\omega_0 t)$

$e^{jn\omega_0 t}$

$e^{j\omega t}$

Fourier series can analyze the spectrum of periodic signals, with clear physical meaning

Fourier transform can analyze the spectral characteristics of both aperiodic and periodic signals, as well as the frequency response characteristics of systems

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Summary

1. Learning the concept of spectrum

2. Understanding frequency-domain analysis as a method of decomposing signals into frequency components, analyzing their responses, and reconstructing them

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Main Content

1. Orthogonal Function Decomposition of Signals
2. Analogy to Vector Orthogonal Decomposition
3. Extraction of Orthogonal Signal Components
4. Real Orthogonal Function Set
5. Complex Orthogonal Function Set
6. Complete Orthogonal Function Set
7. Parseval's Theorem

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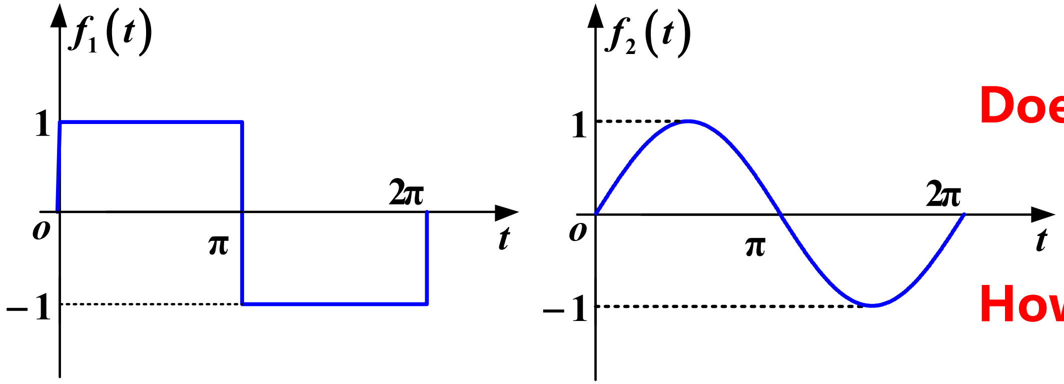
1. Orthogonal Function Decomposition of Signals



Does $f_1(t)$ contain any $f_2(t)$ components?

How much $f_1(t)$ component is present in $f_2(t)$?

$$f_1(t) = c_{12} f_2(t) + f_e(t)$$

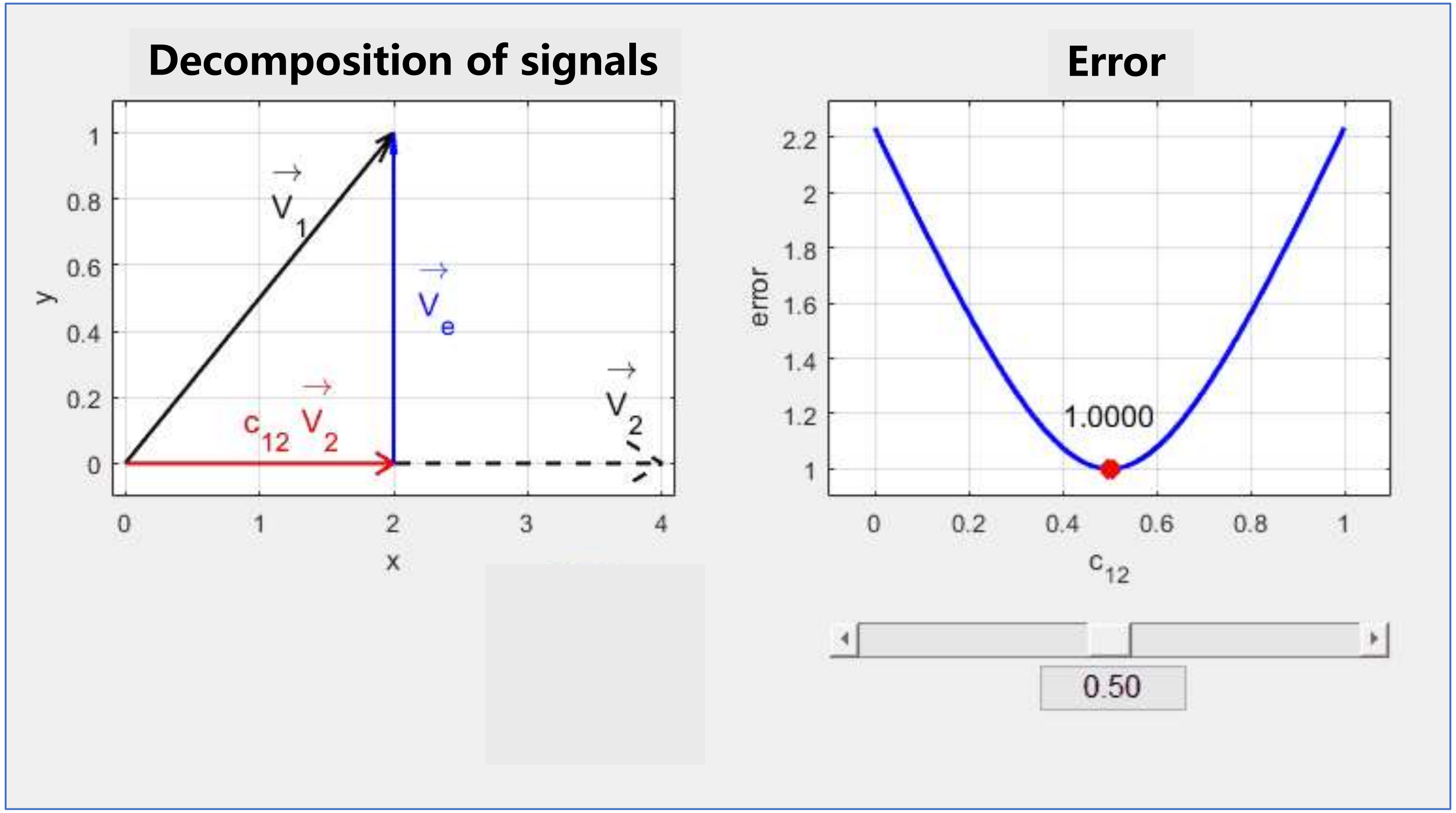
Error Function: $$f_e(t) = f_1(t) - c_{12} f_2(t)$$

Projection Coefficient: c_{12}

To determine the projection coefficients, the method of orthogonal function decomposition must be employed

2. Analogy to Vector Orthogonal Decomposition

Example: Given signal $\overrightarrow{V_1}$and component $\overrightarrow{V_2}$, the signal can be decomposed into $\overrightarrow{V_1}$=$c_{12}\overrightarrow{V_2}$+ $\overrightarrow{V_e}$. How to select coefficient c_{12}?



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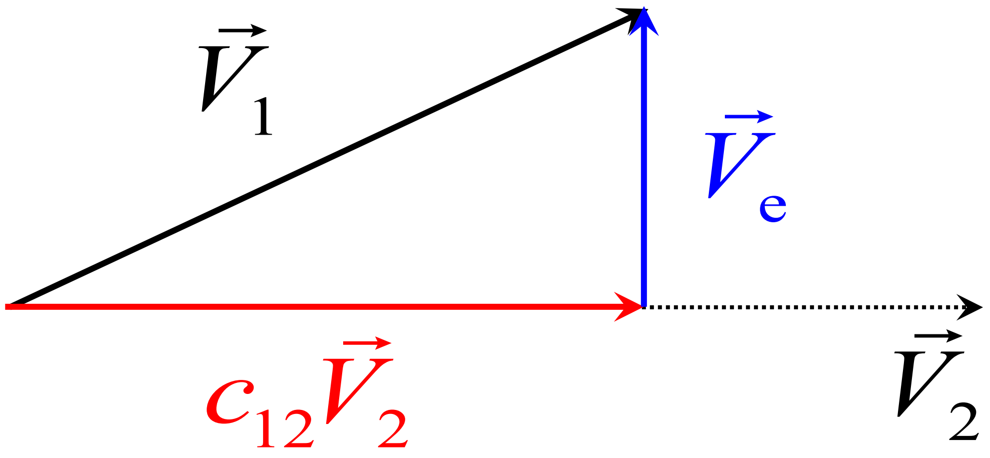
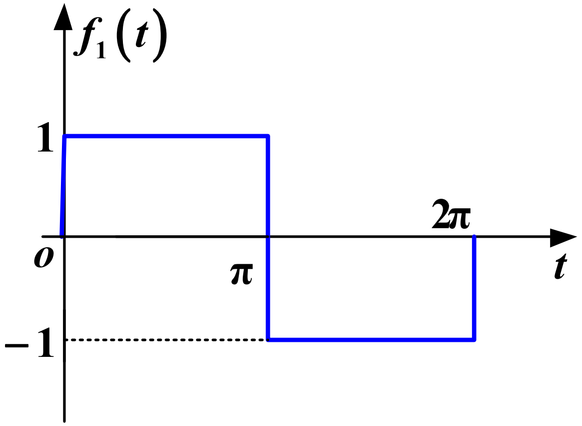
2. Analogy to Vector Orthogonal Decomposition

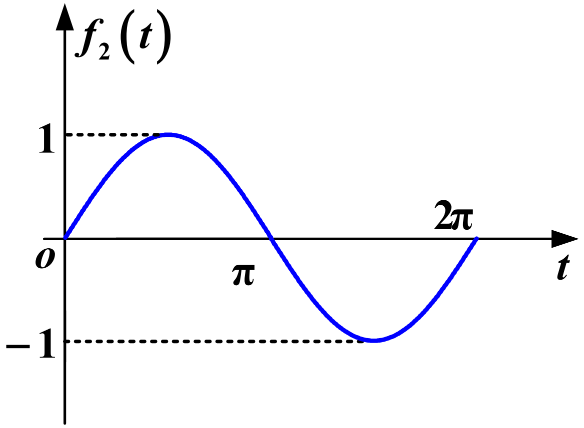
Selection of Projection Coefficients



- When two vectors are orthogonal, the error component is minimized
- The error component no longer contains any $\vec{V}_2$ component

3. Extraction of Orthogonal Signal Components





$$f_1(t) = c_{12} f_2(t) + f_e(t)$$

Error Function:

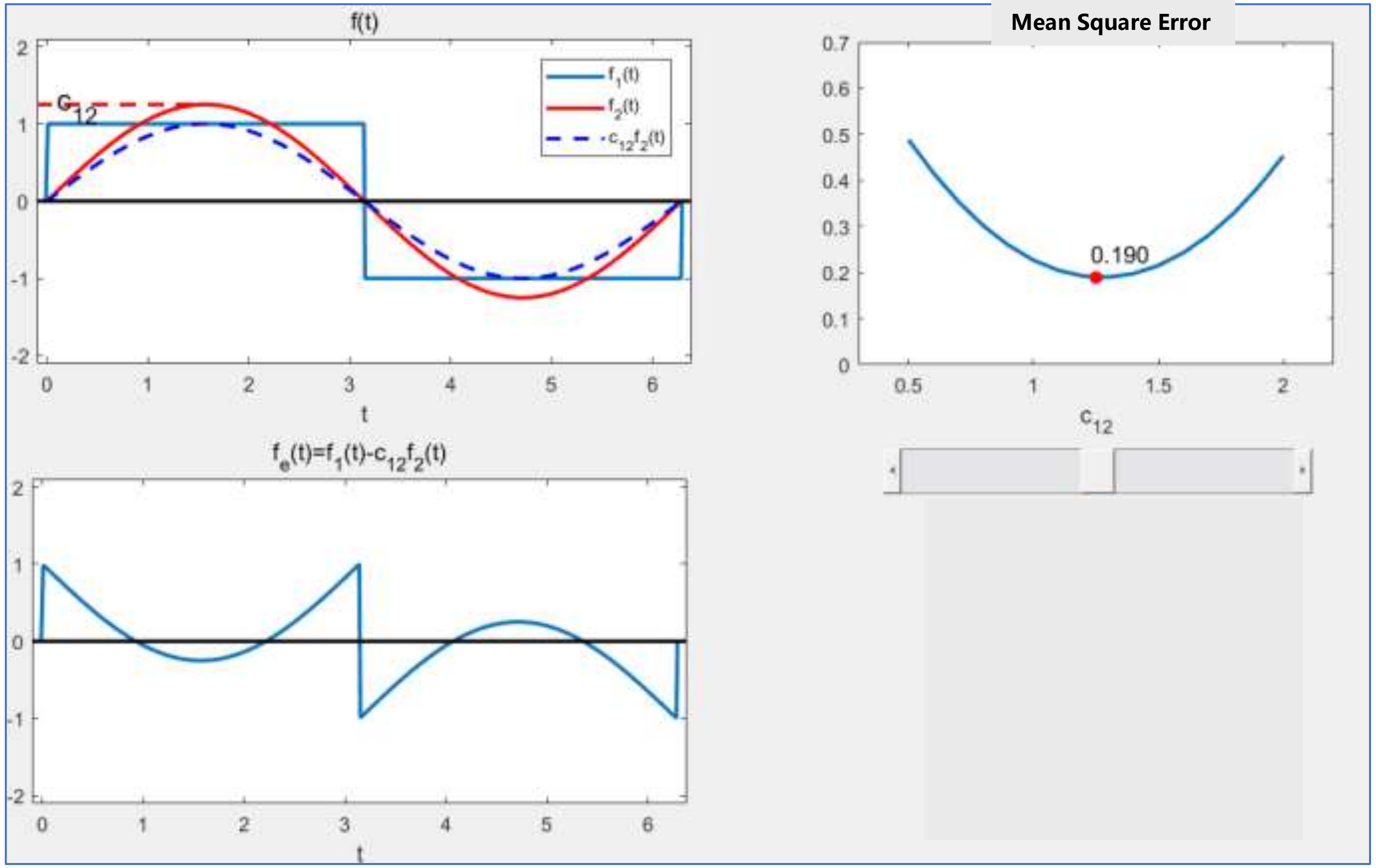
$$f_e(t) = f_1(t) - c_{12} f_2(t)$$

Mean Power (Mean Square Error) of the Error Signal over the Interval (t_1, t_2)

$$\overline{\varepsilon^2} = \overline{f_e^2(t)} = \frac{1}{t_2 - t_1}\int_{t_1}^{t_2}\left[f_1(t) - c_{12} f_2(t)\right]^2 \mathrm{d}t$$

3. Extraction of Orthogonal Signal Components

Determination of Projection Coefficients Based on the Minimum Mean Square Error (MMSE) Criterion



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3. Extraction of Orthogonal Signal Components

$$\frac{\mathrm{d}\,\overline{\varepsilon^2}}{\mathrm{d}\,c_{12}} = 0 \Rightarrow$$

$$\frac{\mathrm{d}}{\mathrm{d}\,c_{12}}\left\{\int_{t_1}^{t_2}\left[f_1(t) - c_{12} f_2(t)\right]^2 \mathrm{d}\,t\right\} = 0$$

Exchange order of differentiation/integration

$$\int_{t_1}^{t_2} \frac{\mathrm{d}}{\mathrm{d}\,c_{12}}\Big[\underbrace{f_1^2(t)}_{(1)} \underbrace{- 2c_{12} f_2(t) f_1(t)}_{(2)} + \underbrace{f_2^2(t) c_{12}^2}_{(3)}\Big]\mathrm{d}\,t = 0$$

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3. Extraction of Orthogonal Signal Components

$$\int_{t_1}^{t_2} \frac{\mathrm{d}}{\mathrm{d}c_{12}} \Big[\underset{(1)}{f_1^2(t)} - \underset{(2)}{2c_{12} f_2(t) f_1(t)} + \underset{(3)}{f_2^2(t) c_{12}^2} \Big] \mathrm{d}t = 0$$

$$(1)\ \frac{\mathrm{d}}{\mathrm{d}c_{12}} f_1^2(t) = 0$$

(Because $f_1(t)$ does not contain c_{12})

$$(2)\ \frac{\mathrm{d}}{\mathrm{d}c_{12}} \left[-2c_{12} f_1(t) \cdot f_2(t) \right] = -2 f_1(t) \cdot f_2(t)$$

$$(3)\ \frac{\mathrm{d}}{\mathrm{d}c_{12}} \left[c_{12}^2 f_2^2(t) \right] = 2c_{12} f_2^2(t)$$

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3. Extraction of Orthogonal Signal Components

$$\int_{t_1}^{t_2}\left[-2 f_1(t)\cdot f_2(t)\right]dt+\int_{t_1}^{t_2} 2c_{12} f_2^2(t)dt=0$$

$$c_{12}=\frac{\int_{t_1}^{t_2} f_1(t)\cdot f_2(t)\,\mathrm{d}t}{\int_{t_1}^{t_2} f_2^2(t)\,\mathrm{d}t}$$

If $c_{12}=0$, then $f_1(t), f_2(t)$ referred to as orthogonal functions, Thus

$$\int_{t_1}^{t_2} f_1(t) f_2(t)\,\mathrm{d}t=0$$

4. Real Orthogonal Function Set

$\{g_r(t)\}$, r=1, 2, 3$\cdots n$

$g_1(t), g_2(t)\cdots g_n(t)$

is called an orthogonal function set if

$$\int_{t_1}^{t_2} g_i(t)\cdot g_j(t)\,\mathrm{d}t = \begin{cases} 0, & i \neq j \\ K_i, & i = j \end{cases} \qquad (0 \le i, j \le n)$$

Definition: All functions in the set must be pairwise orthogonal

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4. Real Orthogonal Function Set

Any real signal $f(t)$ can be approximated by a sum of n orthogonal functions:

$$f(t) \approx c_1 g_1(t) + c_2 g_2(t) + \cdots + c_n g_n(t) \qquad (t_1 < t < t_2)$$

$$= \sum_{r=1}^{n} c_r g_r(t)$$

$$c_r = \frac{\int_{t_1}^{t_2} f(t) g_r(t) \, \mathrm{d}t}{\int_{t_1}^{t_2} g_r^2(t) \, \mathrm{d}t} = \frac{\int_{t_1}^{t_2} f(t) g_r(t) \, \mathrm{d}t}{K_r}$$

$c_1, c_2, \cdots c_n$ **are independent and can be computed in any order (a property unique to orthogonal functions)**

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Example 3

Prove the trigonometric function set is orthogonal over the interval

$$\left\{\sin(k\omega_0 t),\cos(k\omega_0 t)\right\},k=0,1,2\cdots(0,T_0),T_0=\frac{2\pi}{\omega_0}$$

$$\sin(n\omega_0 t)\sin(m\omega_0 t)$$

$$=\frac{1}{2}\left\{-\cos\left[(n+m)\omega_0 t\right]+\cos\left[(n-m)\omega_0 t\right]\right\}$$

The **sum-frequency term** consists of $n+m$ complete sinusoidal waves over interval $(0,T_0)$

The **difference-frequency term** consists of $n-m$ complete sinusoidal waves over interval $(0,T_0)$

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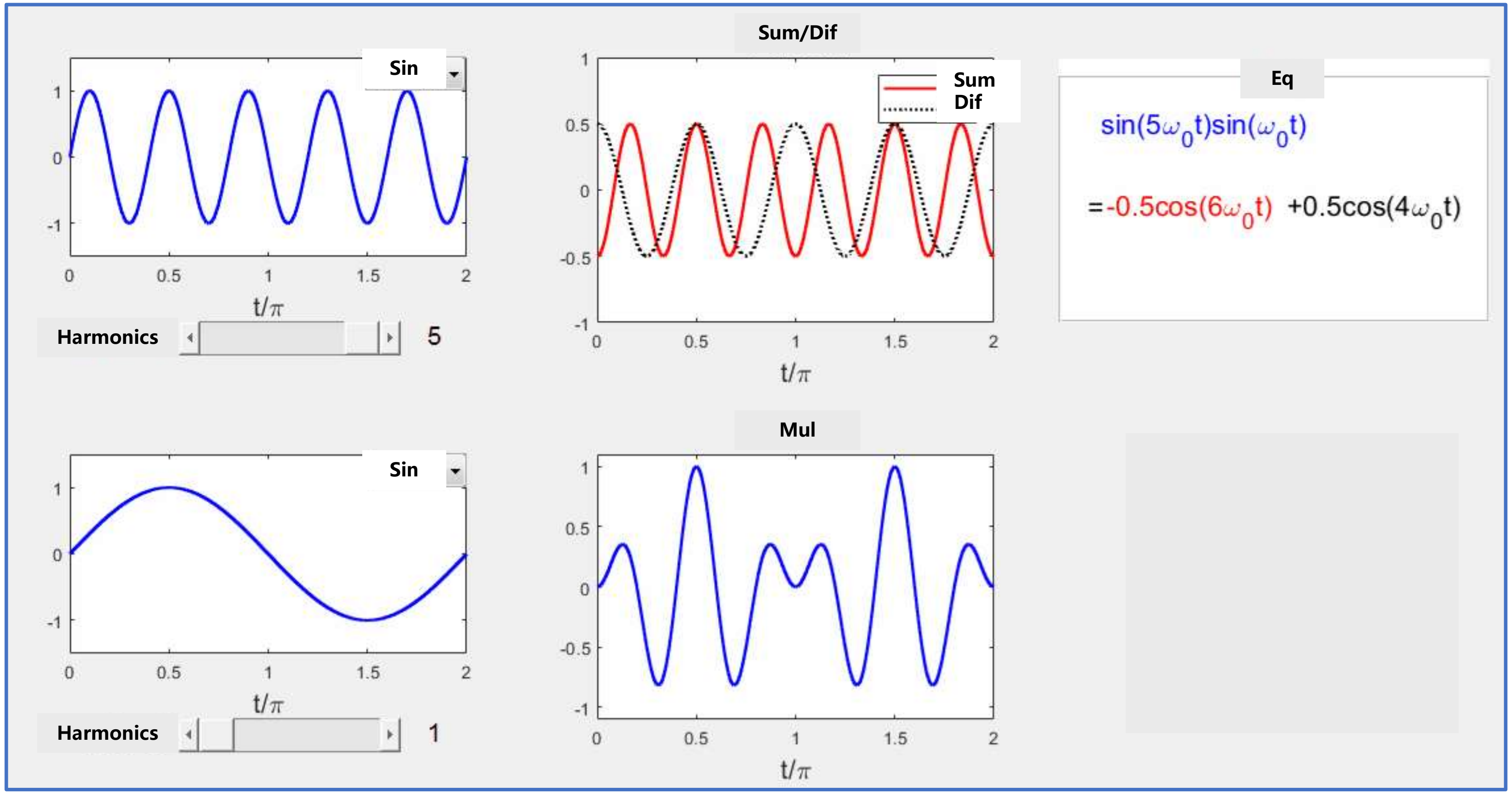
Example 3

$$\int_0^{T_0} \sin(n\omega_0 t)\sin(m\omega_0 t)\,\mathrm{d}t = \begin{cases} 0 & n \neq m \\ \frac{1}{2}T_0 & n = m \end{cases}$$



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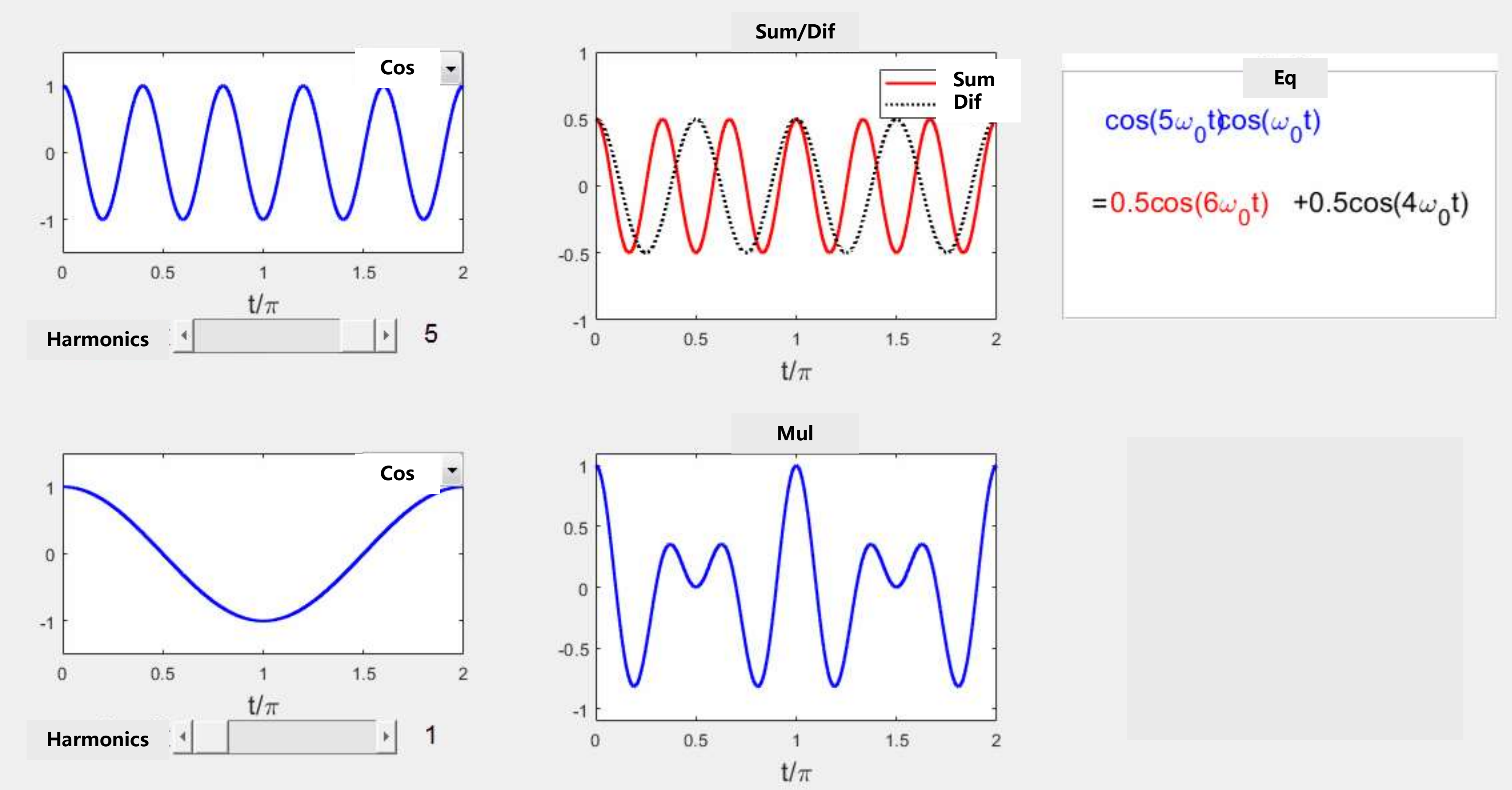
Example 3

$$\int_0^{T_0} \cos(n\omega_0 t)\cos(m\omega_0 t)\,\mathrm{d}t = \begin{cases} 0 & n \neq m \\ \frac{1}{2}T_0 & n = m \end{cases}$$



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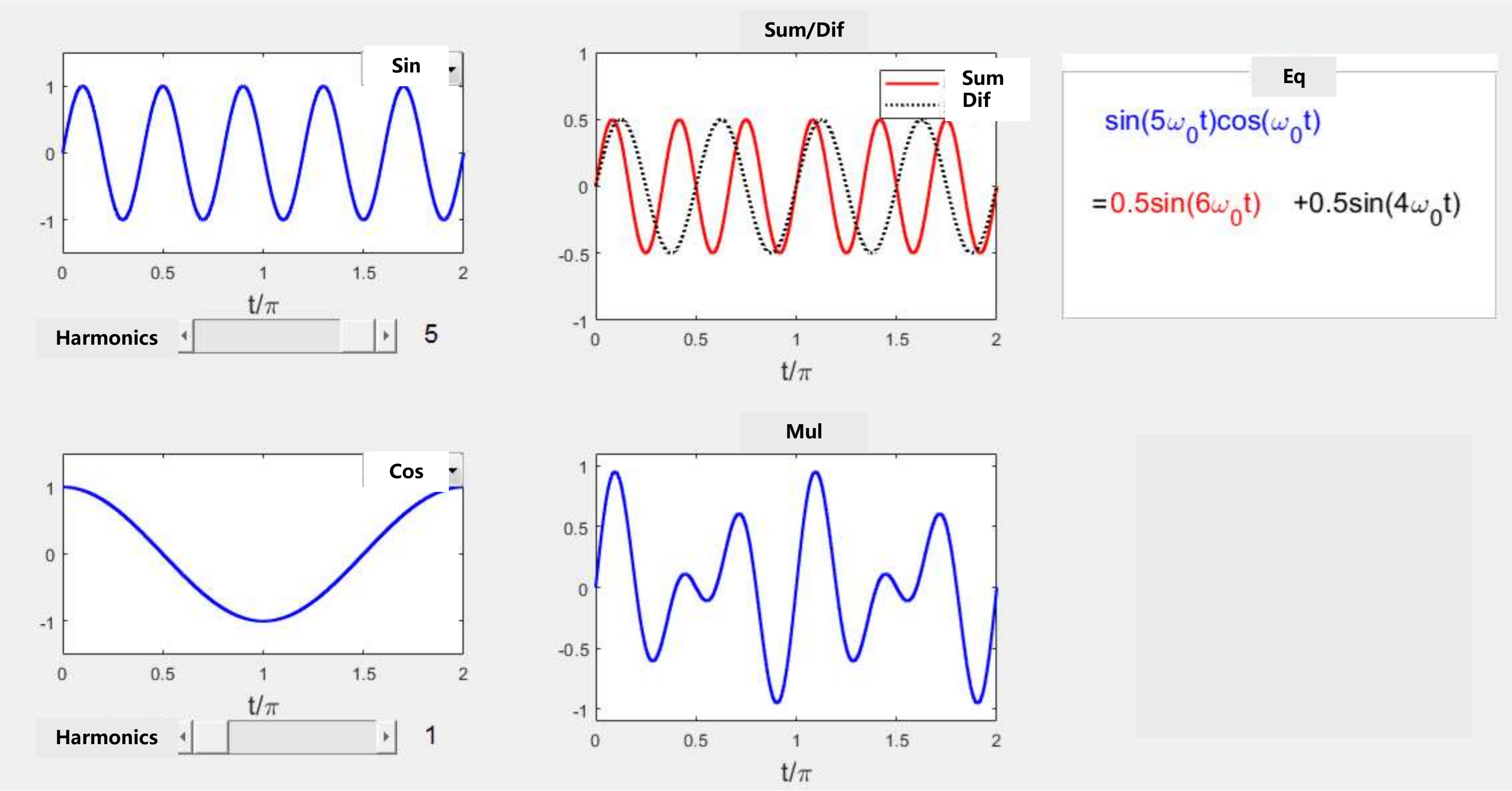
Example 3

$$\int_0^{T_0} \sin(n\omega_0 t)\cos(m\omega_0 t)\,\mathrm{d}t = 0$$



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Example 3

$$\int_0^{T_0} \sin(n\omega_0 t)\sin(m\omega_0 t)\,\mathrm{d}t = \begin{cases} 0 & n \neq m \\ \frac{1}{2}T_0 & n = m \end{cases}$$

$$\int_0^{T_0} \cos(n\omega_0 t)\cos(m\omega_0 t)\,\mathrm{d}t = \begin{cases} 0 & n \neq m \\ \frac{1}{2}T_0 & n = m \end{cases}$$

$$\int_0^{T_0} \sin(n\omega_0 t)\cos(m\omega_0 t)\,\mathrm{d}t = 0$$

It satisfies the conditions for an orthogonal function, so this function is an orthogonal function set

5. Complex Orthogonal Function Set

A set of complex functions $\{g_r(t)\}(r=1,2,\cdots,n)$ is orthogonal over (t_1,t_2)

If: $$\int_{t_1}^{t_2} g_i(t)g_j^*(t)\,\mathrm{d}t=\begin{cases}0 & i\neq j\\ K_i & i=j\end{cases}$$

The signal $f(t)$ can be approximated as:

$|f(t)|^2 = f(t)\cdot f^*(t)$

$$f(t)\approx\sum_{r=1}^{n}c_r g_r(t),\qquad c_r=\frac{\int_{t_1}^{t_2} f(t)g_r^*(t)\,\mathrm{d}t}{\int_{t_1}^{t_2} g_r(t)g_r^*(t)\,\mathrm{d}t}$$

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Example 4

Prove $\left\{e^{jk\omega_1 t}\right\}, k = 0, \pm 1, \pm 2 \cdots$ over the interval $(0, T_1)$, $T_1 = \dfrac{2\pi}{\omega_1}$ is an orthogonal function set

Let n, m be any integer, $g_n(t) = e^{jn\omega_1 t}, g_m(t) = e^{jm\omega_1 t}$, Then

$$\int_0^{T_1} g_n(t) g_m^*(t) \, dt = \int_0^{T_1} e^{jn\omega_1 t} e^{-jm\omega_1 t} \, dt = \int_0^{T_1} e^{j(n-m)\omega_1 t} \, dt$$

$$= \int_0^{T_1} \cos\left[(n-m)\omega_1 t\right] dt + j\int_0^{T_1} \sin\left[(n-m)\omega_1 t\right] dt$$

$$= \begin{cases} 0 & n \neq m \\ T_1 & n = m \end{cases}$$

It satisfies the conditions for an orthogonal function, so this function is indeed an orthogonal function set.

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6. Complete Orthogonal Function Set

$$f(t) \approx \sum_{r=1}^{n} c_r g_r(t)$$

Definition 1:

As n increases, $\overline{\varepsilon^2}$ decreases. If $n \to \infty$, then $\overline{\varepsilon^2} \to 0$. In this case, $g_1(t), g_2(t) \cdots g_r(t) \cdots g_n(t)$ is a complete orthogonal function set

Definition 2:

If a function $x(t)$ satisfies $\int_{t_1}^{t_2} g_r(t) \cdot x(t) dt = 0$ for all r, the set is incomplete (unless $x(t)$ is part of the set)

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7. Parseval’ s Theorem

For a signal represented by a complete orthogonal set:

$$f(t)=\sum_{r=1}^{\infty} c_r g_r(t)$$

$$\int_{t_1}^{t_2} |f(t)|^2 \, \mathrm{d}t = \sum_{r=1}^{\infty} \int_{t_1}^{t_2} |c_r g_r(t)|^2 \, \mathrm{d}t$$

Energy of the signal

Energy of each component

The total energy of $f(t)$ equals the sum of energies of its orthogonal components

Proof: Cross-terms vanish due to orthogonality

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Summary

Orthogonal Decomposition of Signals

$$f_1(t) = c_{12} f_2(t) + f_e(t)$$

Projection Coefficient

$$c_{12} = \frac{\int_{t_1}^{t_2} f_1(t) \cdot f_2(t) \, \mathrm{d}t}{\int_{t_1}^{t_2} f_2^2(t) \, \mathrm{d}t}$$

Two signals are orthogonal over the interval (t_1, t_2) if:

$$\int_{t_1}^{t_2} f_1(t) f_2(t) \, \mathrm{d}t = 0$$

Summary

- Orthogonal Function Set
- Complete Orthogonal Function Set
- Parseval's Theorem

Main Content

1. Trigonometric Form of Fourier Series
2. Exponential Form of Fourier Series
3. Relationship Between Function Symmetry and Fourier Series Coefficients
4. Parseval’s Theorem
5. Finite Fourier Series and Minimum Mean Square Error
6. Spectral Structure of Periodic Rectangular Pulse Trains
7. Gibbs Phenomenon
8. Bandwidth

1. Trigonometric Form of Fourier Series

Periodic signal $f(t)$ Fundamental period T_0 Fundamental angular frequency $\omega_0 = \frac{2\pi}{T_0}$

The signal $f(t)$ can be decomposed into a combination of harmonically related trigonometric functions

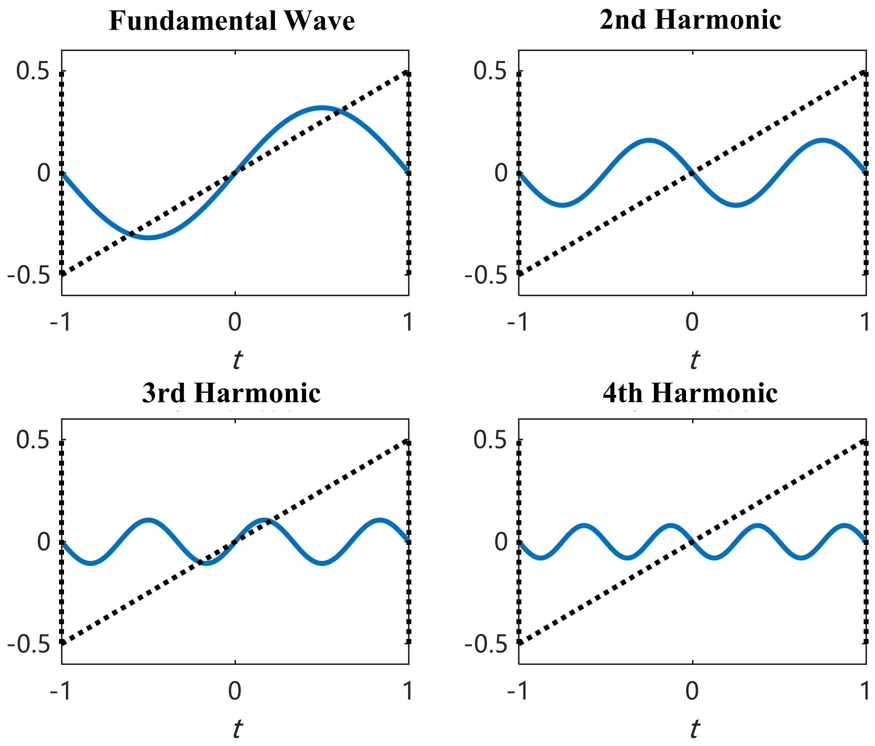
$$f(t) = a_0 + \sum_{k=1}^{\infty}\left[a_k \cos\left(k\omega_0 t\right) + b_k \sin\left(k\omega_0 t\right)\right]$$

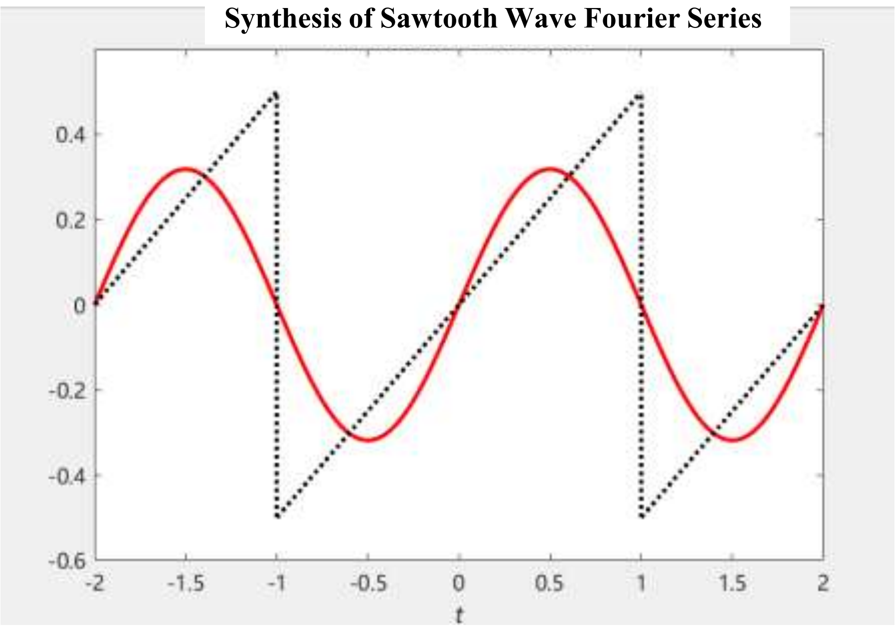
$$\left\{\cos\left(k\omega_0 t\right), \sin\left(k\omega_0 t\right)\right\}, \quad k=0, 1, 2, \cdots$$

This set forms a complete orthogonal function set over one period T_0

$f(t)$ must satisfy the Dirichlet conditions (ensuring the series converges to the signal)

(1) Harmonically Related





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(2) Complete Orthogonal Function Set

$\{\cos(k\omega_0 t), \sin(k\omega_0 t)\}$, $k=0, 1, 2, \cdots$ is orthogonal function set

$$\int_0^{T_0} \sin(n\omega_0 t)\sin(m\omega_0 t)\,\mathrm{d}t = \begin{cases} 0 & n \neq m \\ \frac{1}{2}T_0 & n = m \end{cases}$$

$$\int_0^{T_0} \cos(n\omega_0 t)\cos(m\omega_0 t)\,\mathrm{d}t = \begin{cases} 0 & n \neq m \\ \frac{1}{2}T_0 & n = m \end{cases}$$

$$\int_0^{T_0} \sin(n\omega_0 t)\cos(m\omega_0 t)\,\mathrm{d}t = 0$$

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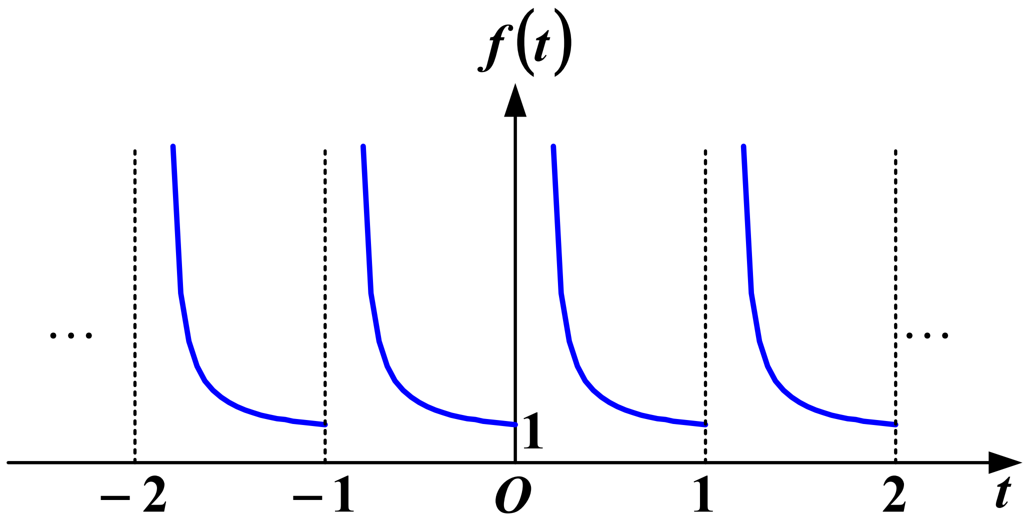
(3) Dirichlet conditions

Condition 1: Absolute Integrability Over One Period

$$\int_{t_0}^{t_0+T_0} \left| f(t) \right| \mathrm{d}t < \infty \qquad (T_0\text{, period})$$

For a signal with a period of 1, if the signal within the interval (0 < t ≤1) is $f(t)=\frac{1}{t}$

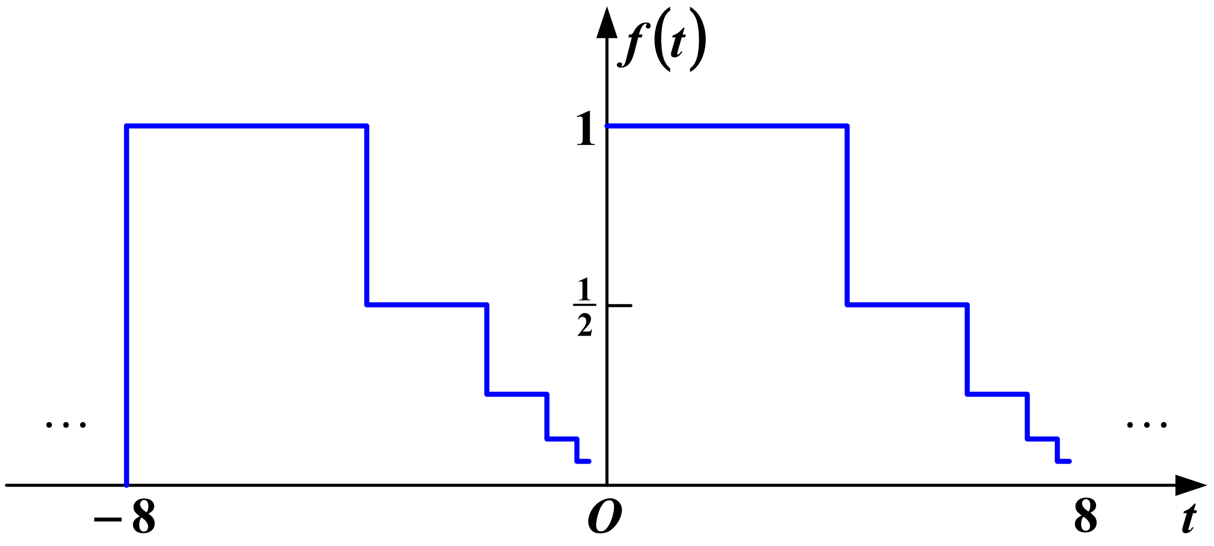


Fails to satisfy the absolute integrability condition

(3) Dirichlet conditions

Condition 2: Within a single period, if discontinuity points exist, their number must be finite

Example violating Condition 2: This signal has a period of 8, where each subsequent step's height and width are half of the previous step. Although its total area within one period does not exceed 8, the number of discontinuity points is infinite

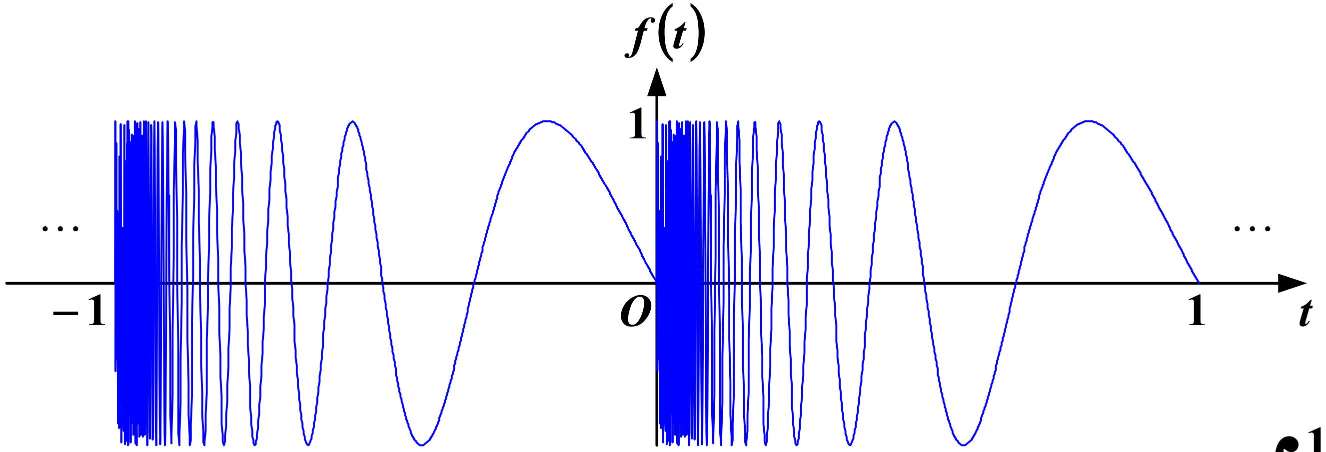


(3) Dirichlet conditions

Condition 3: Within one period, the number of local maxima and minima must be finite

A function violating Condition 3:

$$f(t) = \sin\left(\frac{2\pi}{t}\right), (0 < t \le 1)$$



For comparison, this periodic function (with period 1) satisfies $\int_0^1 |f(t)| \mathrm{d}t < 1$

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(4) How to determine the expansion coefficients?

$$f(t) = a_0 + \sum_{k=1}^{\infty}\left[a_k \cos\left(k\omega_0 t\right) + b_k \sin\left(k\omega_0 t\right)\right]$$

Review: Real Orthogonal Function Decomposition

$$f\left(t\right) = c_1 g_1\left(t\right) + c_2 g_2\left(t\right) + \ldots + c_r g_r\left(t\right) + \cdots$$

Coefficient

$$c_r = \frac{\int_{t_1}^{t_2} f(t) g_r(t) \, \mathrm{d}t}{\int_{t_1}^{t_2} g_r^2(t) \, \mathrm{d}t}$$

3. Fourier Series Analysis of Periodic Signals

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(4) How to determine the expansion coefficients?

$$f(t)=a_0+\sum_{k=1}^{\infty}\left[a_k\cos\left(k\omega_0 t\right)+b_k\sin\left(k\omega_0 t\right)\right]$$

$$a_0=\frac{1}{T_0}\int_{t_0}^{t_0+T_0}f(t)\,\mathrm{d}t$$

The average value of a signal

$$a_k=\frac{\int_{t_0}^{t_0+T_0}f(t)\cos\left(k\omega_0 t\right)\mathrm{d}t}{\int_{t_0}^{t_0+T_0}\cos^2\left(k\omega_0 t\right)\mathrm{d}t}=\frac{\int_{t_0}^{t_0+T_0}f(t)\cos\left(k\omega_0 t\right)\mathrm{d}t}{\int_{t_0}^{t_0+T_0}\frac{1+\cos\left(2k\omega_0 t\right)}{2}\mathrm{d}t}$$

The integral of the double frequency term is 0

$$=\frac{2}{T_0}\int_{t_0}^{t_0+T_0}f(t)\cos\left(k\omega_0 t\right)\mathrm{d}t$$

Similarly $$b_k=\frac{2}{T_0}\int_{t_0}^{t_0+T_0}f(t)\sin\left(k\omega_0 t\right)\mathrm{d}t$$

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(5) Harmonic Form (Frequency-Term Merging)

$$f(t) = a_0 + \sum_{k=1}^{\infty}\left[a_k \cos\left(k\omega_0 t\right) + b_k \sin\left(k\omega_0 t\right)\right]$$

Merged Harmonic Form (Cosine-Phase Representation)

$$f(t) = c_0 + \sum_{k=1}^{\infty} c_k \cos\left(k\omega_0 t + \varphi_k\right)$$

$k = 0, \quad c_0 = a_0$ **DC Component, Average Value**

$k = 1$ **Fundamental wave, also called the 1st harmonic in some literature**

k **k-th harmonic**

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(5) Harmonic Form (Frequency-Term Merging)

$$a_k \cos(k\omega_0 t) + b_k \sin(k\omega_0 t)$$
$$= c_k \cos(k\omega_0 t + \varphi_k)$$
$$= c_k \cos(k\omega_0 t)\cos(\varphi_k) - c_k \sin(k\omega_0 t)\sin(\varphi_k)$$

$$\boxed{\cos(\alpha+\beta) = \cos(\alpha)\cos(\beta) - \sin(\alpha)\sin(\beta)}$$

Compare Coefficients, get $a_k = c_k \cos(\varphi_k)$ $\quad b_k = -c_k \sin(\varphi_k)$

Solve for c_k and φ_k

$$c_k^2 = a_k^2 + b_k^2 \qquad \tan(\varphi_k) = -\frac{b_k}{a_k}$$

$$c_k = \sqrt{a_k^2 + b_k^2} \quad (c_k \ge 0) \qquad \varphi_k = \arctan\left(-\frac{b_k}{a_k}\right)?$$

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Example 1

Given $f_1(t) = \sin\omega_0 t$, $f_2(t) = \sqrt{3}\cos\omega_0 t$, combining the two same frequency terms to get $f(t)=f_1(t)+f_2(t)$

$$a_1 = \sqrt{3}, b_1 = 1, \text{ Thus } c_1 (\geq 0)$$

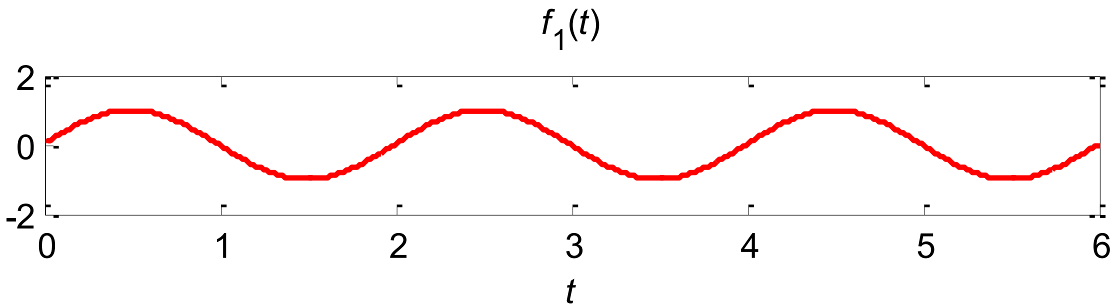
$$c_1 = \sqrt{a_1^2 + b_1^2} = 2$$

$$\varphi_1 = -\arctan\frac{b_1}{a_1} = -\arctan\frac{1}{\sqrt{3}} = -\frac{\pi}{6} \quad (\text{rad/s})$$

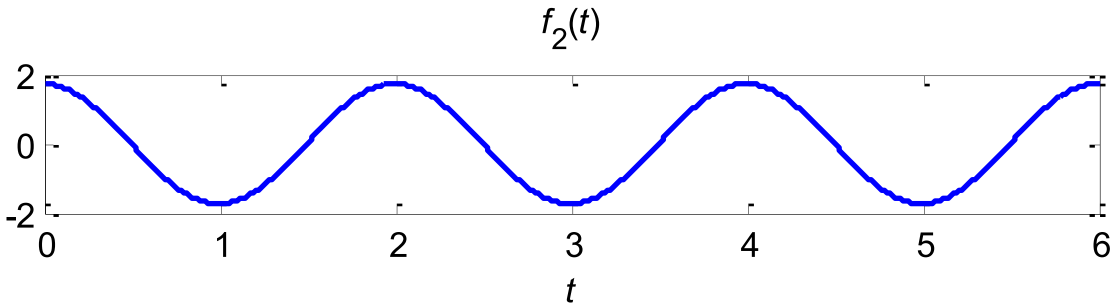
$$f(t) = 2\cos\left(\omega_0 t - \frac{\pi}{6}\right)$$

Example 1

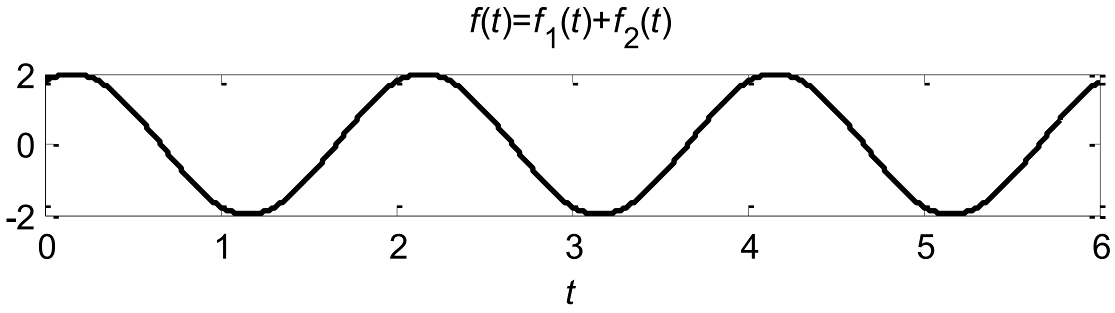
$$f_1(t) = \sin \omega_0 t$$



$$f_2(t) = \sqrt{3} \cos \omega_0 t$$



$$f(t) = f_1(t) + f_2(t)$$



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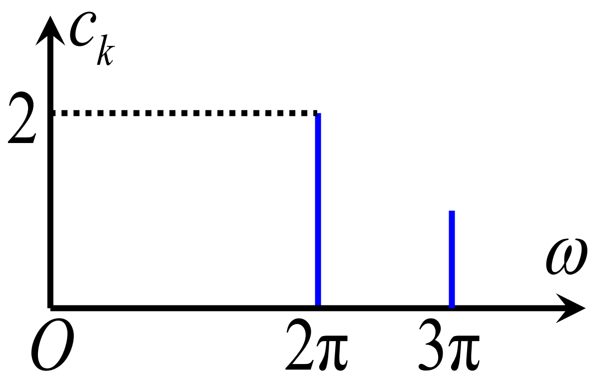
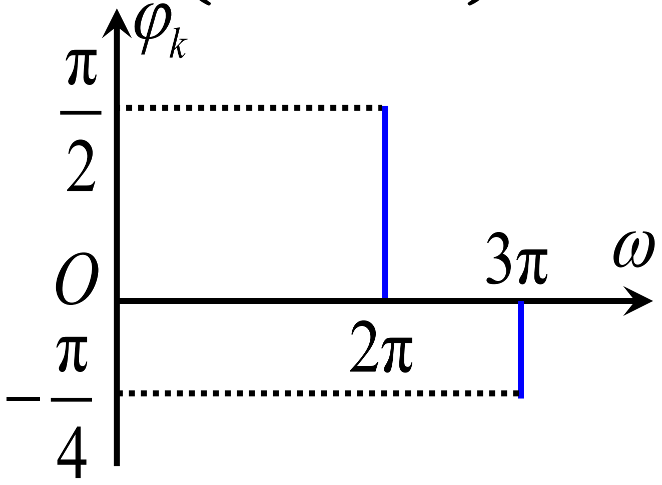
(6) Spectrum Diagram

$$f(t) = 2\cos\left(2\pi t + \frac{\pi}{2}\right) + \cos\left(3\pi t - \frac{\pi}{4}\right)$$





$c_k \sim \omega$ **curve is called the amplitude spectrum diagram**

$\varphi_k \sim \omega$ **curve is called the phase spectrum diagram**

2. Exponential Form of Fourier Series

(1) From Trigonometric Functions to Exponential Functions

$$f(t) = c_0 + \sum_{k=1}^{\infty} c_k \cos\left(k\omega_0 t + \varphi_k\right)$$

$$c_k \cos\left(k\omega_0 t + \varphi_k\right)$$

Euler's formula

$$\frac{1}{2} c_k \left[\mathrm{e}^{\mathrm{j}\left(k\omega_0 t + \varphi_k\right)} + \mathrm{e}^{-\mathrm{j}\left(k\omega_0 t + \varphi_k\right)} \right]$$

$$= \frac{1}{2} c_k \mathrm{e}^{\mathrm{j}\varphi_k} \mathrm{e}^{\mathrm{j}k\omega_0 t} + \frac{1}{2} c_k \mathrm{e}^{-\mathrm{j}\varphi_k} \mathrm{e}^{-\mathrm{j}k\omega_0 t}$$

$$= F_k \mathrm{e}^{\mathrm{j}k\omega_0 t} + F_{-k} \mathrm{e}^{-\mathrm{j}k\omega_0 t}$$

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(2) Exponential Form of Fourier Series Expression

$\left\{\mathrm{e}^{\mathrm{j}k\omega_0 t}\right\}, k = 0, \pm 1, \pm 2 \cdots$ is orthogonal function set

Series form

$$f(t) = \sum_{k=-\infty}^{\infty} F_k \, \mathrm{e}^{\mathrm{j}k\omega_0 t}$$

Fourier series coefficients

It can also be written as $F\left(k\omega_0\right)$

$$F_k = \frac{\int_{t_0}^{t_0+T_1} f(t)\,\mathrm{e}^{-\mathrm{j}k\omega_0 t}\,\mathrm{d}t}{\int_{t_0}^{t_0+T_1} \mathrm{e}^{\mathrm{j}k\omega_0 t}\mathrm{e}^{-\mathrm{j}k\omega_0 t}\,\mathrm{d}t}$$

Projection coefficient

$$= \frac{1}{T_1}\int_{t_0}^{t_0+T_1} f(t)\mathrm{e}^{-\mathrm{j}k\omega_0 t}\mathrm{d}t$$

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(3) Amplitude Spectrum and Phase Spectrum

$$f(t) = \sum_{k=-\infty}^{\infty} F_k \, \mathrm{e}^{\mathrm{j}k\omega_0 t}$$

F_k is a complex coefficient and can be expressed in polar form as $F_k = |F_k| \mathrm{e}^{\mathrm{j}\varphi_k}$

$$f(t) = \sum_{k=-\infty}^{\infty} |F_k| \mathrm{e}^{\mathrm{j}\varphi_k} \, \mathrm{e}^{\mathrm{j}k\omega_0 t} = \sum_{k=-\infty}^{\infty} |F_k| \mathrm{e}^{\mathrm{j}(k\omega_0 t + \varphi_k)}$$

$|F_k| \sim \omega$ Amplitude spectrum

$\varphi_k \sim \omega$ Phase spectrum

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(4) Relationship Between the Two Forms of Fourier Series

Assuming $f(t)$ is a real-valued function and $k>0$

$$F_k = \frac{1}{T_1}\int_{T_1} f(t)\mathrm{e}^{-\mathrm{j}k\omega_0 t}\,\mathrm{d}t$$

$$= \frac{1}{T_1}\int_{T_1} f(t)\cos\left(k\omega_0 t\right)\mathrm{d}t - \mathrm{j}\frac{1}{T_1}\int_{T_1} f(t)\sin\left(k\omega_0 t\right)\mathrm{d}t$$

$$= \frac{1}{2}\left(a_k - \mathrm{j}b_k\right)$$

$$F_{-k} = \frac{1}{T_1}\int_{T_1} f(t)\cos\left(k\omega_0 t\right)\mathrm{d}t + \mathrm{j}\frac{1}{T_1}\int_{T_1} f(t)\sin\left(k\omega_0 t\right)\mathrm{d}t$$

$$= \frac{1}{2}\left(a_k + \mathrm{j}b_k\right)$$

(4) Relationship Between the Two Forms of Fourier Series

$$\begin{cases} F_k = \dfrac{1}{2}\left(a_k - \mathrm{j} b_k\right) \\ F_{-k} = \dfrac{1}{2}\left(a_k + \mathrm{j} b_k\right) \end{cases} \qquad \begin{cases} F_k = \left|F_k\right| \mathrm{e}^{\mathrm{j}\varphi_k} \\ F_{-k} = \left|F_{-k}\right| \mathrm{e}^{\mathrm{j}\varphi_{-k}} \end{cases}$$

Amplitude-Frequency Characteristic

$$\left|F_k\right| = \left|F_{-k}\right| = \frac{1}{2}\sqrt{a_k^2 + b_k^2} = \frac{1}{2} c_k$$

Phase-Frequency Characteristic

$$\varphi_k = \arctan\left(-\frac{b_k}{a_k}\right)$$

$$\varphi_k = -\varphi_{-k}$$

$$\varphi_{-k} = \arctan\left(\frac{b_k}{a_k}\right)$$

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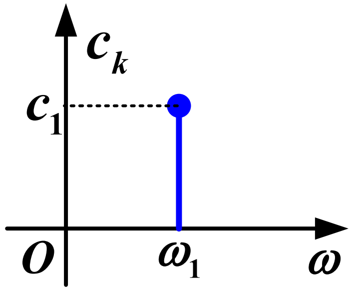
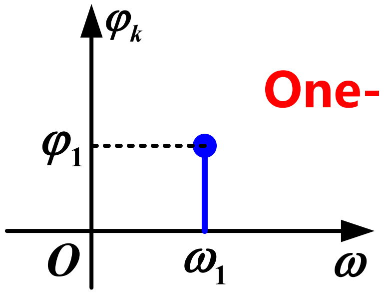
Example 2

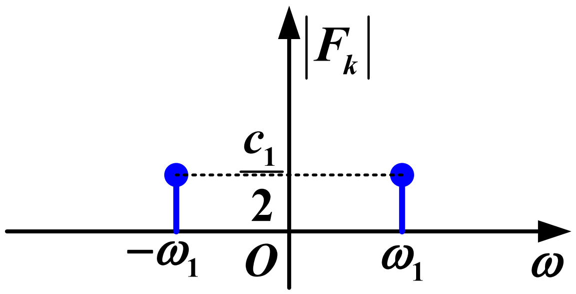
$$f(t)=c_1\cos(\omega_1 t+\varphi_1)=\frac{1}{2}c_1\left[e^{-j(\omega_1 t+\varphi_1)}+e^{j(\omega_1 t+\varphi_1)}\right]$$

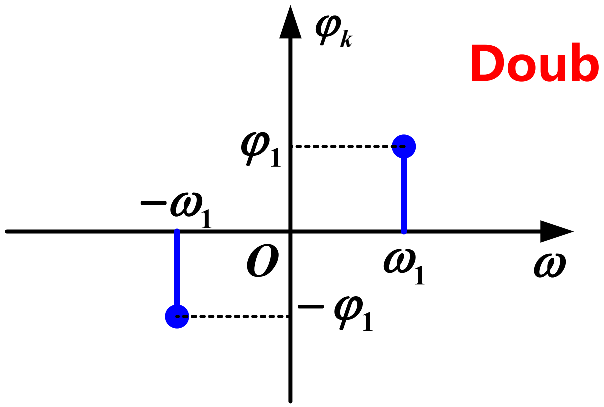




One-Sided Spectrum

$$f(t)=\frac{1}{2}c_1 e^{-j\varphi_1}e^{-j\omega_1 t}+\frac{1}{2}c_1 e^{j\varphi_1}e^{j\omega_1 t}$$





Double-Sided Spectrum

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Example 3

$$f(t) = 1 + \sin\omega_0 t + \sqrt{3}\cos\omega_0 t - \cos\left(2\omega_0 t + \frac{5\pi}{4}\right)$$

Plot the Amplitude spectrum and phase spectrum

$$\text{Let } f_1(t) = \sin\omega_0 t + \sqrt{3}\cos\omega_0 t$$

Merge terms with the same frequency and convert to cosine form

Utilizing the result from Example 1

$$f_1(t) = 2\cos\left(\omega_0 t - \frac{\pi}{6}\right)$$

$$-\cos\left(2\omega_0 t + \frac{5\pi}{4}\right)$$

$$= \cos\left(2\omega_0 t + \frac{5\pi}{4} - \pi\right)$$

$$= \cos\left(2\omega_0 t + \frac{\pi}{4}\right)$$

3. Fourier Series Analysis of Periodic Signals

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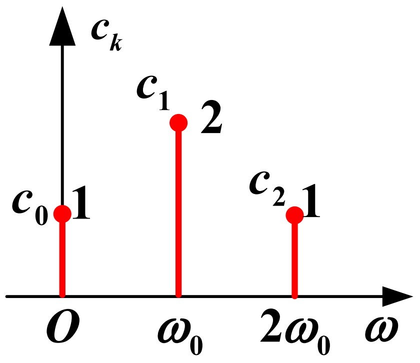
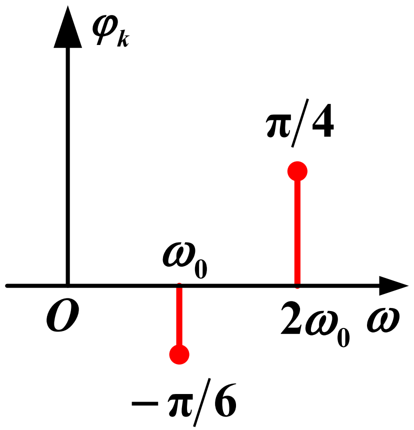
Example 3

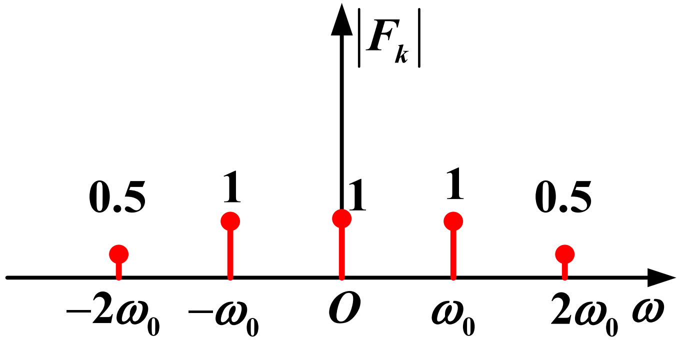
$$f(t) = 1 + 2\cos\left(\omega_0 t - \frac{\pi}{6}\right) + \cos\left(2\omega_0 t + \frac{\pi}{4}\right)$$

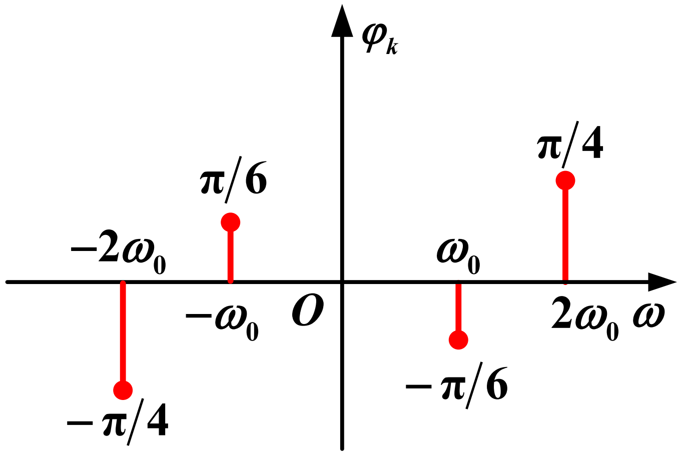
One-Sided Spectrum





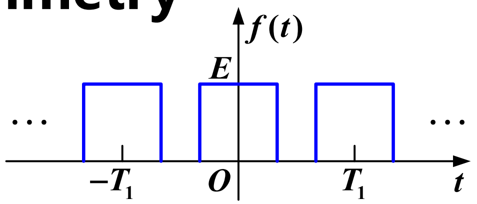
Double-Sided Spectrum



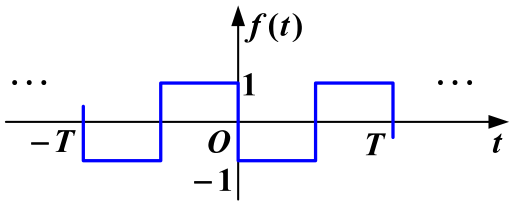


3. Relationship Between Function Symmetry and Fourier Series Coefficients

Whole-cycle symmetry

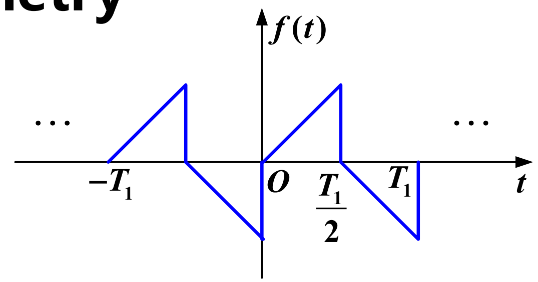


$$f(t) = f(-t)$$

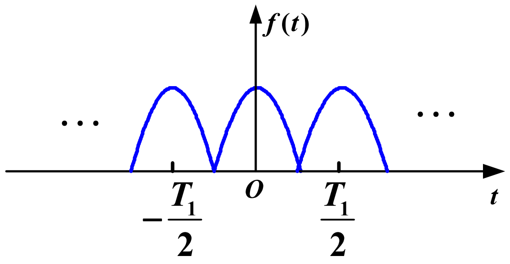


$$f(t) = -f(-t)$$

Half-cycle symmetry



$$f(t) = -f\left(t \pm \frac{T_1}{2}\right)$$

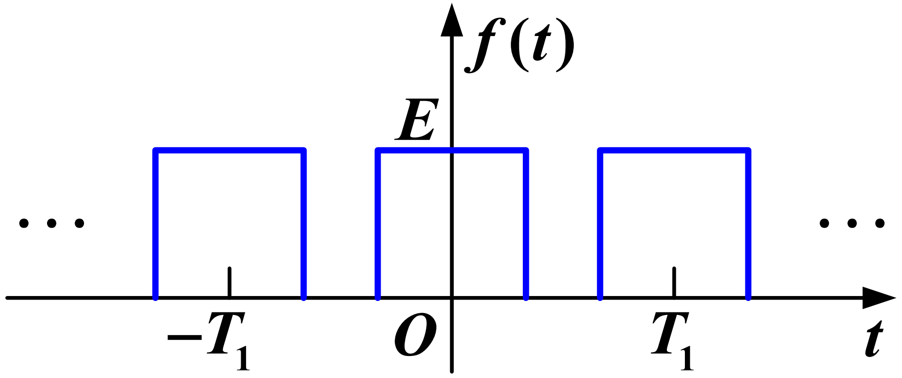


$$f(t) = f\left(t \pm \frac{T_1}{2}\right)$$

(1) Even Function

A signal waveform is symmetric with respect to the y-axis

$$f(t) = f(-t)$$

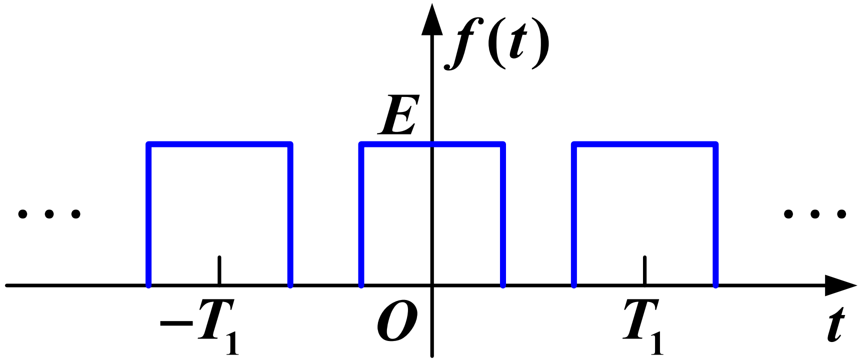


The sine function is an odd function, so even functions do not have odd function components

$$b_k = 0$$

$$a_k = \frac{4}{T_1}\int_0^{\frac{T_1}{2}} f(t)\cos(k\omega_0 t)\,\mathrm{d}t \neq 0$$

(1) Even Function



$$b_k = 0 \qquad a_k \neq 0$$

$$F_k = \frac{1}{2}\left(a_k - \mathrm{j}b_k\right) = \frac{1}{2}a_k$$

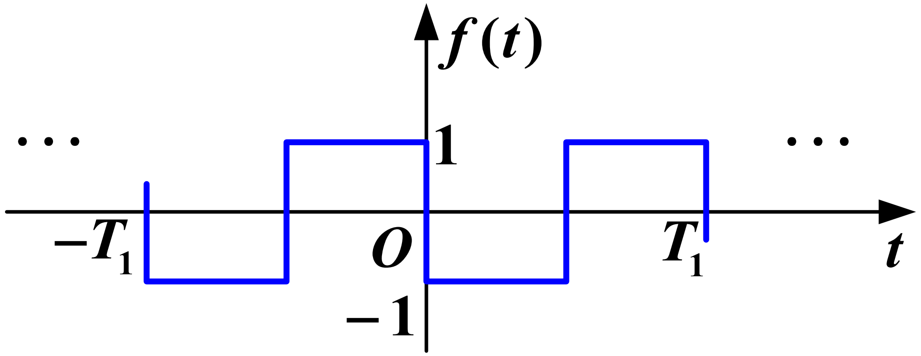
F_k is a **real** function

Time-domain even function → Frequency-domain real function

(2) Odd Function

The waveform is antisymmetric about the vertical axis

$$f(t) = -f(-t)$$



$$a_0 = \frac{1}{T_1}\int_{-\frac{T_1}{2}}^{\frac{T_1}{2}} f(t)\,\mathrm{d}t = 0$$

$$a_k = \frac{2}{T_1}\int_{-\frac{T_1}{2}}^{\frac{T_1}{2}} f(t)\cos(k\omega_0 t)\,\mathrm{d}t = 0$$ **No cosine term**

$$b_k = \frac{4}{T_1}\int_{0}^{\frac{T_1}{2}} f(t)\sin(k\omega_0 t)\,\mathrm{d}t \neq 0$$

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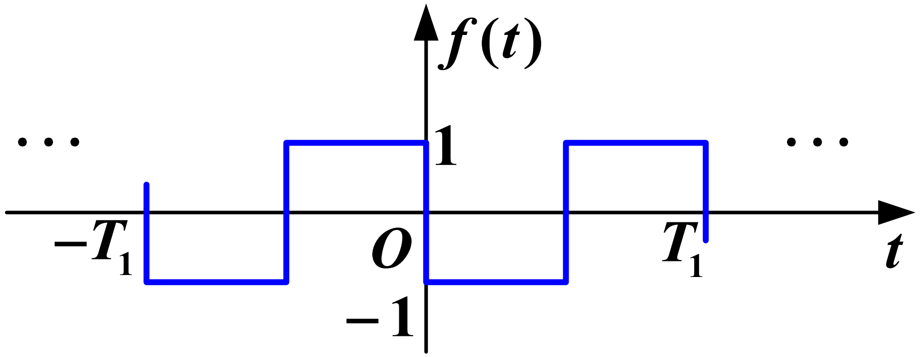
(2) Odd Function

The waveform is antisymmetric about the vertical axis $f(t) = -f(-t)$



$$a_0 = 0 \qquad a_k = 0 \qquad b_k \neq 0$$

$$F_k = \frac{1}{2}(a_k - \mathrm{j}b_k) = -\frac{1}{2}\mathrm{j}b_k$$

F_k is an **imaginary function**

Time-domain odd function → frequency-domain imaginary function

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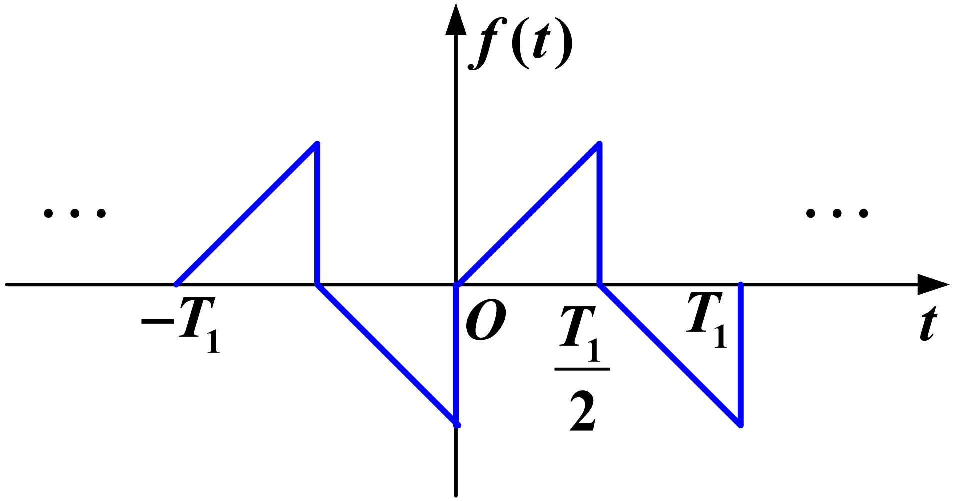
(3) Odd-Harmonic Function

A function is odd-harmonic (or possesses half-wave symmetry) if shifting it by half a period and inverting its amplitude leaves the waveform unchanged:

$$f(t) = -f\left(t \pm \frac{T_1}{2}\right)$$



$$a_0 = 0$$

$$k = 2, 4, 6 \cdots \qquad a_k = b_k = 0$$

Without even harmonic terms

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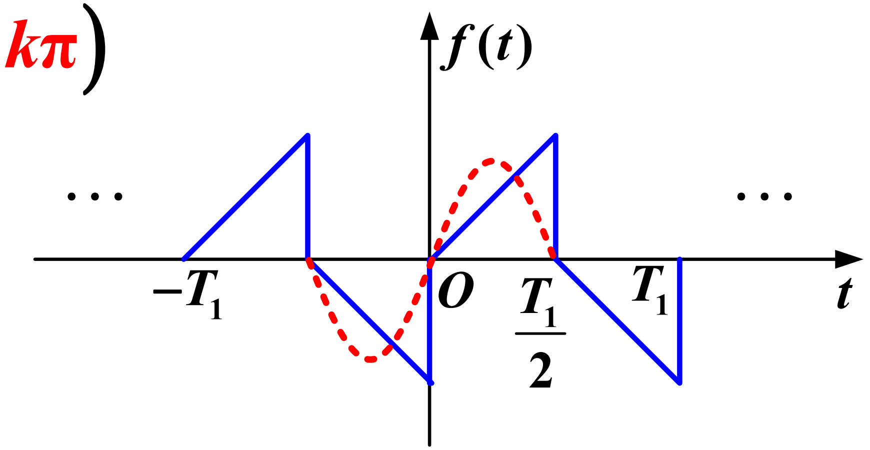
(3) Odd-Harmonic Function

Taking the sine terms as an example, the cosine-coefficient calculation is similar

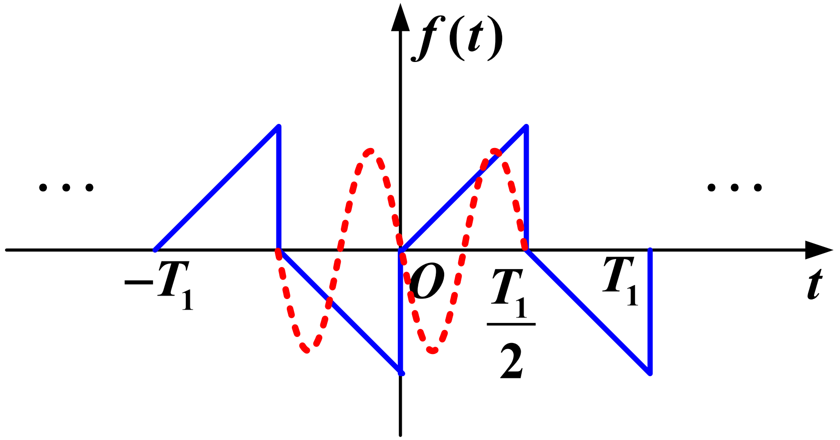
$$\sin\left[k\omega_0\left(t\pm\frac{T}{2}\right)\right]=\sin\left(k\omega_0 t\pm k\pi\right)$$

$$=\begin{cases}-\sin\left(k\omega_0 t\right) & k \text{ odd}\\ \sin\left(k\omega_0 t\right) & k \text{ even}\end{cases}$$



$$f\left(t\pm\frac{T}{2}\right)\sin\left[k\omega_0\left(t\pm\frac{T}{2}\right)\right]$$

$$=\begin{cases}f\left(t\right)\sin\left(k\omega_0 t\right) & k \text{ odd}\\ -f\left(t\right)\sin\left(k\omega_0 t\right) & k \text{ even}\end{cases}$$



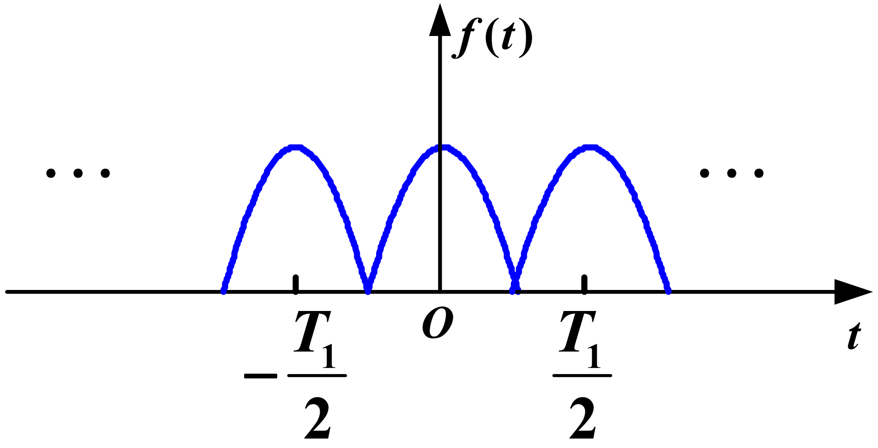
Without even harmonic terms

(4) Even-Harmonic Function

Effective Period is Half the Labeled Period

A function is even-harmonic if shifting it by half its labeled period $\pm\frac{T_1}{2}$ results in an identical waveform

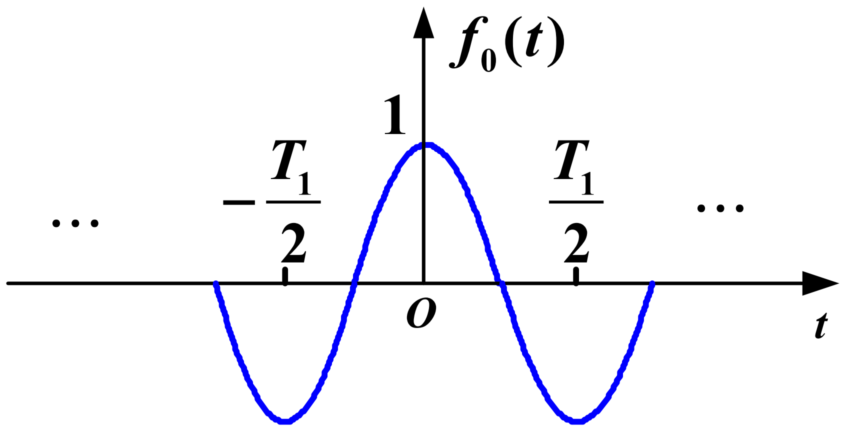
$$f(t) = f\left(t \pm \frac{T_1}{2}\right) \qquad \omega_1 = \frac{2\pi}{T_1}$$

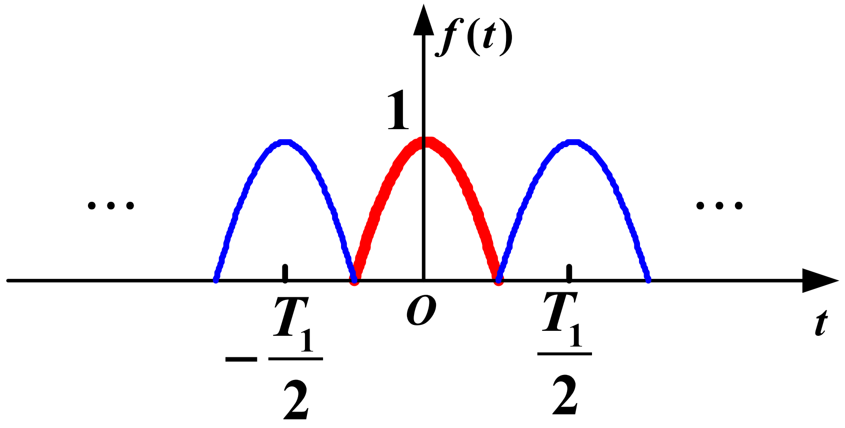


$$\text{When } k = 1,3,5\cdots \qquad a_k = b_k = 0$$

(4) Even-Harmonic Function

Effective Period is Half the Labeled Period





- T_1 **Labeled period**

$$\omega_1 = \frac{2\pi}{T_1}$$ **Fundamental angular frequency**

- $\frac{T_1}{2}$ **Actual period**

The actual frequency components included are integer multiples of

$$\frac{2\pi}{\frac{T_1}{2}} = 2\omega_1$$

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4. Parseval’ s Theorem

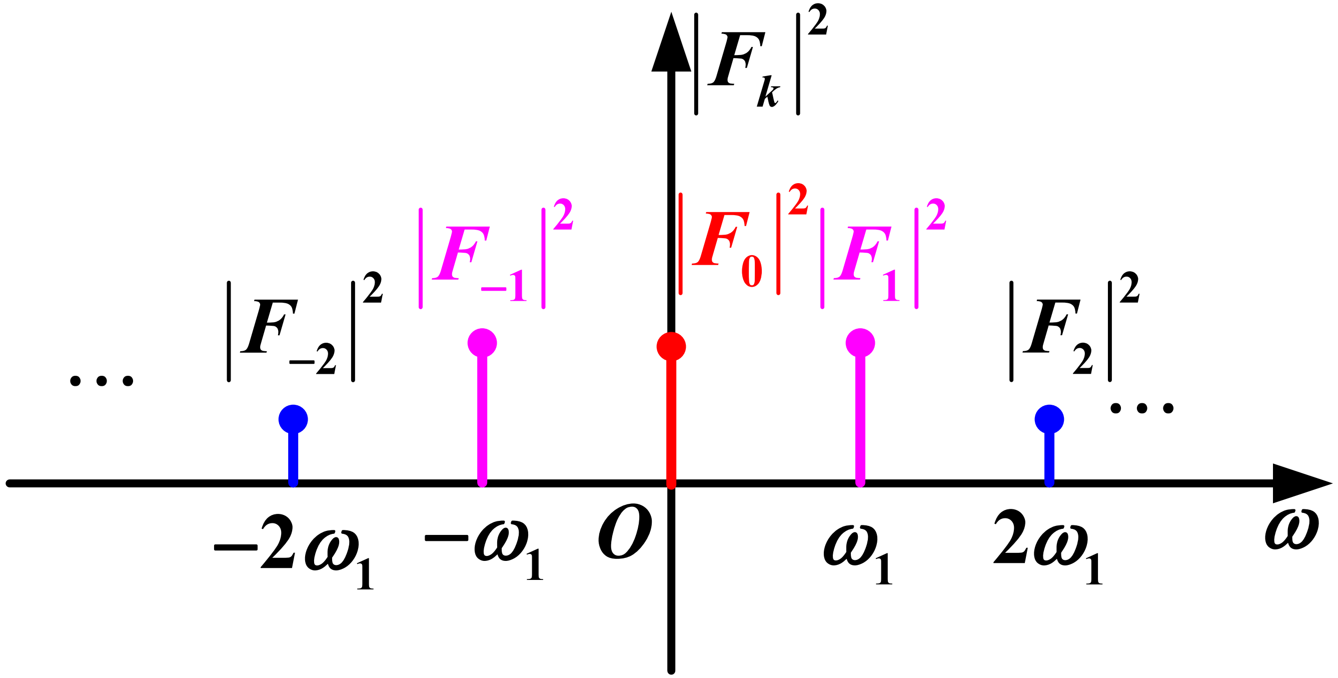
$$P = \frac{1}{T_1}\int_0^{T_1} \left|f(t)\right|^2 \mathrm{d}t = \begin{cases} a_0^2 + \frac{1}{2}\sum_{k=1}^{\infty}\left(a_k^2 + b_k^2\right) & \text{Trigonometric Form} \\ c_0^2 + \frac{1}{2}\sum_{k=1}^{\infty} c_k^2 & \text{Harmonic Form} \\ \sum_{k=-\infty}^{\infty}\left|F\left(k\omega_0\right)\right|^2 & \text{Exponential Form} \end{cases}$$

This is a specific manifestation of Parseval's theorem in the context of Fourier series

It indicates that: the average power of a periodic signal equals the sum of the squares of the effective values of the DC component, fundamental wave, and each harmonic component; in other words, energy is conserved between the time domain and the frequency domain

4. Parseval's Theorem

$|F_k|^2 \sim \omega$ **: the distribution of the average power of each harmonic component with respect to frequency is called the power spectral coefficients**



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5. Finite Fourier Series and Minimum Mean Square Error

$$f(t) = a_0 + \sum_{k=1}^{\infty}\left[a_k \cos(k\omega_0 t) + b_k \sin(k\omega_0 t)\right]$$

Approximate $f(t)$ by taking the first N harmonic terms

$$S_N(t) = a_0 + \sum_{k=1}^{N}\left[a_k \cos(k\omega_0 t) + b_k \sin(k\omega_0 t)\right]$$

Error function

$$\varepsilon_N(t) = f(t) - S_N(t)$$

Mean Square Error

$$E_N = \overline{\varepsilon_N^{\ 2}(t)} = P - \left[a_0^{\ 2} + \frac{1}{2}\sum_{k=1}^{N}\left(a_k^{\ 2} + b_k^{\ 2}\right)\right]$$

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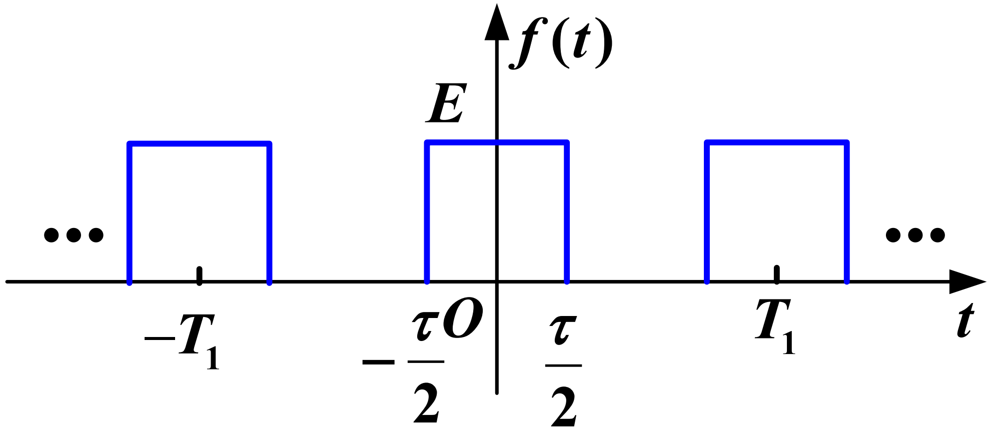
6. Spectral Structure of Periodic Rectangular Pulse Trains

(1) Spectral Coefficients in Trigonometric Form



Pulse width τ

Period T_1

Amplitude E

$$a_0 = E\frac{\tau}{T_1} \qquad \left(\frac{\tau}{T_1}: \text{Duty Cycle}\right)$$

$$f(t) \text{ is an even function} \rightarrow b_k = 0 \quad F_k = \frac{1}{2}(a_k - \mathrm{j}b_k) = \frac{1}{2}a_k$$

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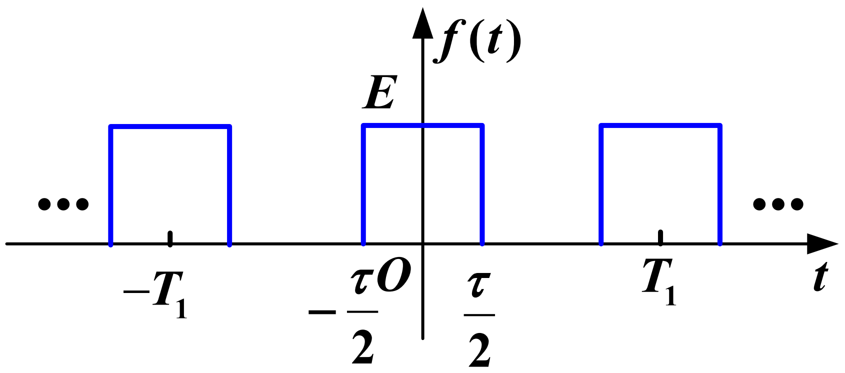
(2) Spectral Coefficients in Exponential Form

$$F_k = \frac{1}{T_1}\int_{-\frac{T_1}{2}}^{\frac{T_1}{2}} f(t)\mathrm{e}^{-\mathrm{j}k\omega_1 t}\,\mathrm{d}t$$

$$= \frac{1}{T_1}\int_{-\frac{\tau}{2}}^{\frac{\tau}{2}} E\mathrm{e}^{-\mathrm{j}k\omega_1 t}\,\mathrm{d}t$$

$$= \frac{E}{T_1}\frac{1}{-\mathrm{j}k\omega_1}\mathrm{e}^{-\mathrm{j}k\omega_1 t}\Big|_{-\frac{\tau}{2}}^{\frac{\tau}{2}}$$

$$= \frac{-E}{\mathrm{j}k\omega_1 T_1}\left(\mathrm{e}^{-\mathrm{j}k\omega_1\frac{\tau}{2}} - \mathrm{e}^{\mathrm{j}k\omega_1\frac{\tau}{2}}\right)$$



$$\omega_1 = \frac{2\pi}{T_1}$$

(2) Spectral Coefficients in Exponential Form

$$F_k = \frac{-E}{\mathrm{j}k\omega_1 T_1}\left(\mathrm{e}^{-\mathrm{j}k\omega_1\frac{\tau}{2}} - \mathrm{e}^{\mathrm{j}k\omega_1\frac{\tau}{2}}\right)$$

$$= \frac{2E}{k\omega_1 T_1}\sin\left(k\omega_1\frac{\tau}{2}\right)$$

$$= \frac{E\tau}{T_1}\frac{\sin\left(k\omega_1\frac{\tau}{2}\right)}{k\omega_1\frac{\tau}{2}}$$

$$= \frac{E\tau}{T_1}\mathrm{Sa}\left(k\omega_1\frac{\tau}{2}\right)$$

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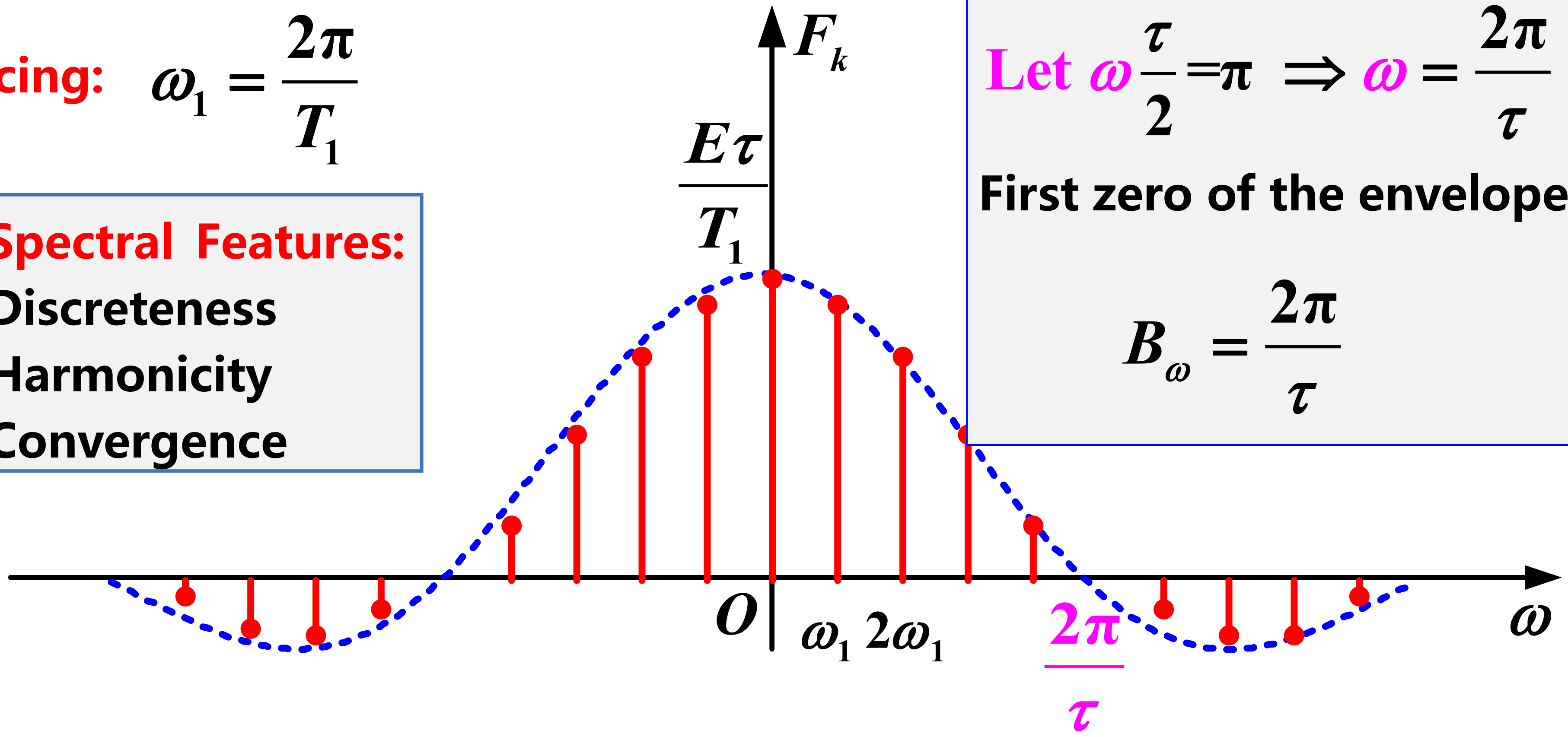
(3) Spectrum and Its Characteristics $(T_1 = 5\tau)$

$$F_k = \frac{E\tau}{T_1}\mathrm{Sa}\left(k\omega_1\frac{\tau}{2}\right)$$

Spectral Coefficients: $\frac{E\tau}{T_1}\mathrm{Sa}\left(\omega\frac{\tau}{2}\right)$

Line spacing: $\omega_1 = \frac{2\pi}{T_1}$

Spectral Features:
Discreteness
Harmonicity
Convergence

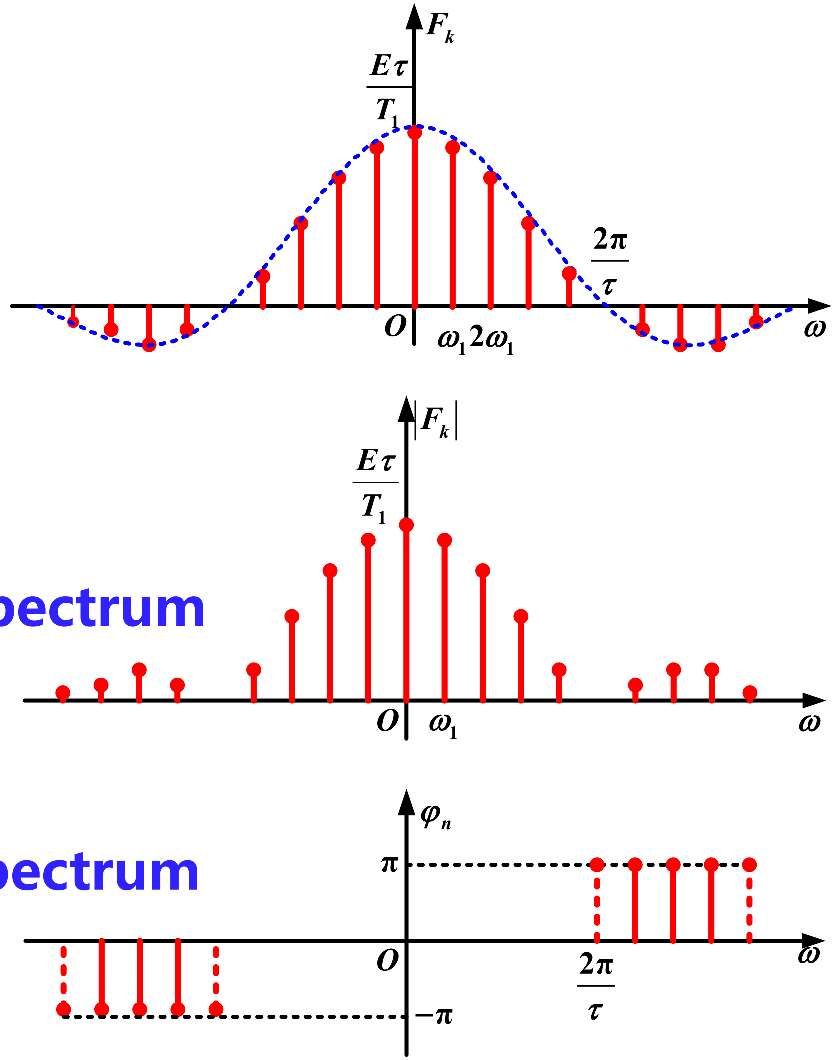


Let $\omega\frac{\tau}{2}=\pi \Rightarrow \omega = \frac{2\pi}{\tau}$

First zero of the envelope:

$$B_\omega = \frac{2\pi}{\tau}$$

(3) Spectrum and Its Characteristics



Amplitude spectrum

Phase spectrum

- $F_k > 0$:

$$F_k = |F_k| \cdot 1 = |F_k| \cdot e^{j0}$$

Phase is 0

- $F_k < 0$:

$$F_k = |F_k| \cdot (-1) = |F_k| e^{j(\pm\pi)}$$

Phase is $\pm\pi$

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(3) Spectrum and Its Characteristics

$$F_k = E\frac{\tau}{T_1}\mathrm{Sa}\left(k\omega_1\frac{\tau}{2}\right)$$

- The spectral coefficients F_k are directly proportional to E
- Pulse width τ: The first zero of the envelope occurs at: $B_\omega = \frac{2\pi}{\tau}$

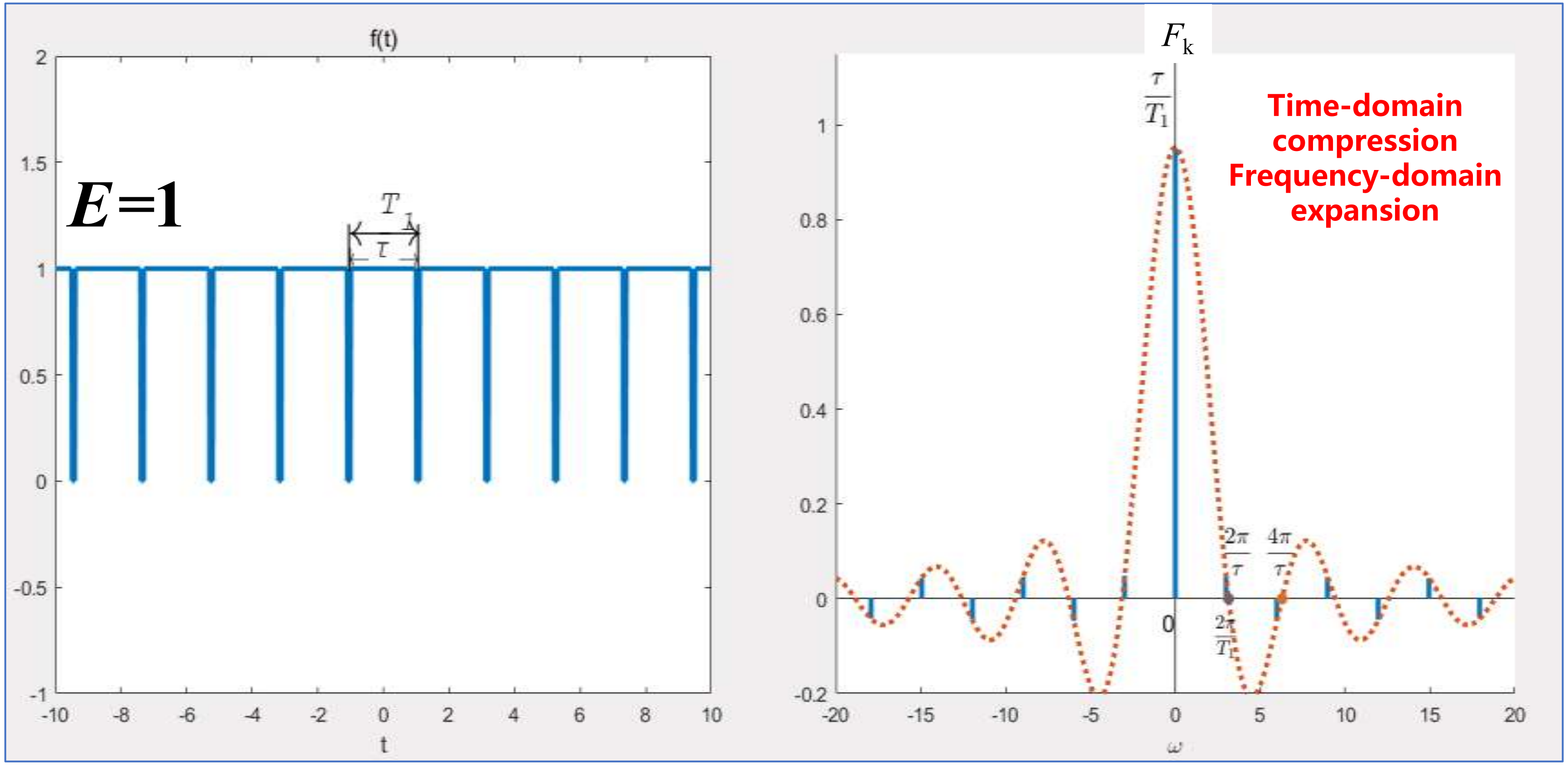
  This bandwidth B_ω is inversely proportional to τ

  Time-domain compression → Frequency-domain expansion
- Period T_1 : The spectral line spacing is determined by: $\omega_1 = \frac{2\pi}{T_1}$

  As T_1 increases, the spectral line spacing decreases

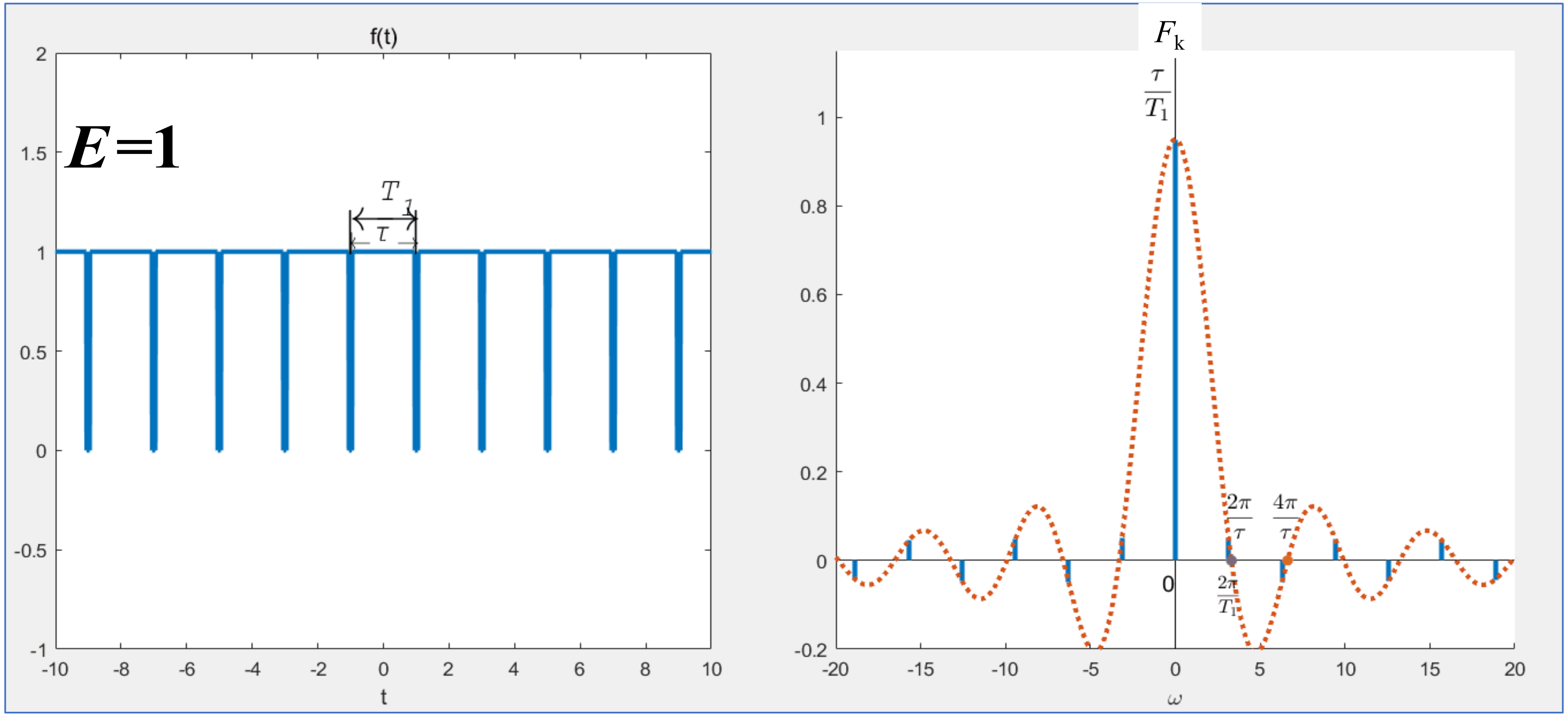
(3) Spectrum and Its Characteristics

$$F_k = E\frac{\tau}{T_1}\text{Sa}\left(k\omega_1\frac{\tau}{2}\right)$$ T_1 unchanged, τ decreases $\Rightarrow$ $\omega_1 = \frac{2\pi}{T_1}$ unchanged, $B_\omega = \frac{2\pi}{\tau}$ increases



(3) Spectrum and Its Characteristics

$F_k = E\frac{\tau}{T_1}\mathrm{Sa}\left(k\omega_1\frac{\tau}{2}\right)$ τ **unchanged,** T_1 **increases** $\Rightarrow$ $\omega_1 = \frac{2\pi}{T_1}$ **decreases,** $B_\omega = \frac{2\pi}{\tau}$ **unchanged**



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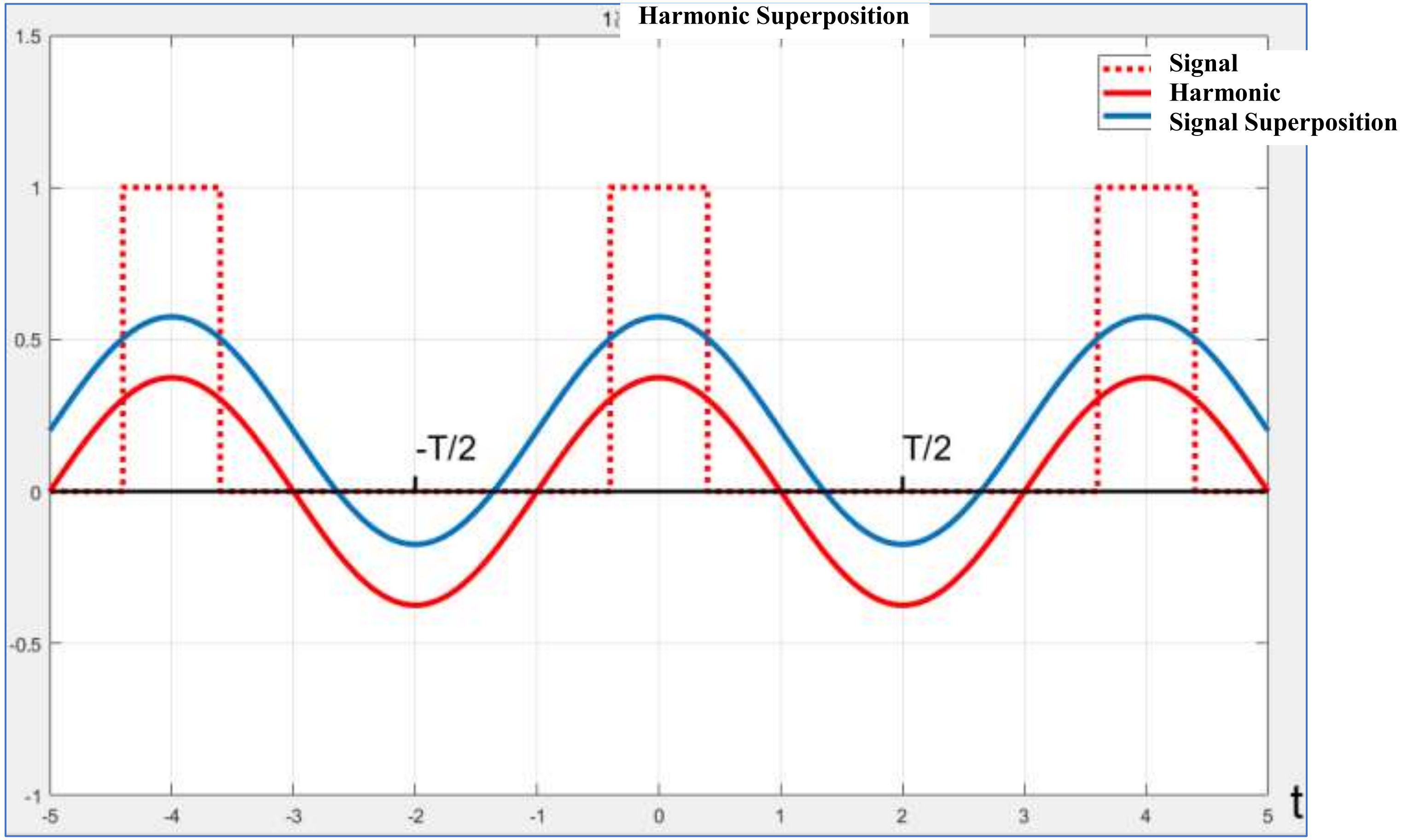
(4) Fourier Series Synthesis

$$f(t)=\sum_{k=-\infty}^{\infty} E\frac{\tau}{T_1}\mathrm{Sa}\left(k\omega_1\frac{\tau}{2}\right)\mathrm{e}^{\mathrm{j}k\omega_1 t}\quad\left(\omega_1=\frac{2\pi}{T_1}\right)$$

$$=E\frac{\tau}{T_1}+2E\frac{\tau}{T_1}\sum_{k=1}^{\infty}\mathrm{Sa}\left(k\omega_1\frac{\tau}{2}\right)\cos(k\omega_1 t)$$

(4) Fourier Series Synthesis

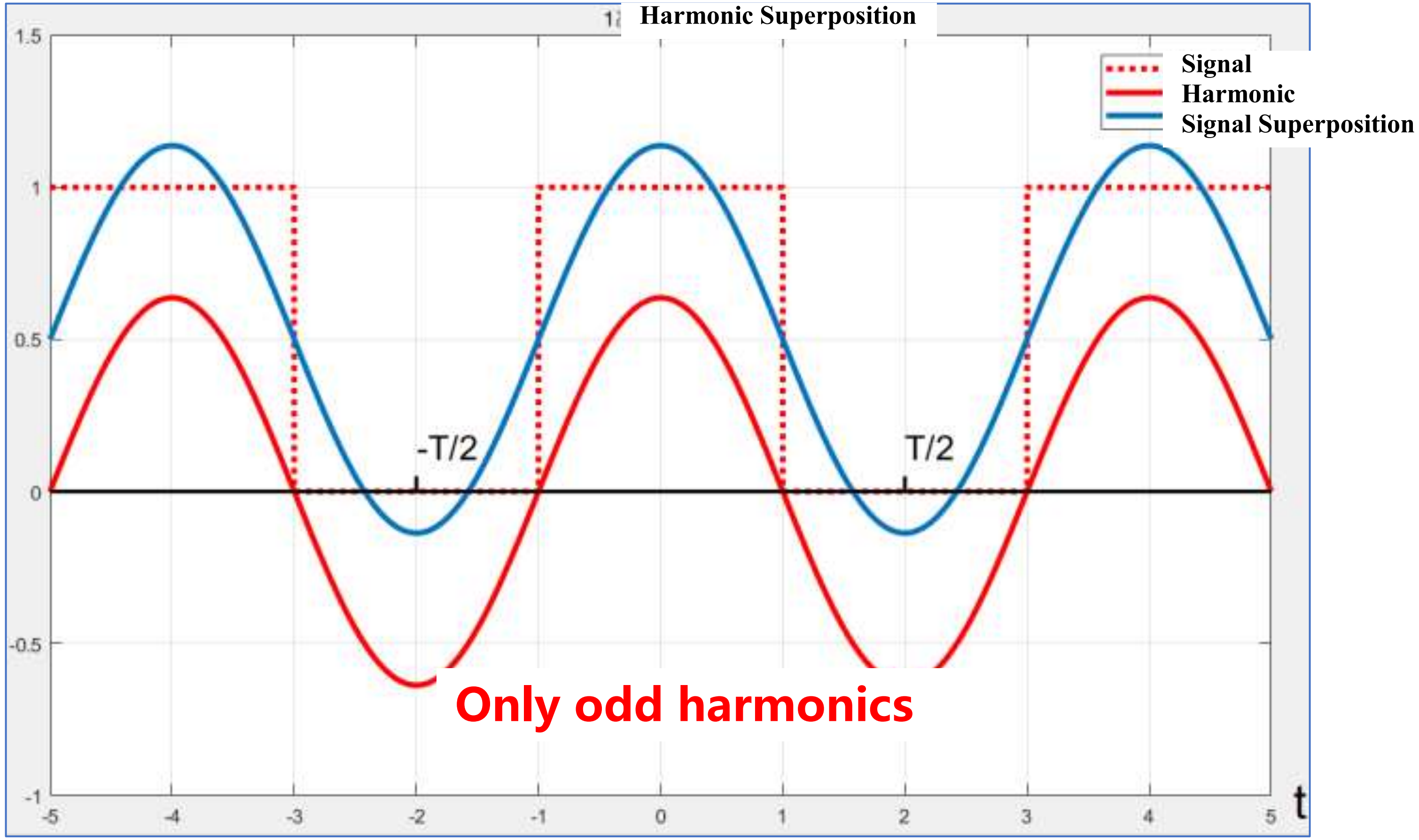
duty cycle of 20%



$$f(t) = E\frac{\tau}{T_1} + 2E\frac{\tau}{T_1}\sum_{k=1}^{\infty}\text{Sa}\left(k\omega_1\frac{\tau}{2}\right)\cos\left(k\omega_1 t\right)$$

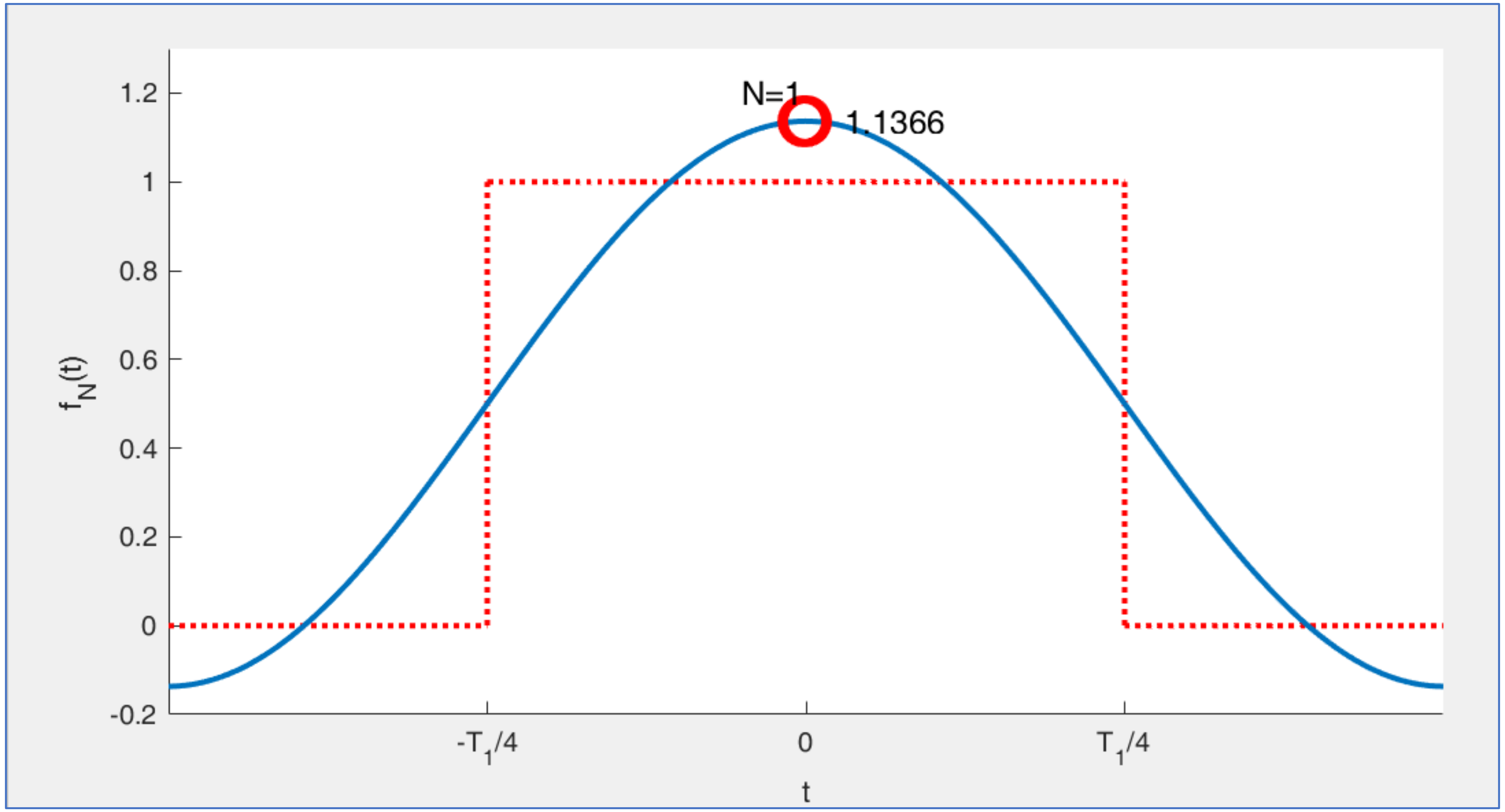
(4) Fourier Series Synthesis

duty cycle of 50%



$$f(t)=\frac{E}{2}+\frac{2E}{\pi}\left[\cos(\omega_1 t)-\frac{1}{3}\cos(3\omega_1 t)+\frac{1}{5}\cos(5\omega_1 t)+\cdots\right]$$

7. Gibbs Phenomenon



The more terms are included, the smaller the mean square error becomes, while the peak of the fluctuations approaches a constant value - approximately 9% of the total jump discontinuity. This is known as the Gibbs phenomenon

7. Gibbs Phenomenon

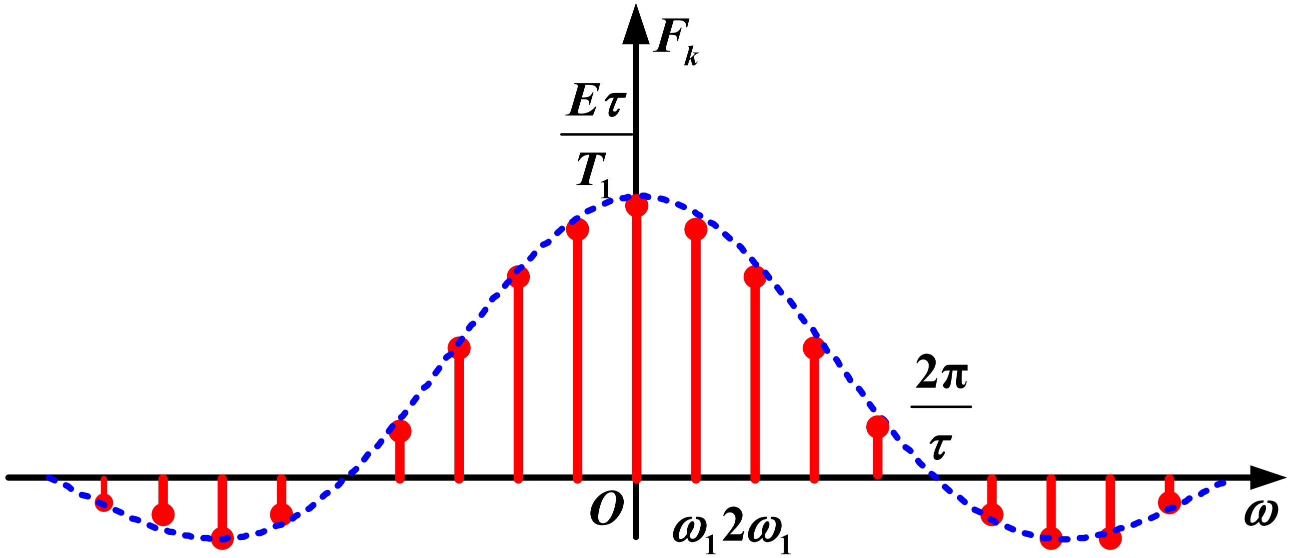
Fourier series decomposition follows the principle of minimizing mean square error in signal approximation. However, for signals with discontinuity points, the series fails to converge precisely at these discontinuities

In 1898, American physicist Albert Michelson observed this phenomenon while testing square wave signals using his self-designed harmonic analyzer. When attempting to synthesize signals, he noted persistent oscillations near discontinuities despite increasing the number of harmonic terms

The renowned mathematical physicist Josiah Gibbs investigated these findings and published his formal analysis in 1899

To understand its origin, we examine the ideal low-pass filter's behavior

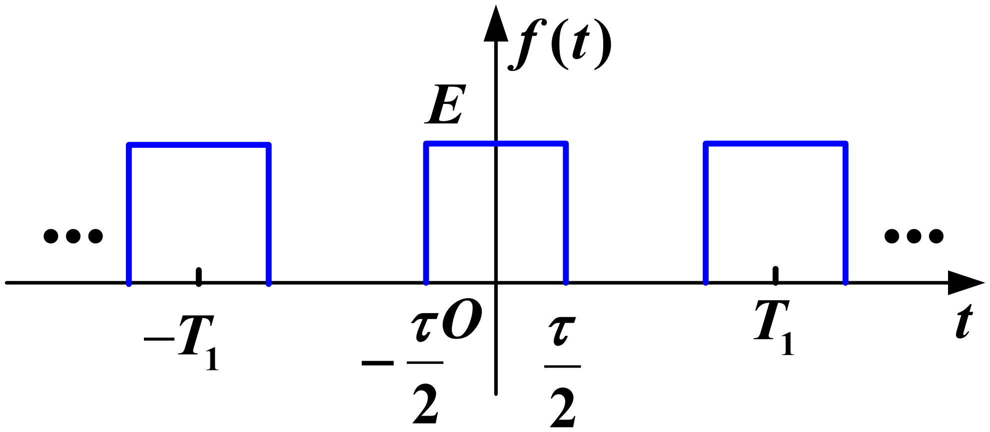
8. Bandwidth



Under specified distortion conditions, a signal can be represented by components within a certain frequency range. This frequency range is called the bandwidth

$B_\omega = \frac{2\pi}{\tau}$ **:The bandwidth is defined by the first zero-crossing point of the signal**

8. Bandwidth



Example: $\tau = \frac{1}{20}\text{s}, T_1 = \frac{1}{4}\text{s}$ the power of the signal is

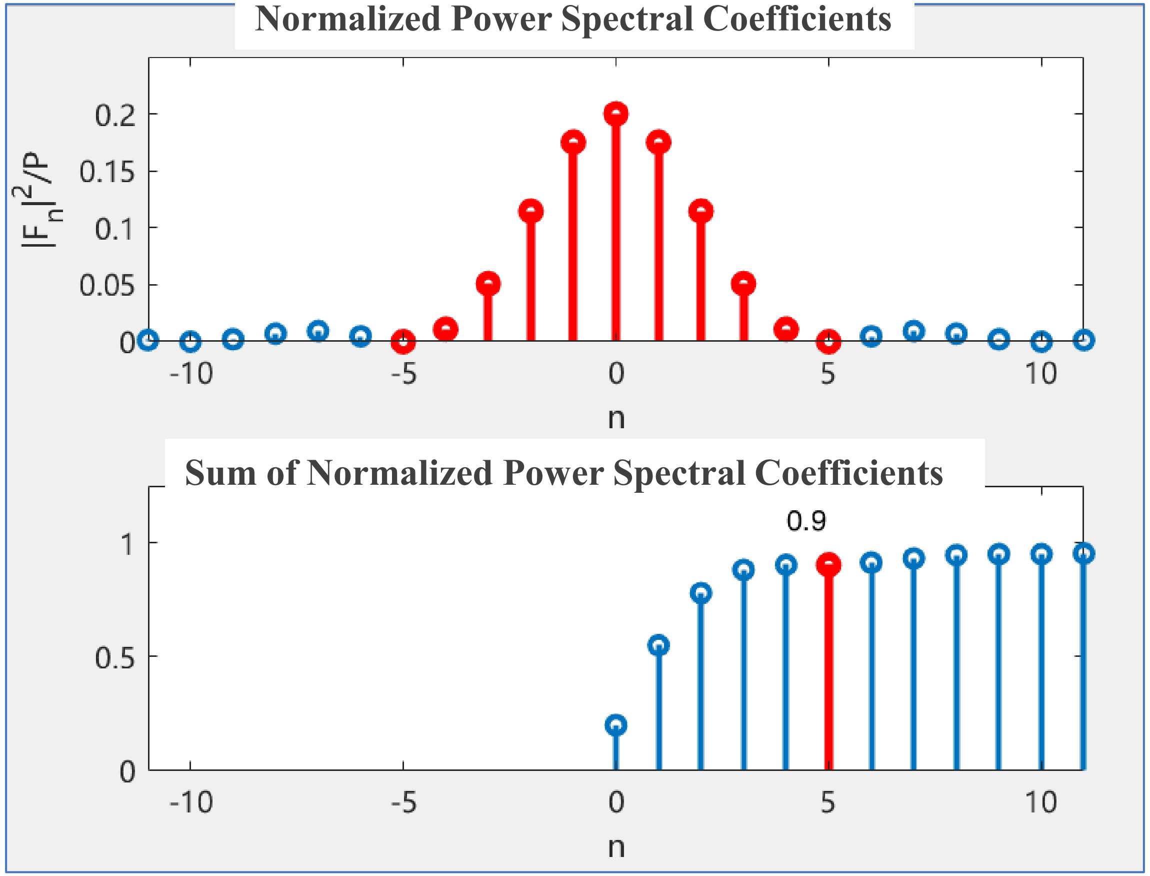
$$P = \frac{1}{T_1}\int_{-\frac{T_1}{2}}^{\frac{T_1}{2}} f^2(t)\mathrm{d}t$$

$$= \frac{1}{T_1}\int_{-\frac{\tau}{2}}^{\frac{\tau}{2}} E^2\mathrm{d}t = 0.2E^2$$

8. Bandwidth

The power of the signal within the main lobe of the spectrum is $P_5 = \sum_{n=-5}^{5} |F_k|^2 \approx 0.18E^2$

$$\frac{P_5}{P} \approx 90\%$$



8. Bandwidth

System's bandwidth > Signal's bandwidth, to avoid distortion

Speech Signal	300~3400Hz
Music Signal	50~15,000Hz
Loudspeaker	15~20,000Hz

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Summary

(1) The Fourier series of a periodic signal $f(t)$ has two forms

Trigonometric Form

$$f(t) = a_0 + \sum_{k=1}^{\infty}\left[a_k \cos\left(k\omega_1 t\right) + b_k \sin\left(k\omega_1 t\right)\right]$$

$$= c_0 + \sum_{k=1}^{\infty} c_k \cos\left(k\omega_1 t + \varphi_k\right)$$

Exponential Form

$$f(t) = \sum_{k=-\infty}^{\infty} F_k \mathrm{e}^{\mathrm{j}k\omega_1 t}$$

Summary

(2) Relationship Between the Two Forms of Fourier Series

- The trigonometric form: $c_k \sim \omega, \quad \varphi_k \sim \omega$ One-sided spectrum
- The exponential form: $\left|F_k\right| \sim \omega, \quad \varphi_k \sim \omega$ Double-sided spectrum

Relationship $\left|F_k\right| = \frac{1}{2} c_k \left(k \neq 0\right)$ $F_0 = c_0 = a_0$

- In the exponential form, the amplitude spectrum is an even function

$$\left|F_k\right| = \left|F_{-k}\right|$$

- The phase spectrum is an odd function

$$\varphi_k = -\varphi_{-k}$$

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Summary

(3) Three Properties

Discreteness: Spectral lines

Harmonicity: Frequency components occur only at $k\omega_1$

Convergence: $|k|\uparrow, \quad |F_k|\downarrow$

Note: These properties are fundamental to analyzing periodic signals but may not hold for non-periodic or singular signals (e.g., ideal impulse trains)

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Summary

(4) Introduction of Negative Frequencies

In two-sided spectra, negative frequencies ($n\omega_1$) are mathematical constructs without direct physical interpretation

Why Include Negative Frequencies?

$f(t)$ is a real function, decomposed into complex exponentials, and it must have a conjugate pair of $e^{jk\omega_1}$ and $e^{-jk\omega_1}$ to preserve the properties of being a real function

$$c_k \cos(k\omega_1 t + \varphi_k) = \frac{1}{2} c_k \left[e^{j(k\omega_1 t + \varphi_k)} + e^{-j(k\omega_1 t + \varphi_k)} \right]$$

Summary

(5) Spectrum of Typical Signals

- Spectrum of Periodic Rectangular Pulse Trains
- Gibbs Phenomenon
- Bandwidth

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Main Content

- Derivation of Fourier Transform
- Inverse Fourier Transform
- Representations of Fourier Transform
- Physical Interpretation
- Existence Conditions

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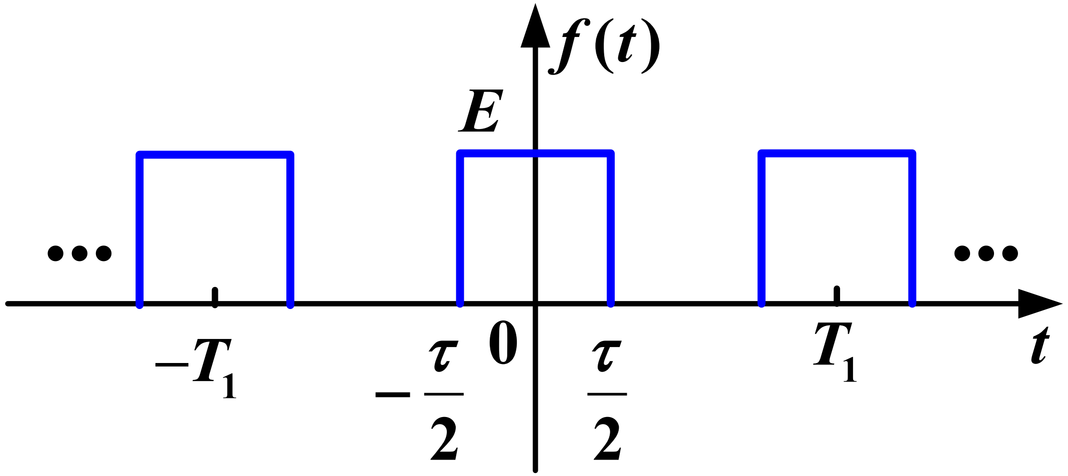
1. Derivation of Fourier Transform



$$f(t) = \sum_{n=-\infty}^{\infty} F_k e^{jk\omega_1 t} \quad \left(\omega_1 = \frac{2\pi}{T_1}\right)$$

$$F_k = F(k\omega_1) = \frac{1}{T_1}\int_{-\frac{T_1}{2}}^{\frac{T_1}{2}} f(t) e^{-jk\omega_1 t} \, dt$$

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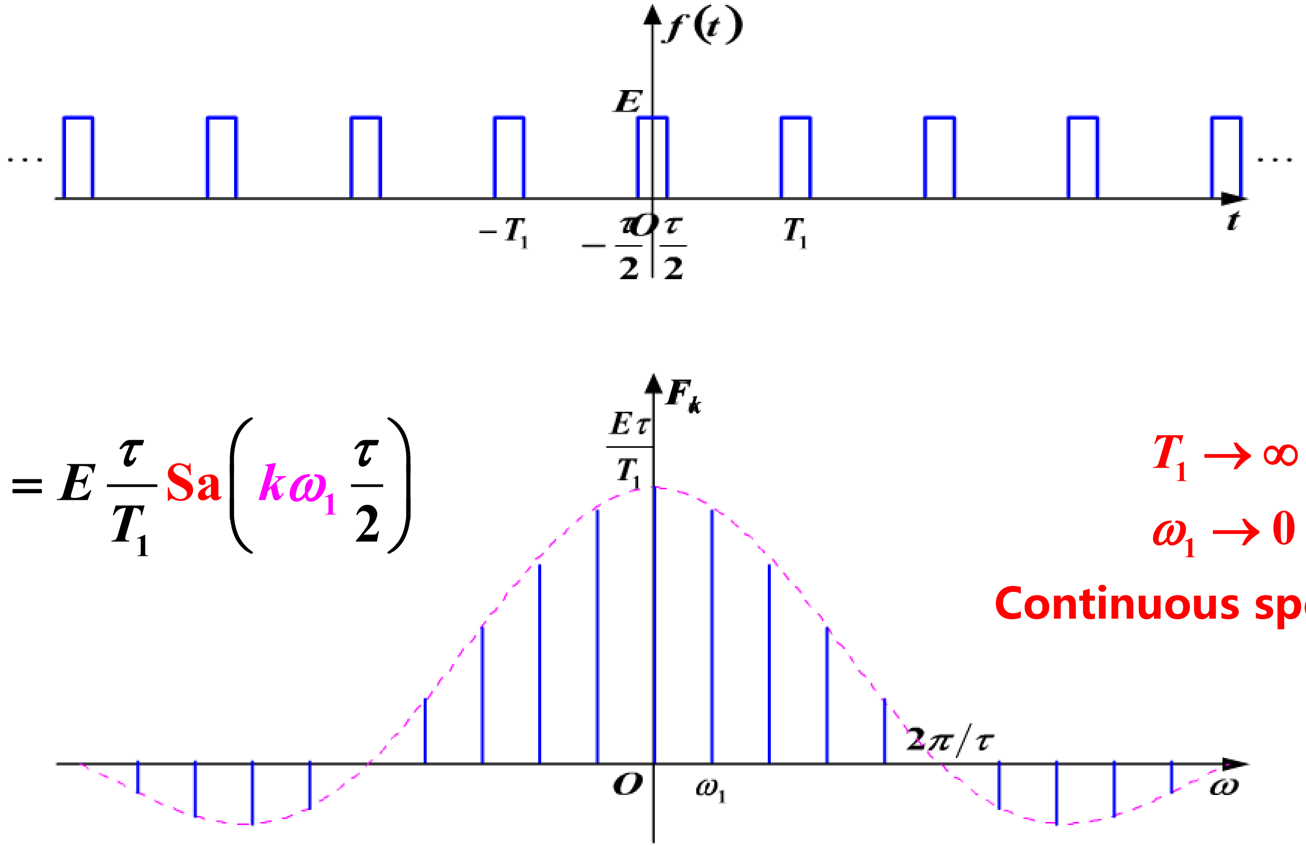
1. Derivation of Fourier Transform

Fourier series of a periodic rectangular pulse signal

τ remains unchanged, T_1 ↑



$$F_k = E\frac{\tau}{T_1}\mathrm{Sa}\left(k\omega_1\frac{\tau}{2}\right)$$

$T_1 \to \infty$

$\omega_1 \to 0$

Continuous spectrum

$\omega_1 = 2\pi/T_1$

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1. Derivation of Fourier Transform

$$F(k\omega_1) = \frac{1}{T_1}\int_{-\frac{T_1}{2}}^{\frac{T_1}{2}} f(t)\mathrm{e}^{-\mathrm{j}k\omega_1 t}\,\mathrm{d}t$$

Multiply both sides by T_1, let $T_1 \to \infty$

$$\lim_{T_1\to\infty} T_1 F(k\omega_1) = \int_{-\frac{T_1}{2}}^{\frac{T_1}{2}} f(t)\mathrm{e}^{-\mathrm{j}k\omega_1 t}\,\mathrm{d}t$$

$$= \int_{-\infty}^{\infty} f(t)\mathrm{e}^{-\mathrm{j}\omega t}\,\mathrm{d}t$$

$$= F(\omega)$$

Spectral Density Function
Abbreviated as
Spectrum Function

$$\omega_1 = \frac{2\pi}{T_1} \to 0$$ Discrete

$$k\omega_1 \to \omega$$ Continuous

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2. Inverse Fourier Transform

From the complex exponential form of the Fourier series

$$f(t)=\sum_{k=-\infty}^{\infty}F(k\omega_1)\mathrm{e}^{\mathrm{j}k\omega_1 t}$$

$$=\sum_{k=-\infty}^{\infty}\frac{F(k\omega_1)}{\omega_1}\cdot\mathrm{e}^{\mathrm{j}k\omega_1 t}\cdot\omega_1$$

$$=\frac{1}{2\pi}\sum_{k=-\infty}^{\infty}T_1F(k\omega_1)\cdot\mathrm{e}^{\mathrm{j}k\omega_1 t}\cdot\omega_1$$

When $T_1\to\infty$

$$f(t)=\frac{1}{2\pi}\int_{-\infty}^{\infty}F(\omega)\cdot\mathrm{e}^{\mathrm{j}\omega t}\,\mathrm{d}\omega$$

$$T_1F(k\omega_1)\to F(\omega)$$

$$k\omega_1\to\omega$$

$$\omega_1\to\mathrm{d}\omega$$

$$\sum_{k=-\infty}^{\infty}\to\int_{-\infty}^{\infty}$$

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3. Representations of Fourier Transform

$$F(\omega)=\int_{-\infty}^{\infty} f(t)\mathrm{e}^{-\mathrm{j}\omega t}\,\mathrm{d}t=\mathscr{F}\left[f(t)\right]$$

$$f(t)=\frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega)\mathrm{e}^{\mathrm{j}\omega t}\,\mathrm{d}\omega=\mathscr{F}^{-1}\left[F(\omega)\right]$$

$$f(t)\xleftrightarrow{\mathrm{FT}}F(\omega)$$

Original function

Fourier transform function

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3. Representations of Fourier Transform

$$F(\omega) = R(\omega) + \mathrm{j}X(\omega)$$

$$F(\omega) = |F(\omega)|\mathrm{e}^{\mathrm{j}\varphi(\omega)}$$

$$|F(\omega)| \sim \omega$$ Amplitude spectrum

$$\varphi(\omega) \sim \omega$$ Phase spectrum

$|F(\omega)|$ Represents the relative magnitudes of various frequency components in the signal

$\varphi(\omega)$ Represents the phase relationships between the various frequency components of the signal

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4. Physical Interpretation

$$f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega) e^{j\omega t}\, d\omega$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} |F(\omega)| e^{j\varphi(\omega)} e^{j\omega t}\, d\omega$$

$$= \int_{-\infty}^{\infty} \left\{ \frac{|F(\omega)|}{2\pi} e^{j[\omega t + \varphi(\omega)]}\, d\omega \right\}$$

An infinite sum of continuous exponential signals with infinitesimal amplitudes

Frequency (ω) range: $-\infty \rightarrow \infty$

5. Existence Conditions

The Dirichlet conditions (sufficient conditions)

1. **Absolute Integrability**, $\int_{-\infty}^{\infty}|f(t)|\,\mathrm{d}t<\infty$, ensures the Fourier integral converges
2. **Finite Extrema**, $f(t)$ must have a finite number of maxima and minima in any finite interval
3. **Finite Discontinuities**, $f(t)$ can have only a finite number of discontinuities in any finite interval. At each discontinuity, the function value must be finite (no infinite jumps)

Additional Notes:

- **Energy-Limited Signals**: The Fourier transform exists for signals with **finite energy**
- **Generalized Functions**: By introducing the Dirac delta function $\delta(\omega)$, the scope of functions admitting a Fourier transform expands significantly (e.g., constants, periodic signals)

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Summary

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} \, dt$$

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} \, d\omega$$

$$= \int_{-\infty}^{\infty} \left\{ \frac{|F(\omega)|}{2\pi} e^{j[\omega t + \varphi(\omega)]} \, d\omega \right\}$$

Decomposing $f(t)$ into an infinite sum of continuous exponential signals with infinitesimal amplitudes

Main Content

- Rectangular Pulse Signal
- One-sided Exponential Signal
- DC (Direct Current) Signal
- Signum Function (Sign Function)
- Unit Impulse Signal (Delta Function)
- Doublet Impulse Signal (Impulse Pair)

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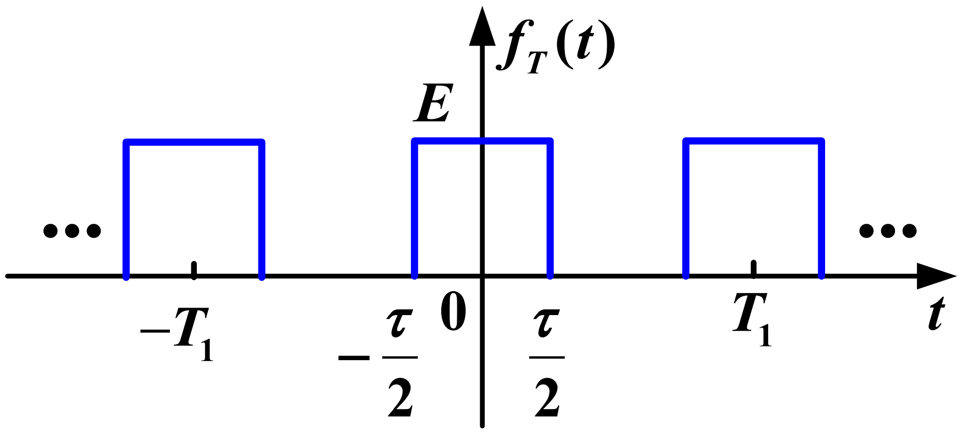
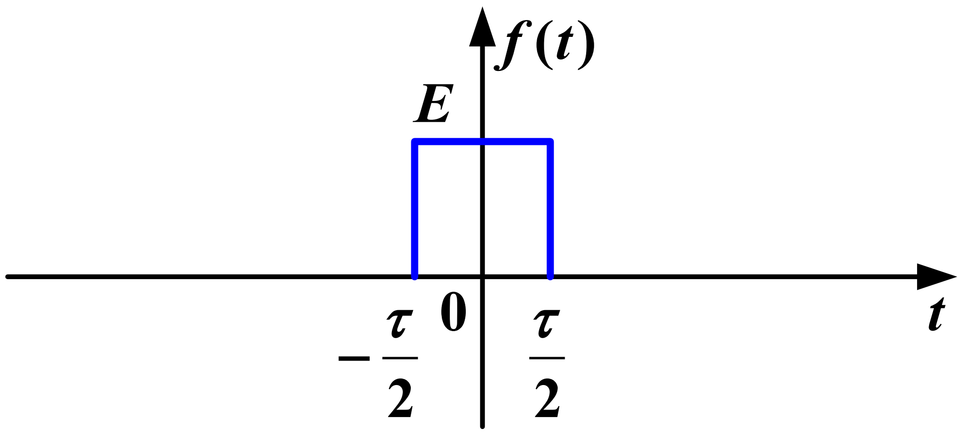
1. Rectangular Pulse Signal



$$F(k\omega_1) = \frac{E\tau}{T_1}\mathrm{Sa}\left(\frac{k\omega_1\tau}{2}\right)$$

Introduction to Fourier transform
Multiply both sides by T_1



$$F(\omega) = \lim_{T_1\to\infty} T_1 F(k\omega_1)$$

$$= E\tau\,\mathrm{Sa}\left(\frac{\omega\tau}{2}\right)$$

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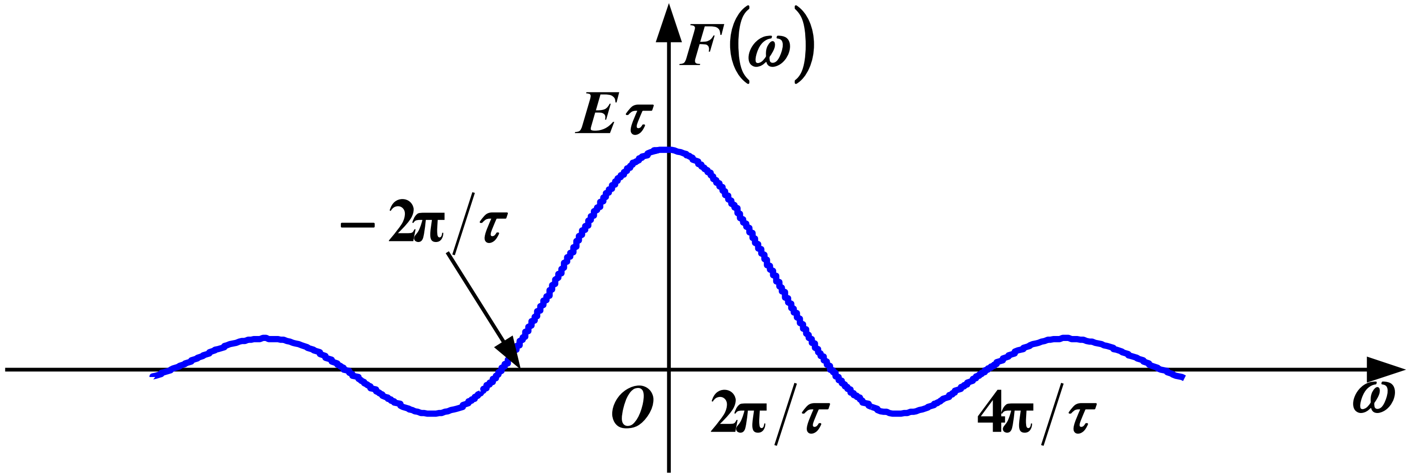
1. Rectangular Pulse Signal

$$F(\omega) = E\tau \, \mathrm{Sa}\left(\frac{\tau}{2} \cdot \omega\right)$$

$$\frac{\mathrm{Sin}(\frac{\tau}{2}\omega)}{\frac{\tau}{2}\omega}$$

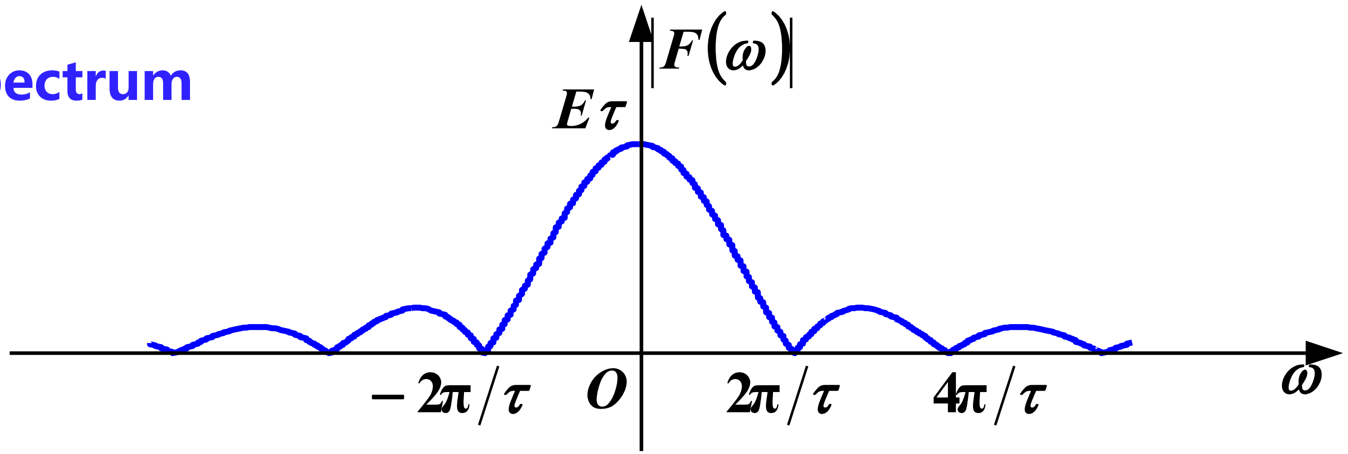
$$\frac{\tau}{2}\omega = K\pi$$



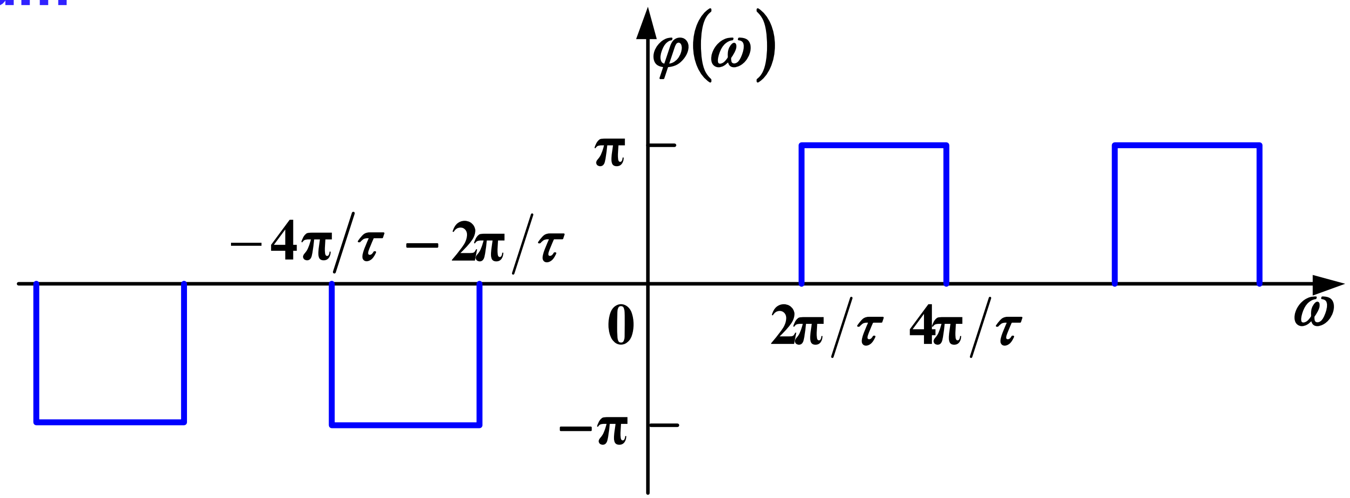
Bandwidth: $\frac{2\pi}{\tau}$

1. Rectangular Pulse Signal

Amplitude spectrum



Phase spectrum



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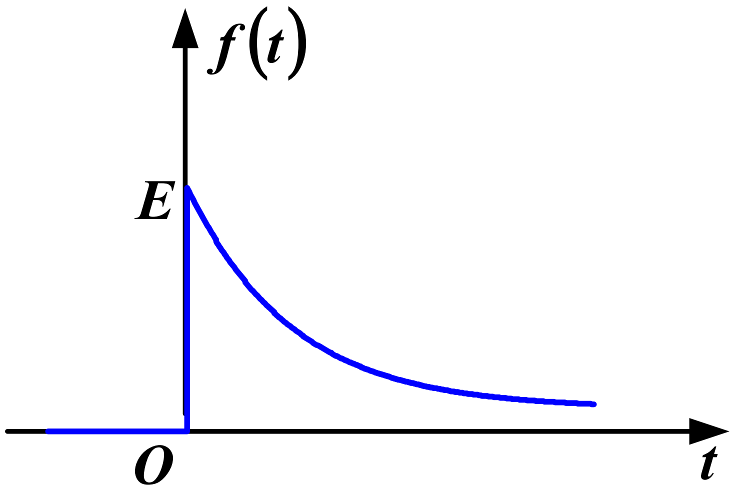
2. One-sided Exponential Signal

$$f(t) = E\mathrm{e}^{-\alpha t}u(t) \quad (\alpha > 0)$$



$$F(\omega) = \int_{-\infty}^{\infty} E\mathrm{e}^{-\alpha t}u(t)\mathrm{e}^{-\mathrm{j}\omega t}\mathrm{d}t$$

$$= \int_{0}^{\infty} E\mathrm{e}^{-(\alpha+\mathrm{j}\omega)t}\,\mathrm{d}t$$

$$= \frac{E}{-(\alpha+\mathrm{j}\omega)}\mathrm{e}^{-(\alpha+\mathrm{j}\omega)t}\Big|_{0}^{\infty}$$

$$= \frac{E}{\alpha+\mathrm{j}\omega}$$

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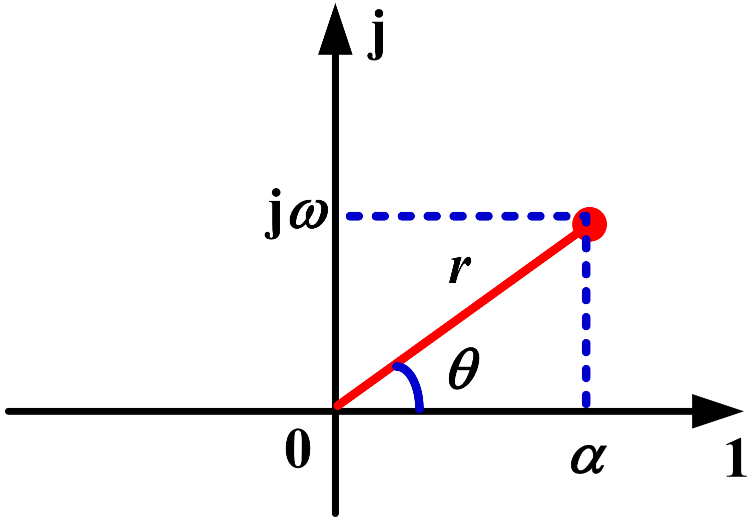
2. One-sided Exponential Signal

$$F(\omega)=\frac{E}{\alpha+\mathrm{j}\omega}=\frac{E\mathrm{e}^{\mathrm{j}0}}{r\mathrm{e}^{\mathrm{j}\theta}}$$



$$|F(\omega)|=\frac{E}{\sqrt{\alpha^2+\omega^2}}$$

$$\varphi(\omega)=-\arctan\frac{\omega}{\alpha}$$

$$r=\sqrt{\alpha^2+\omega^2}$$

$$\tan\theta=\frac{\omega}{\alpha}$$

$$\theta=\arctan\frac{\omega}{\alpha}\quad(\alpha>0)$$

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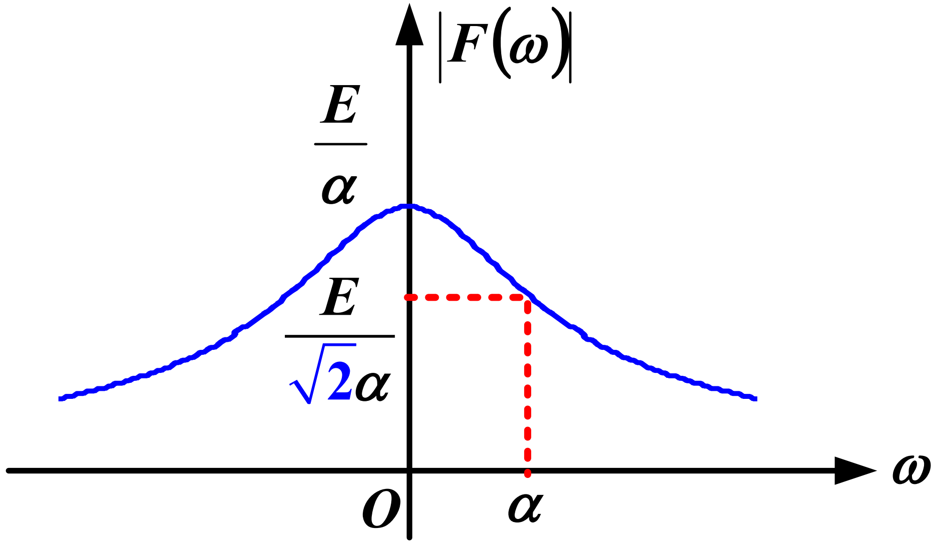
2. One-sided Exponential Signal

Amplitude spectrum

$$|F(\omega)| = \frac{E}{\sqrt{\alpha^2 + \omega^2}}$$

- $\omega = 0$ $\quad |F(\omega)| = \dfrac{E}{\alpha}$
- $\omega \to \pm\infty$ $\quad |F(\omega)| \to 0$
- $\omega = \alpha$ $\quad |F(\omega)| = \dfrac{E}{\sqrt{2}\alpha}$



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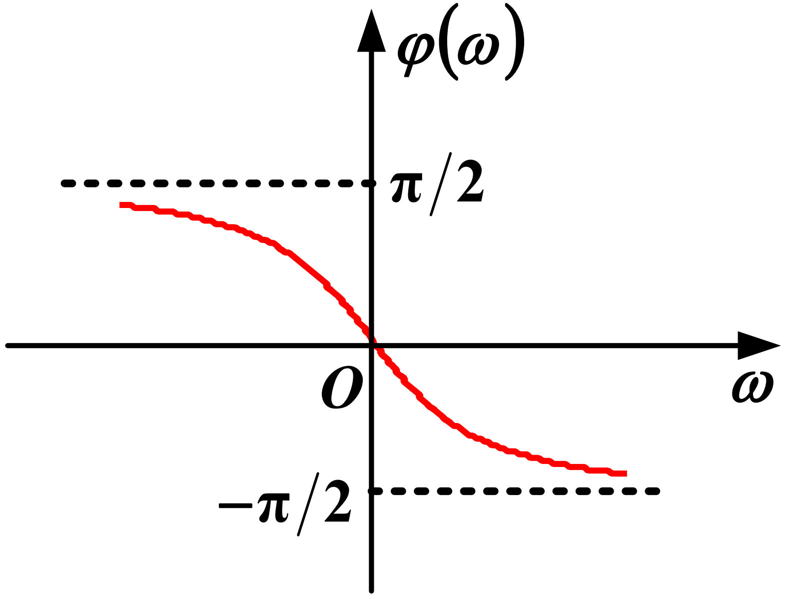
2. One-sided Exponential Signal

Phase spectrum

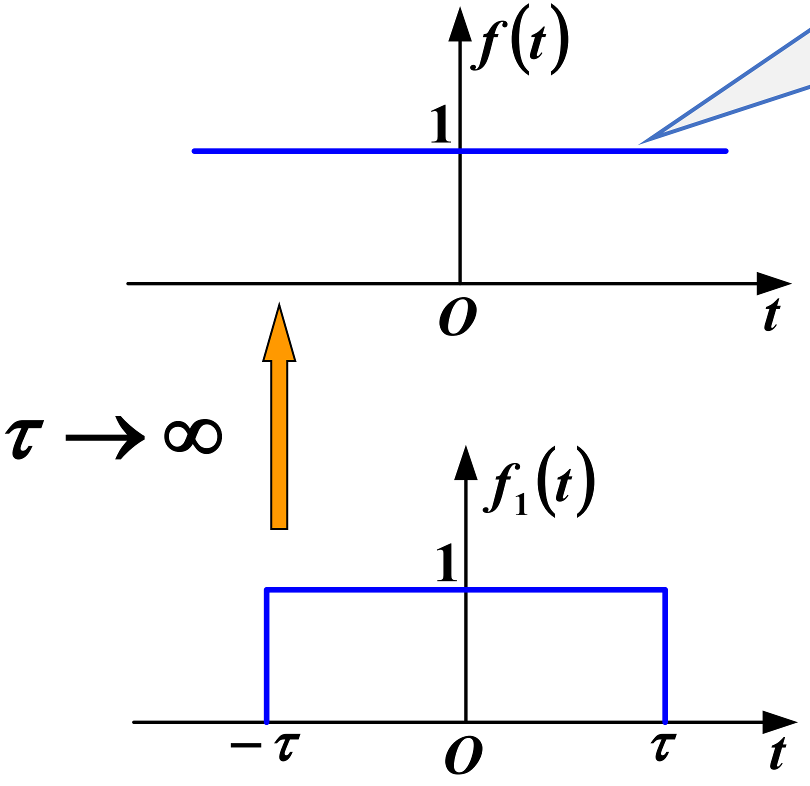
$$\varphi(\omega) = -\arctan\frac{\omega}{\alpha}$$

$$\begin{cases} \omega \to 0, & \varphi(\omega) = 0 \\ \omega \to +\infty, & \varphi(\omega) \to -\pi/2 \\ \omega \to -\infty, & \varphi(\omega) \to \pi/2 \end{cases}$$



3. DC (Direct Current) Signal

$$f(t)=1, \quad -\infty < t < +\infty$$



Does not satisfy the absolute integrability condition and cannot be directly evaluated using the definition

$$F_1(\omega)=2\tau\,\mathrm{Sa}(\omega\tau)$$

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3. DC (Direct Current) Signal

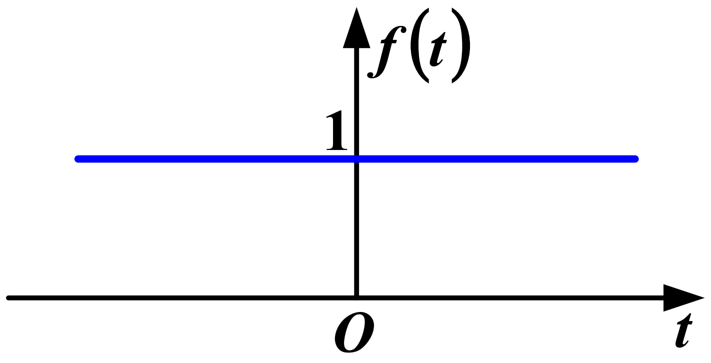
$$F(\omega) = \lim_{\tau \to \infty} F_1(\omega)$$

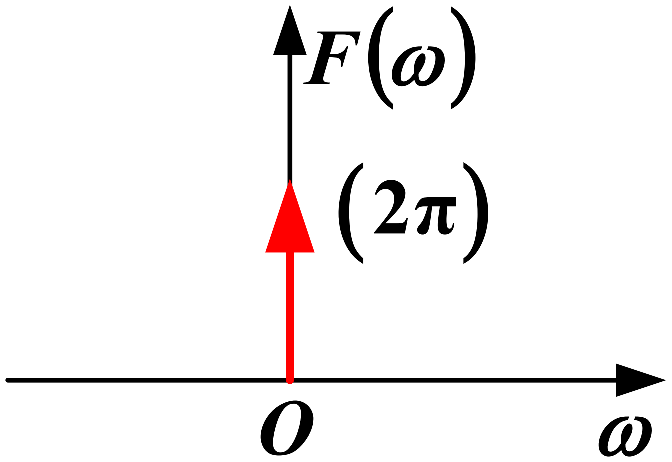
$$= 2\pi \lim_{\tau \to \infty} \frac{\tau}{\pi} \mathrm{Sa}(\omega\tau)$$

$$= 2\pi\delta(\omega)$$

$$= 2\pi\delta(2\pi f)$$

$$= \delta(f)$$

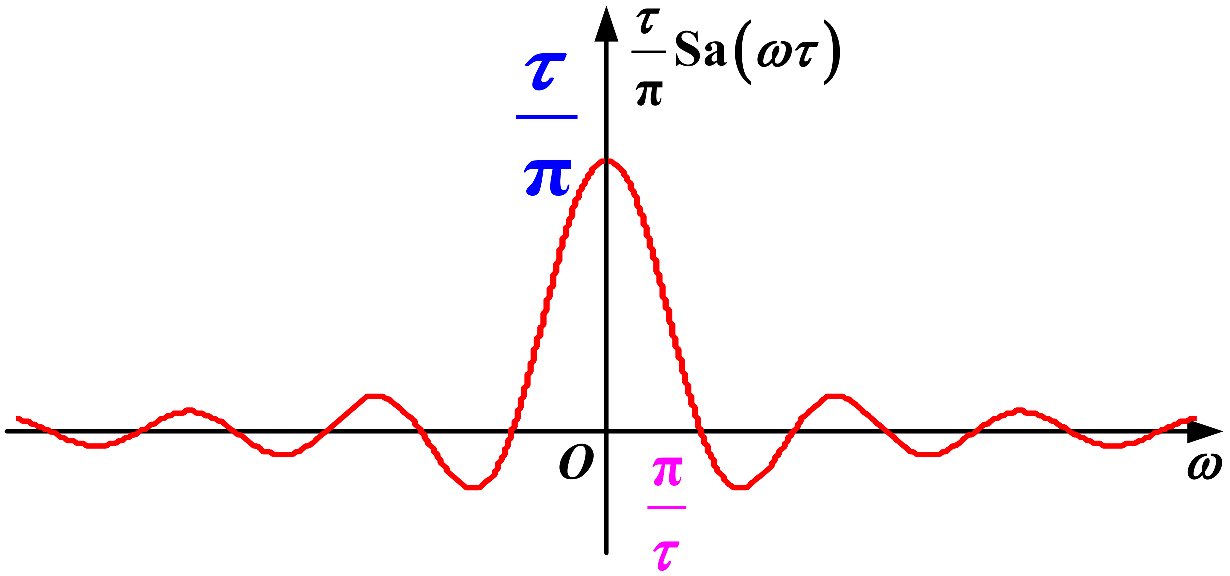




$$\delta(at) = \frac{1}{|a|}\delta(t)$$

3. DC (Direct Current) Signal

Explanation



$\tau\uparrow, \frac{\pi}{\tau}\downarrow$ waveform compression, $\frac{\tau}{\pi}\uparrow, \omega=0$ point value increases

$\tau\to\infty$, energy compressed to $\omega=0$, area $\frac{\tau}{\pi}\frac{\pi}{\tau}=1$ always the same

$$\lim_{\tau\to\infty}\frac{\tau}{\pi}\mathrm{Sa}(\omega\tau)=\delta(\omega)$$

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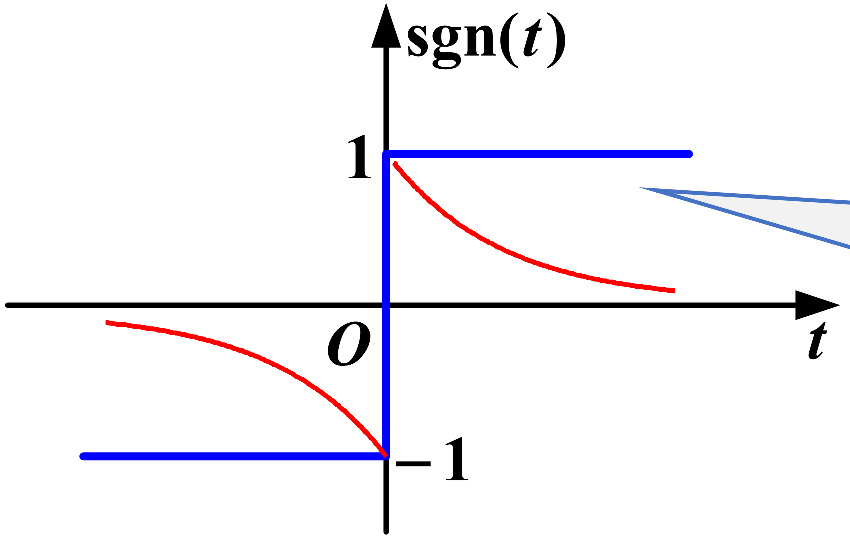
4. Signum Function (Sign Function)

$$f(t) = \mathrm{sgn}(t)$$



does not satisfy the absolute integrability condition

Processing method: constructing a two-sided exponential function

$$f_1(t) = \mathrm{sgn}(t)\mathrm{e}^{-\alpha|t|}$$

$$F(\omega) = \lim_{\alpha \to 0} F_1(\omega)$$

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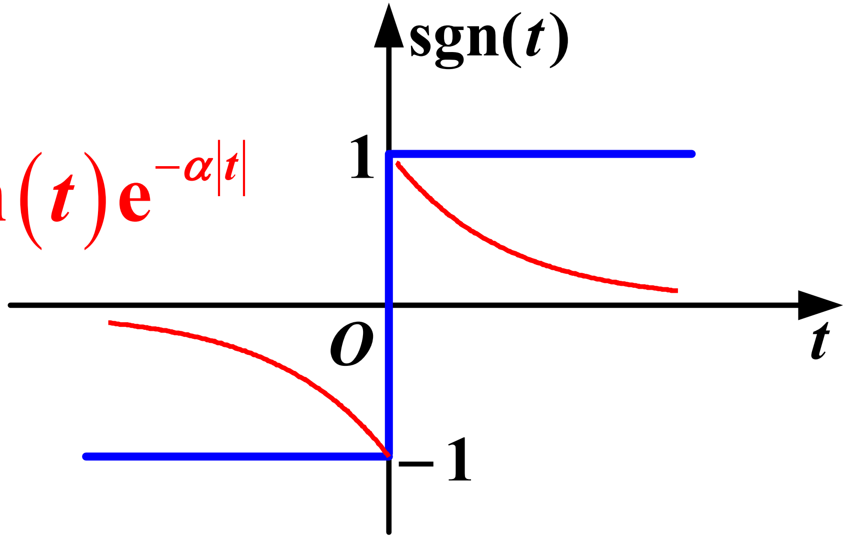
4. Signum Function (Sign Function)

$$f_1(t) = \mathrm{sgn}(t)\,\mathrm{e}^{-\alpha|t|}$$



$$F_1(\omega) = \int_{-\infty}^{0} -\mathrm{e}^{\alpha t}\mathrm{e}^{-\mathrm{j}\omega t}\,\mathrm{d}t + \int_{0}^{\infty} \mathrm{e}^{-\alpha t}\mathrm{e}^{-\mathrm{j}\omega t}\,\mathrm{d}t$$

$$= \frac{-1}{\alpha - \mathrm{j}\omega} + \frac{1}{\alpha + \mathrm{j}\omega}$$

$$= \frac{-\mathrm{j}2\omega}{\alpha^2 + \omega^2}$$

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4. Signum Function (Sign Function)

$$F(\omega)=\lim_{\alpha\to 0}F_1(\omega)$$

$$=\lim_{\alpha\to 0}\frac{-\mathrm{j}2\omega}{\alpha^2+\omega^2}$$

$$=\frac{2}{\mathrm{j}\omega}$$

$$=\begin{cases}\dfrac{2}{|\omega|}\mathrm{e}^{-\mathrm{j}\frac{\pi}{2}} & \omega>0\\[2ex] \dfrac{2}{|\omega|}\mathrm{e}^{\mathrm{j}\frac{\pi}{2}} & \omega<0\end{cases}$$

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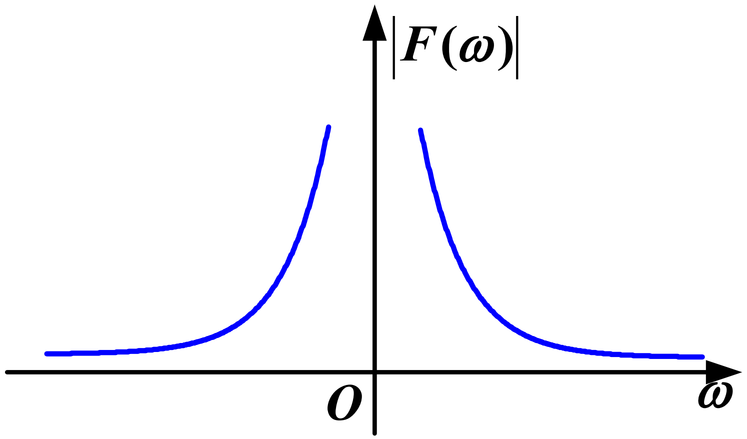
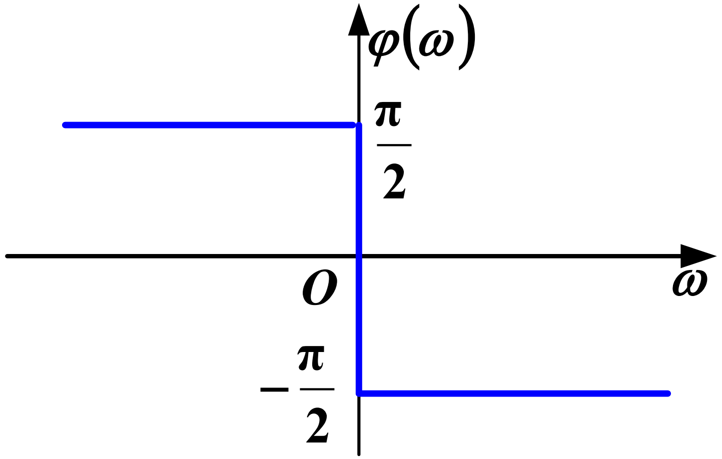
4. Signum Function (Sign Function)

$$F(\omega)=\lim_{\alpha\to 0}\frac{-\mathrm{j}2\omega}{\alpha^2+\omega^2}=\frac{2}{\mathrm{j}\omega}$$

Amplitude spectrum $$|F(\omega)|=\frac{2}{|\omega|}$$

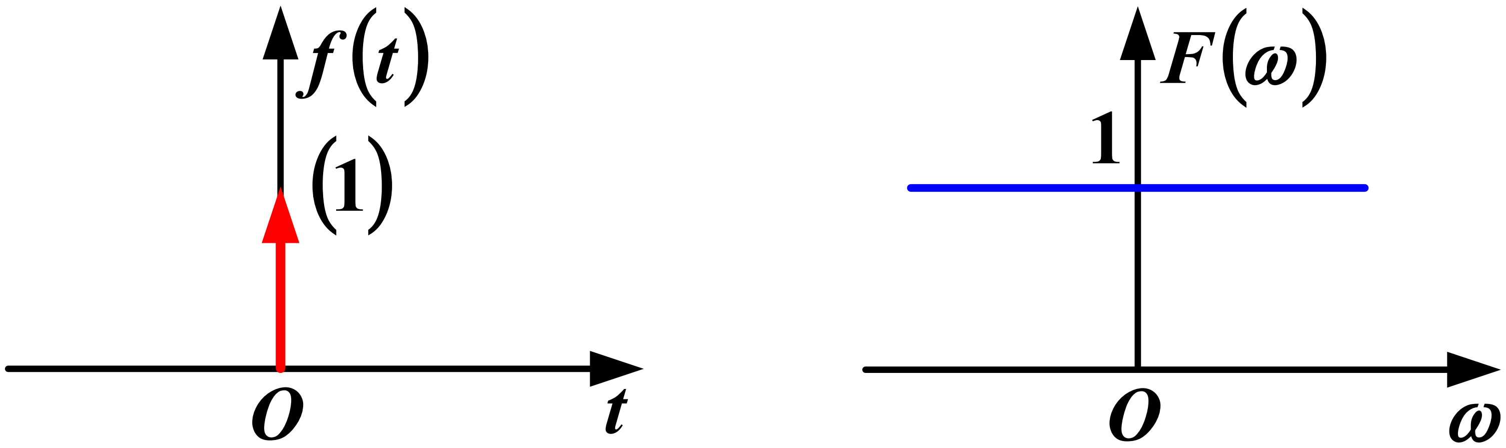


Phase spectrum $$\begin{cases}-\dfrac{\pi}{2}, & \omega>0\\ \dfrac{\pi}{2}, & \omega<0\end{cases}$$



5. Unit Impulse Signal (Delta Function)

$$\mathscr{F}\left[\delta(t)\right]=\int_{-\infty}^{\infty}\delta(t)\mathrm{e}^{-\mathrm{j}\omega t}\,\mathrm{d}t=1$$



**Infinitesimally narrow in the time domain,
infinitely wide in the frequency domain**

6. Doublet Impulse Signal (Impulse Pair)

$$\mathscr{F}\left[\delta'(t)\right] = \int_{-\infty}^{\infty} \delta'(t) \mathrm{e}^{-\mathrm{j}\omega t} \, \mathrm{d}t$$

$$= -\left(-\mathrm{j}\omega \mathrm{e}^{-\mathrm{j}\omega t}\right)\Big|_{t=0}$$

$$= \mathrm{j}\omega$$

$$\boxed{\int_{-\infty}^{\infty} f(t)\delta'(t)\mathrm{d}t = -f'(0)}$$

Summary

- Rectangular Pulse Signal
- One-sided Exponential Signal
- DC (Direct Current) Signal
- Signum Function (Sign Function)
- Unit Impulse Signal (Delta Function)
- Doublet Impulse Signal (Impulse Pair)

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Main Content

The Fourier transform is unique. The properties of the Fourier transform reveal the deterministic intrinsic relationship between the time-domain and frequency-domain characteristics of signals. The purpose of discussing the properties of the Fourier transform is to:

- **Understand the intrinsic connections of these properties**
- **Utilize the properties to derive the Fourier transform of certain signals**
- **Understand the applications of the Fourier transform in communication systems**

1. Linearity

If $\mathscr{F}\left[f_i(t)\right]=F_i(\omega)\left(i=1,2,\cdots n\right)$, Then

$$\mathscr{F}\left[\sum_{i=1}^{n}a_i f_i(t)\right]=\sum_{i=1}^{n}a_i F_i(\omega)$$

a_i are constants, n is a positive integer

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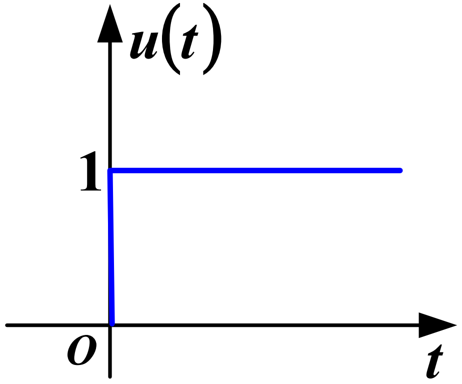
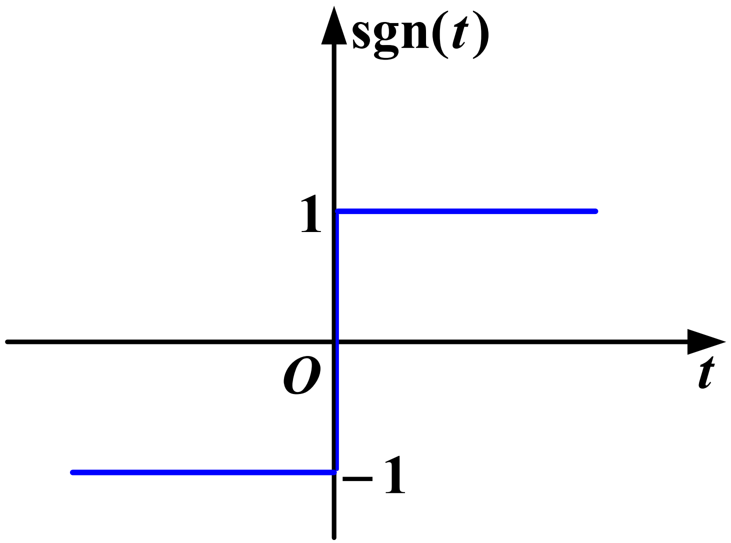
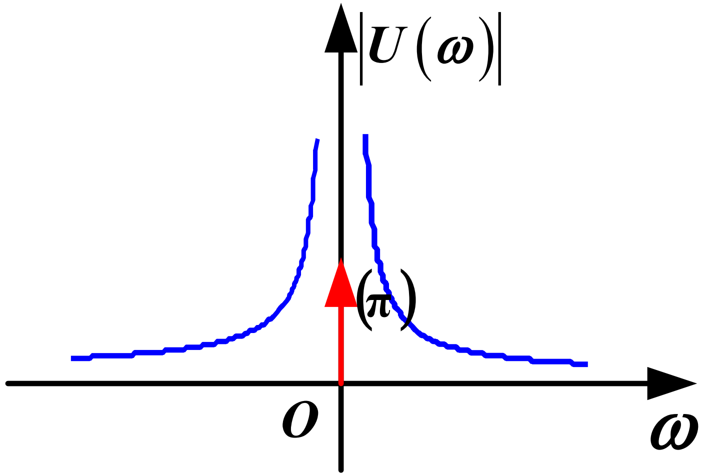
Example 1 Fourier transform of a unit step signal



$$u(t)=\frac{1}{2}+\frac{1}{2}\operatorname{sgn}(t)$$

$$\xleftrightarrow{\text{FT}}\pi\delta(\omega)+\frac{1}{\mathrm{j}\omega}$$





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2. Duality

(1) Statement

$$\text{IF}\ \mathscr{F}\left[f(t)\right]=F(\omega),\ \text{Then}\ \mathscr{F}\left[F(t)\right]=2\pi f(-\omega)$$

(2) Analysis

$$\begin{cases} \text{FT:}\ \ F(\omega)=\int_{-\infty}^{\infty} f(t)\mathrm{e}^{-\mathrm{j}\omega t}\mathrm{d}t \\ \text{IFT:}\ \ f(t)=\dfrac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega)\mathrm{e}^{\mathrm{j}\omega t}\mathrm{d}\omega \end{cases}$$

The forward and inverse transforms are structurally similar but not identical. The duality property formalizes this symmetry

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Example 2

$$\text{Given } f(t) = \delta(t),\ F(\omega) = 1$$

Use the duality property to find the Fourier transform of $F(t) = 1$

$$\mathscr{F}\left[F(t)\right] = 2\pi f(-\omega)$$

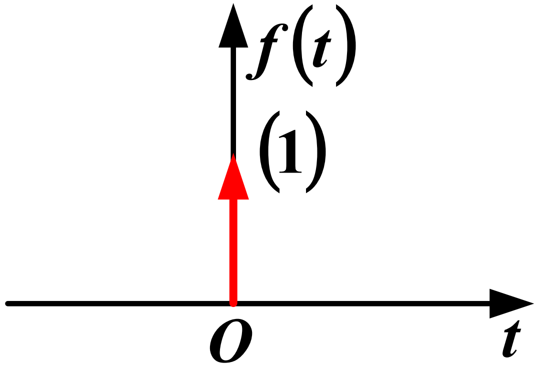
$$= 2\pi\delta(-\omega) \qquad \left(\delta(\omega)\text{is even function}\right)$$

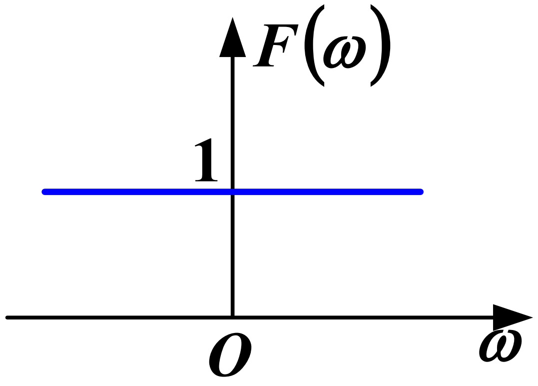
$$= 2\pi\delta(\omega)$$

6. Basic Properties of Fourier Transform

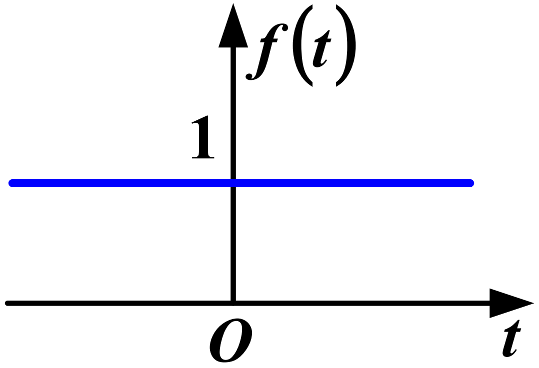
Example 2

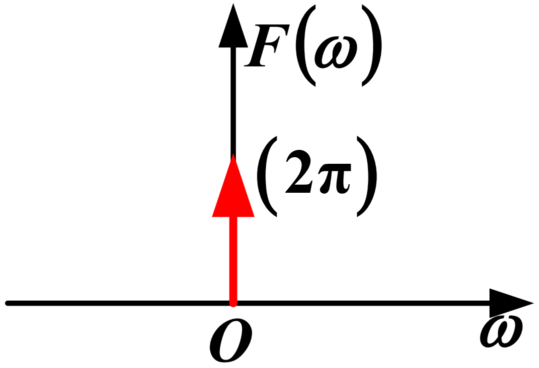
$$\delta(t) \leftrightarrow 1$$





$$1 \leftrightarrow 2\pi\delta(\omega)$$





Duality Property

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Example 4

Find the Fourier transform of the sampling signal $\mathrm{Sa}(\omega_C t)$

Analysis

Using the Fourier transform definition:

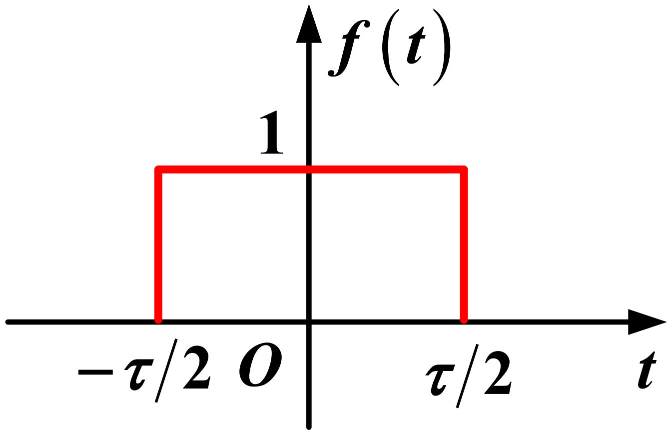
$$F_1(\omega)=\int_{-\infty}^{\infty}\mathrm{Sa}(\omega_C t)\mathrm{e}^{-\mathrm{j}\omega t}\mathrm{d}t$$

$$\int_{-\infty}^{\infty}\mathrm{Sa}(\omega_C t)\mathrm{e}^{-\mathrm{j}\omega t}\mathrm{d}t$$

This integral is challenging to solve directly

Therefore, we attempt to solve it using duality property

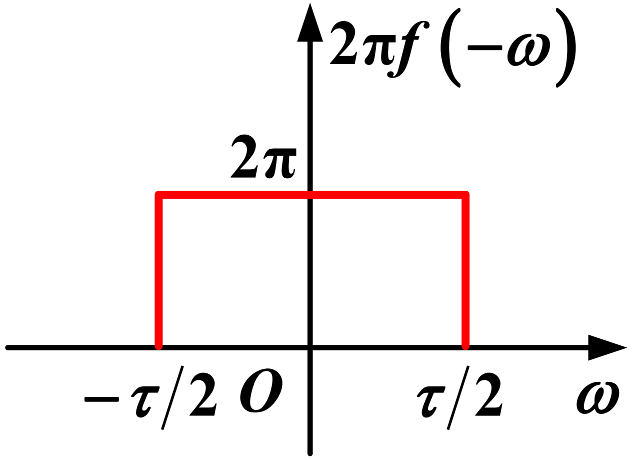
Example 4



$$\leftrightarrow \quad F(\omega) = \tau\,\mathrm{Sa}\left(\frac{\tau}{2}\omega\right)$$

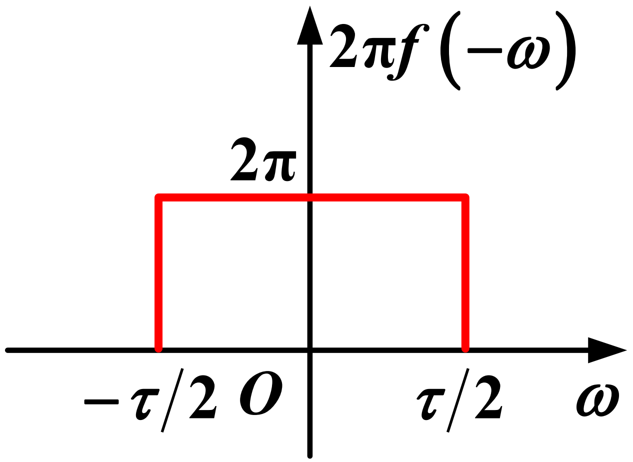
Obtain a new transform pair using duality property

$$F(t) = \tau\,\mathrm{Sa}\left(\frac{\tau}{2}t\right) \quad \leftrightarrow$$



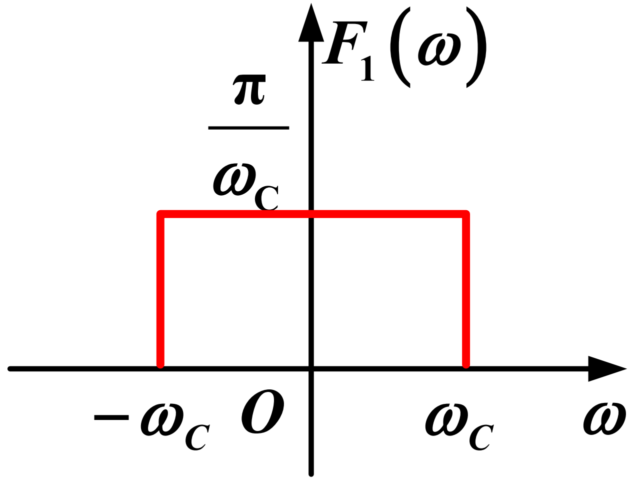
Example 4

$$F(t) = \tau \,\mathrm{Sa}\left(\frac{\tau}{2} t\right) \quad \leftrightarrow$$



Let $\dfrac{\tau}{2} = \omega_C$, divide both sides by $2\omega_C$

$$\mathrm{Sa}(\omega_C t) \quad \leftrightarrow$$



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3. Conjugation Symmetry

$$F(\omega)=\int_{-\infty}^{\infty} f(t)\mathrm{e}^{-\mathrm{j}\omega t}\,\mathrm{d}t$$

$$F(\omega)=|F(\omega)|\mathrm{e}^{\mathrm{j}\varphi(\omega)}=R(\omega)+\mathrm{j}X(\omega)$$

Magnitude Phase Real part Imaginary part

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(1) Symmetry of the Spectrum for Real-Valued Time-Domain Functions

$$f(t) = f_e(t) + f_o(t)$$

Real signal Even Odd

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} \, dt$$

Euler formula

$$= \int_{-\infty}^{\infty} \left[f_e(t) + f_o(t) \right] \cdot \left[\cos(\omega t) - j \sin(\omega t) \right] dt$$

$$= 2\int_{0}^{\infty} f_e(t) \cos(\omega t) \, dt - j2\int_{0}^{\infty} f_o(t) \sin(\omega t) \, dt$$

Real part Imaginary part

$$= R(\omega) + jX(\omega)$$

The integral of an odd function is 0

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(1) Symmetry of the Spectrum for Real-Valued Time-Domain Functions

$$R(\omega)=2\int_0^{\infty} f_e(t)\cos(\omega t)\mathrm{d}t$$

Even function of ω

$$X(\omega)=-2\int_0^{\infty} f_o(t)\sin(\omega t)\mathrm{d}t$$

Odd function of ω

$$|F(\omega)|=\sqrt{[R(\omega)]^2+[X(\omega)]^2}$$

Even function of ω

$$\varphi(\omega)=\mathrm{atan2}\left[X(\omega),\ R(\omega)\right]$$

Odd function of ω

$$F(-\omega)=R(-\omega)+\mathrm{j}X(-\omega)=R(\omega)-\mathrm{j}X(\omega)$$

$$=F^*(\omega)$$

Conjugate symmetry

Four-quadrant arctangent

6. Basic Properties of Fourier Transform

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(1) Symmetry of the Spectrum for Real-Valued Time-Domain Functions

$$R(\omega)=2\int_0^{\infty} f_e(t)\cos(\omega t)\,\mathrm{d}t$$ Even function of ω

$$X(\omega)=-2\int_0^{\infty} f_o(t)\sin(\omega t)\,\mathrm{d}t$$ Odd function of ω

$f(t)$	$F(\omega)=\mathscr{F}[f(t)]$	
Even function: odd component is zero	Real even function: phase is 0 or π	$F(\omega)=R(\omega)$
Odd function: even component is zero	Imaginary odd function: phase is $\pm\pi/2$	$F(\omega)=\mathrm{j}X(\omega)$

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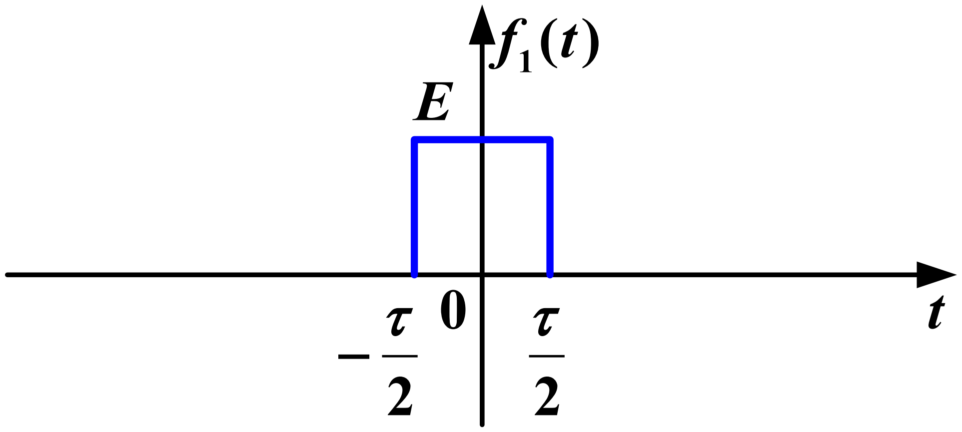
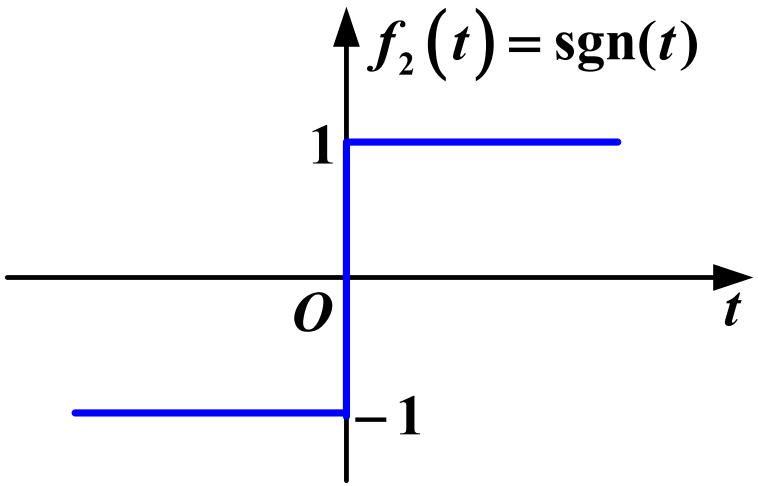
(1) Symmetry of the Spectrum for Real-Valued Time-Domain Functions



$$F_1(\omega) = E\tau\,\mathrm{Sa}\left(\frac{\omega\tau}{2}\right)$$



$$F_2(\omega) = \frac{2}{\mathrm{j}\omega}$$

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(2) Spectral Properties of Complex Time-Domain Signals

The Fourier transform of $f^*(t)$

$$\int_{-\infty}^{\infty} f^*(t)\mathrm{e}^{-\mathrm{j}\omega t}\,\mathrm{d}t \qquad (1)$$

Taking the complex conjugate twice on (1)

$$\left[\int_{-\infty}^{\infty} f(t)\mathrm{e}^{\mathrm{j}\omega t}\,\mathrm{d}t\right]^* \qquad (2)$$

Comparing with the forward Fourier transform, (2) equals $F^*(-\omega)$

$$F\left[f^*(t)\right] = F^*(-\omega) \qquad (3)$$

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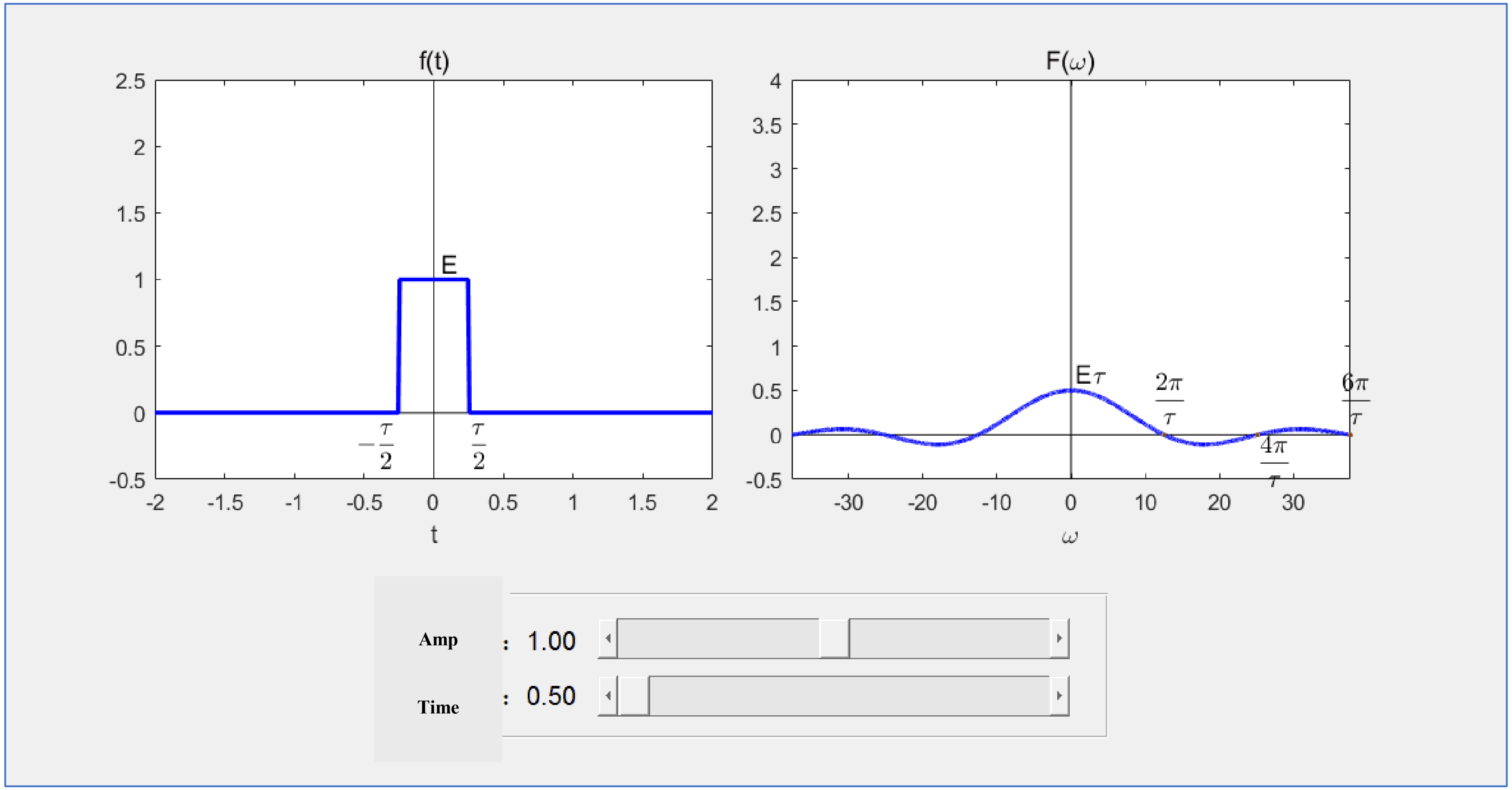
4. Time-Scaling Property

$$\text{If } \mathscr{F}[f(t)] = F(\omega), \text{ then } \mathscr{F}\left[f\left(at\right)\right] = \frac{1}{|a|} F\left(\frac{\omega}{a}\right),$$

Where a is a non-zero real constant

$$\text{When } a = -1 \quad f\left(t\right) \rightarrow f\left(-t\right), \; F\left(\omega\right) \rightarrow F\left(-\omega\right)$$

4. Time-Scaling Property



Time duration and bandwidth are inversely proportional

5. Time-Shifting Property

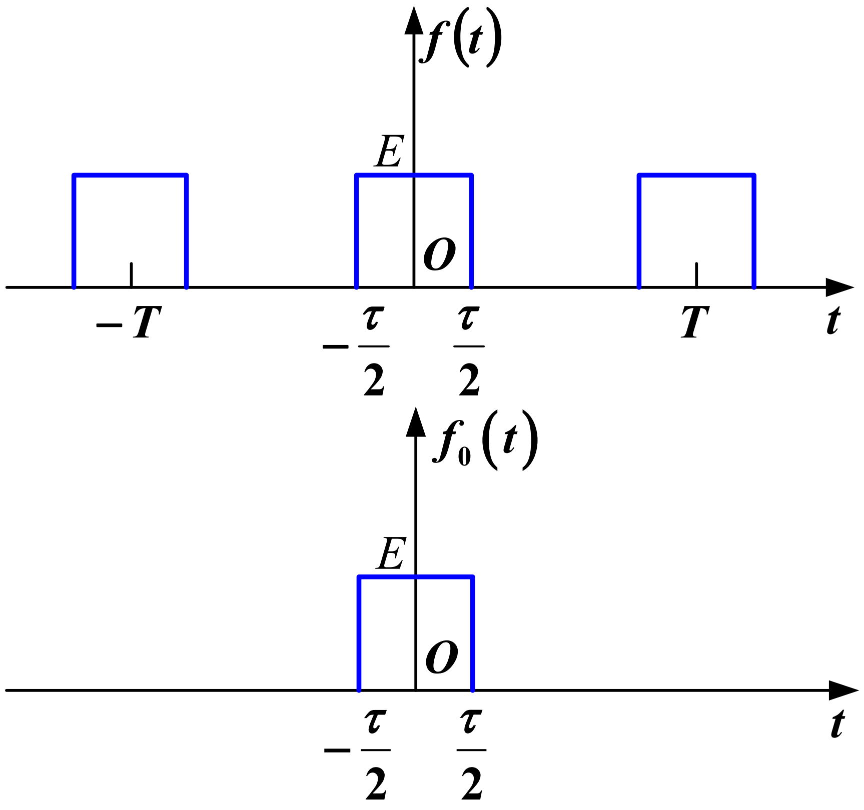
$$\text{If } \mathscr{F}\left[f(t)\right]=F(\omega),\text{Then } \mathscr{F}\left[f(t-t_0)\right]=F(\omega)\mathrm{e}^{-\mathrm{j}\omega t_0};$$

In Polar Form $$\mathscr{F}\left[f(t-t_0)\right]=\left|F(\omega)\right|\cdot \mathrm{e}^{\mathrm{j}\left[\varphi(\omega)-\omega t_0\right]}$$

Magnitude spectrum remains unchanged, only affecting the phase spectrum

Example 5

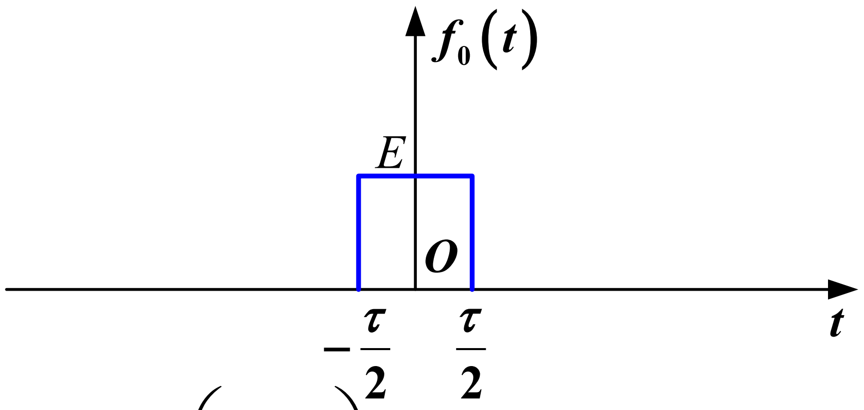
Find the frequency spectrum of the three-pulse signal



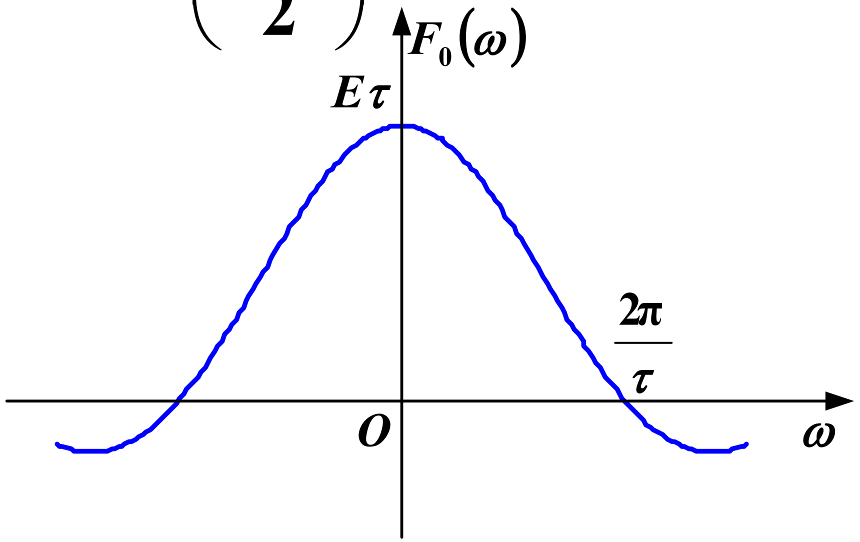
$$f(t) = f_0(t) + f_0(t+T) + f_0(t-T)$$

6. Basic Properties of Fourier Transform

Example 5



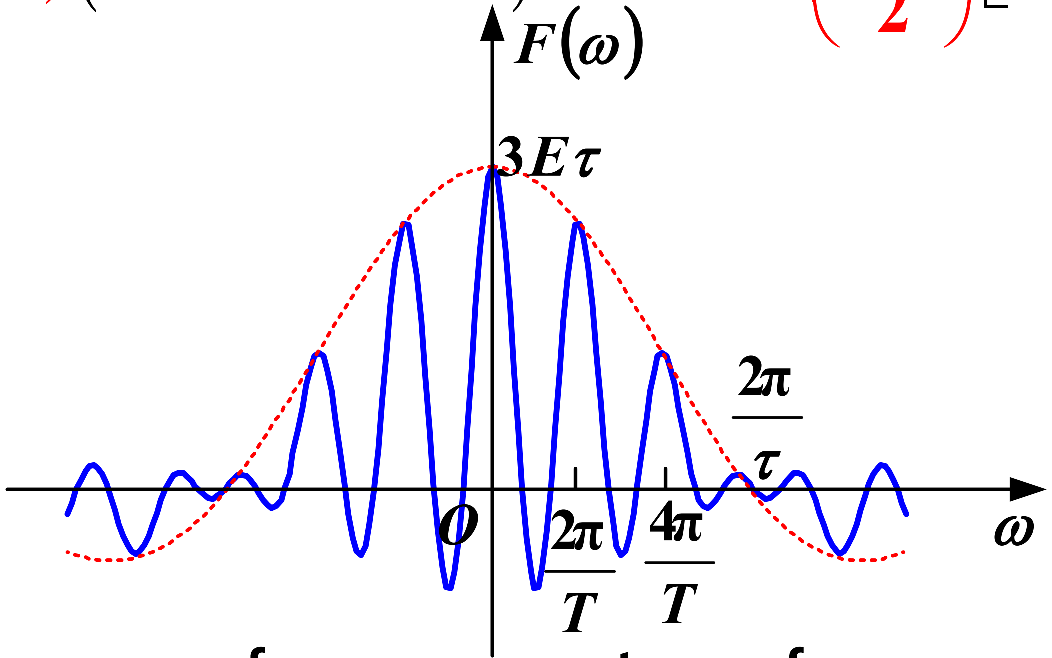
$$F_0(\omega) = E\tau \cdot \mathrm{Sa}\left(\frac{\omega\tau}{2}\right)$$



Example 5

$$f(t) = f_0(t) + f_0(t+T) + f_0(t-T)$$

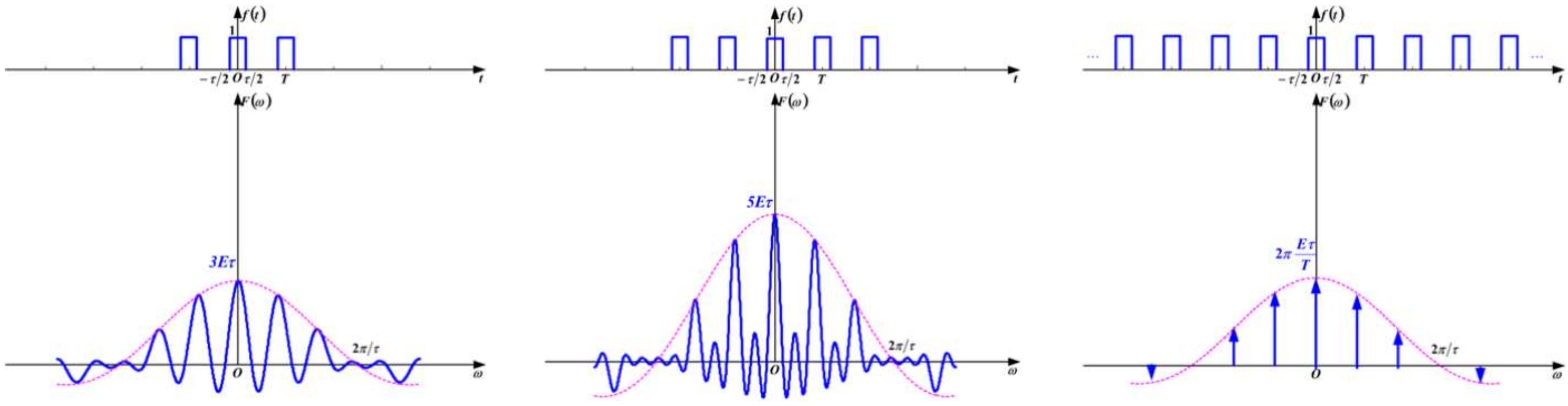
$$F(\omega) = F_0(\omega)\left(1 + e^{j\omega T} + e^{-j\omega T}\right) = E\tau \cdot \mathrm{Sa}\left(\frac{\omega\tau}{2}\right)\left[1 + 2\cos(\omega T)\right]$$



frequency spectrum of
the three-pulse signal

The envelope of the spectrum is determined by the shape of a single pulse

Example 5



As the number of pulses increases, the spectral envelope remains unchanged, and the bandwidth stays the same

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5. Time-Shifting Property

Time-Shifting and Scaling Combined

$$\text{If } \mathscr{F}\left[f(t)\right] = F(\omega), \text{ then } \mathscr{F}\left[f\left(at+b\right)\right] = ?$$

$$\mathscr{F}\left[f\left(at+b\right)\right] = \mathscr{F}\left\{f\left[a\left(t+\frac{b}{a}\right)\right]\right\} = \frac{1}{|a|}F\left(\frac{\omega}{a}\right)\cdot e^{j\omega\frac{b}{a}}$$

Given $\mathscr{F}\left[f(t)\right] = F(\omega)$, power spectral density function of $f\left(2t-5\right)$?

$$f\left(2t\right) \leftrightarrow \frac{1}{2}F\left(\frac{\omega}{2}\right)$$

$$f\left(2t-5\right) = f\left[2\left(t-\frac{5}{2}\right)\right] \leftrightarrow \frac{1}{2}F\left(\frac{\omega}{2}\right)\cdot e^{-j\omega\frac{5}{2}}$$

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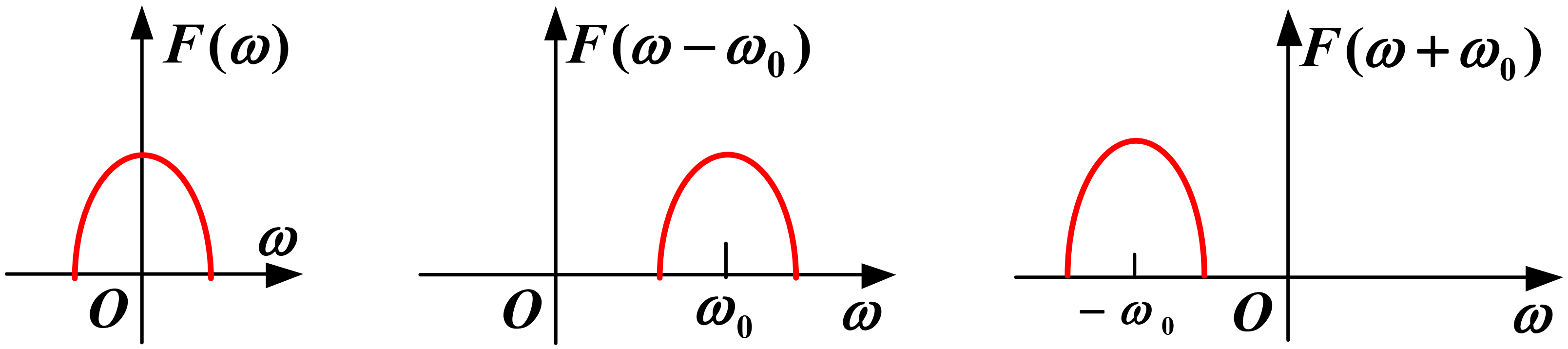
6. Frequency-Shifting Property

Property

If $\mathscr{F}\left[f(t)\right] = F(\omega) = \int_{-\infty}^{\infty} f(t)\mathrm{e}^{-\mathrm{j}\omega t}\,\mathrm{d}t$

Thus $\begin{cases} \mathscr{F}\left[f(t)\mathrm{e}^{\mathrm{j}\omega_0 t}\right] = F\left(\omega - \omega_0\right) \\ \mathscr{F}\left[f(t)\mathrm{e}^{-\mathrm{j}\omega_0 t}\right] = F\left(\omega + \omega_0\right) \end{cases}$

ω_0 is a constant, note: + or -

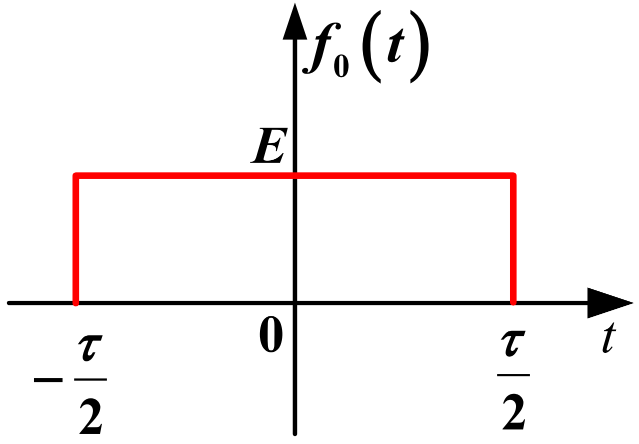


Example 6

Find the frequency spectrum of the signal $f(t) = f_0(t)\cos(\omega_0 t)$

Apply Euler's Formula:

$$f(t) = \frac{1}{2} f_0(t)\left(e^{j\omega_0 t} + e^{-j\omega_0 t}\right)$$



Use the Frequency-Shifting Property:

$$f(t) \to F(\omega)$$

$$F(\omega) = \frac{1}{2} F_0(\omega - \omega_0) + \frac{1}{2} F_0(\omega + \omega_0)$$

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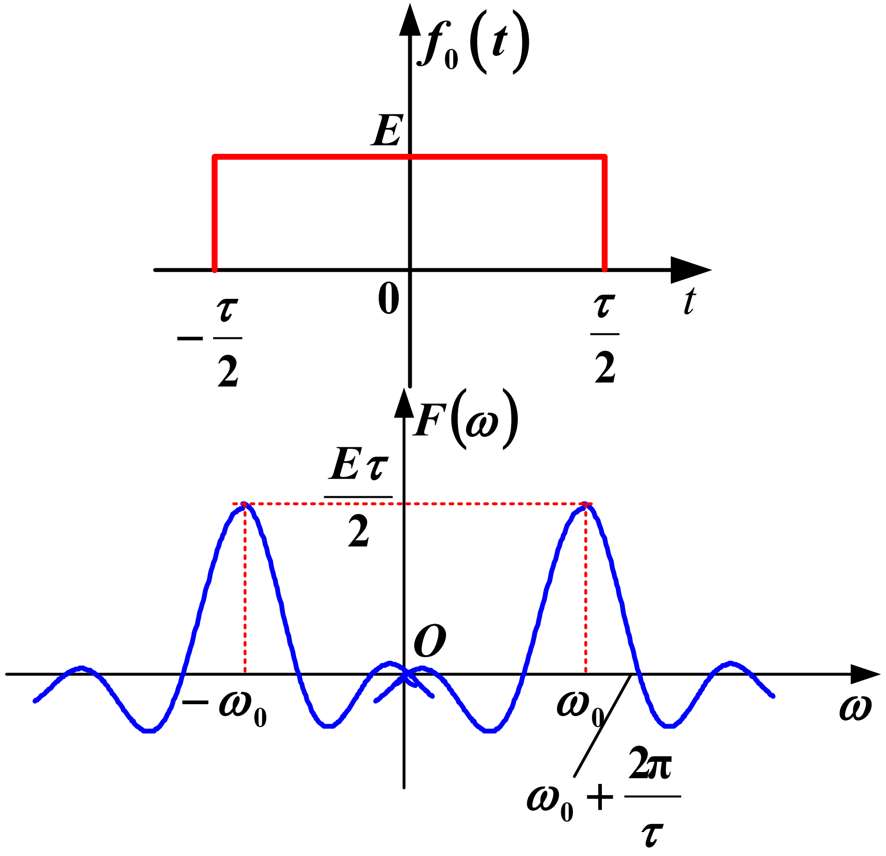
Example 6

Given the frequency spectrum $F_0(\omega)$ of the rectangular pulse $f_0(t)$

$$F_0(\omega) = E\tau \cdot \mathrm{Sa}\left(\frac{\omega\tau}{2}\right)$$

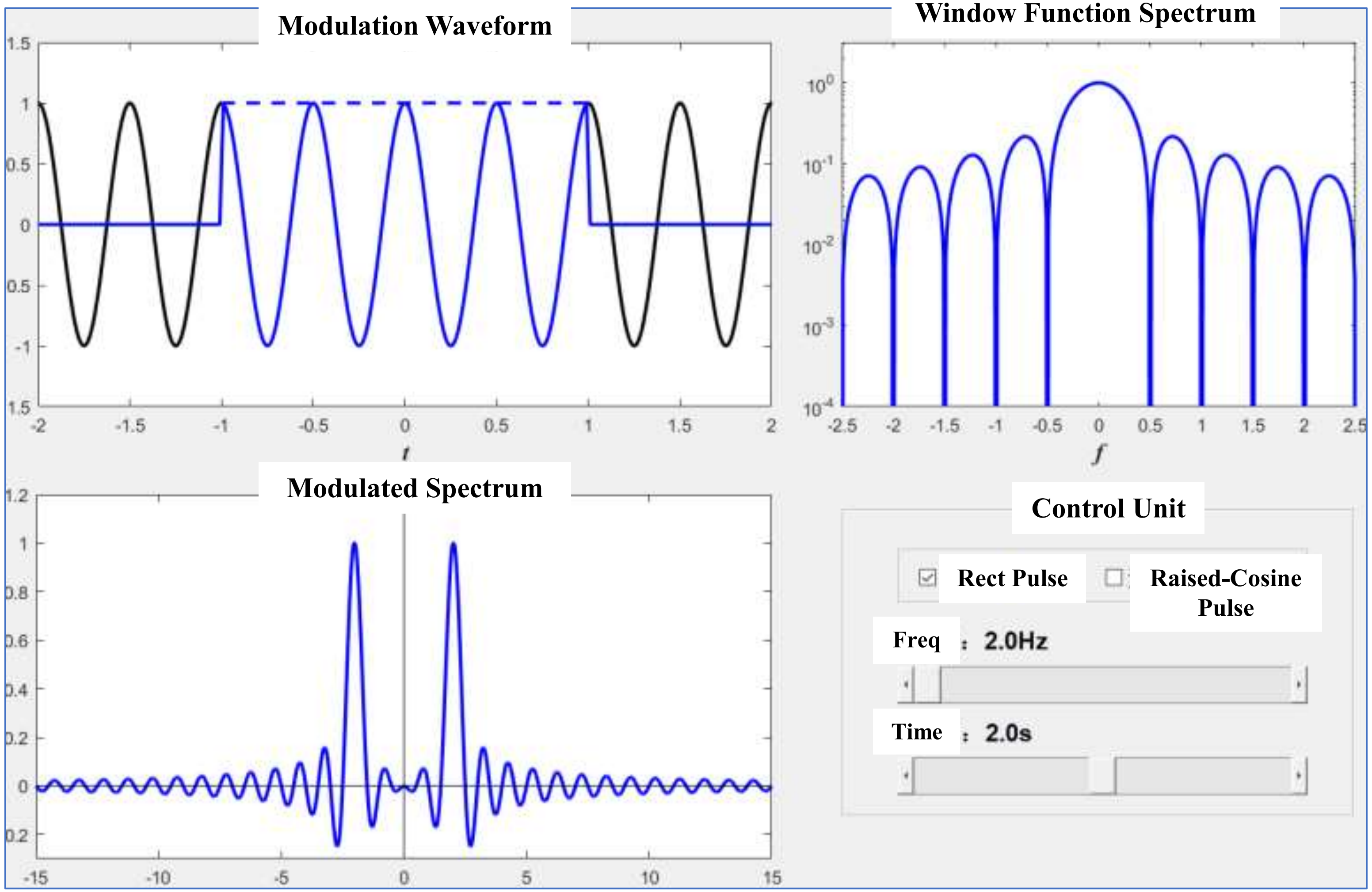
$$F(\omega) = \frac{1}{2}F_0(\omega - \omega_0) + \frac{1}{2}F_0(\omega + \omega_0)$$



The original spectrum $F_0(\omega)$ (a sinc function centered at $\omega_0 = 0$) is split into two copies, shifted to $\omega = \omega_0$ and $\omega = -\omega_0$, scaled by ½ in amplitude

6. Basic Properties of Fourier Transform

Example 6



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7. Time-Domain Differentiation Property

$$\text{Given } \mathscr{F}\left[f(t)\right]=F(\omega), \quad \text{then } \mathscr{F}\left[f'(t)\right]=\mathrm{j}\omega F(\omega)$$

$$\begin{cases} \text{The magnitude spectrum is multiplied by } |\omega| \\ \text{phase added, } \mathrm{j}=e^{\mathrm{j}\frac{\pi}{2}} \end{cases}$$

Generalization for n-th Derivative: $f^{(n)}(t) \leftrightarrow (\mathrm{j}\omega)^n F(\omega)$

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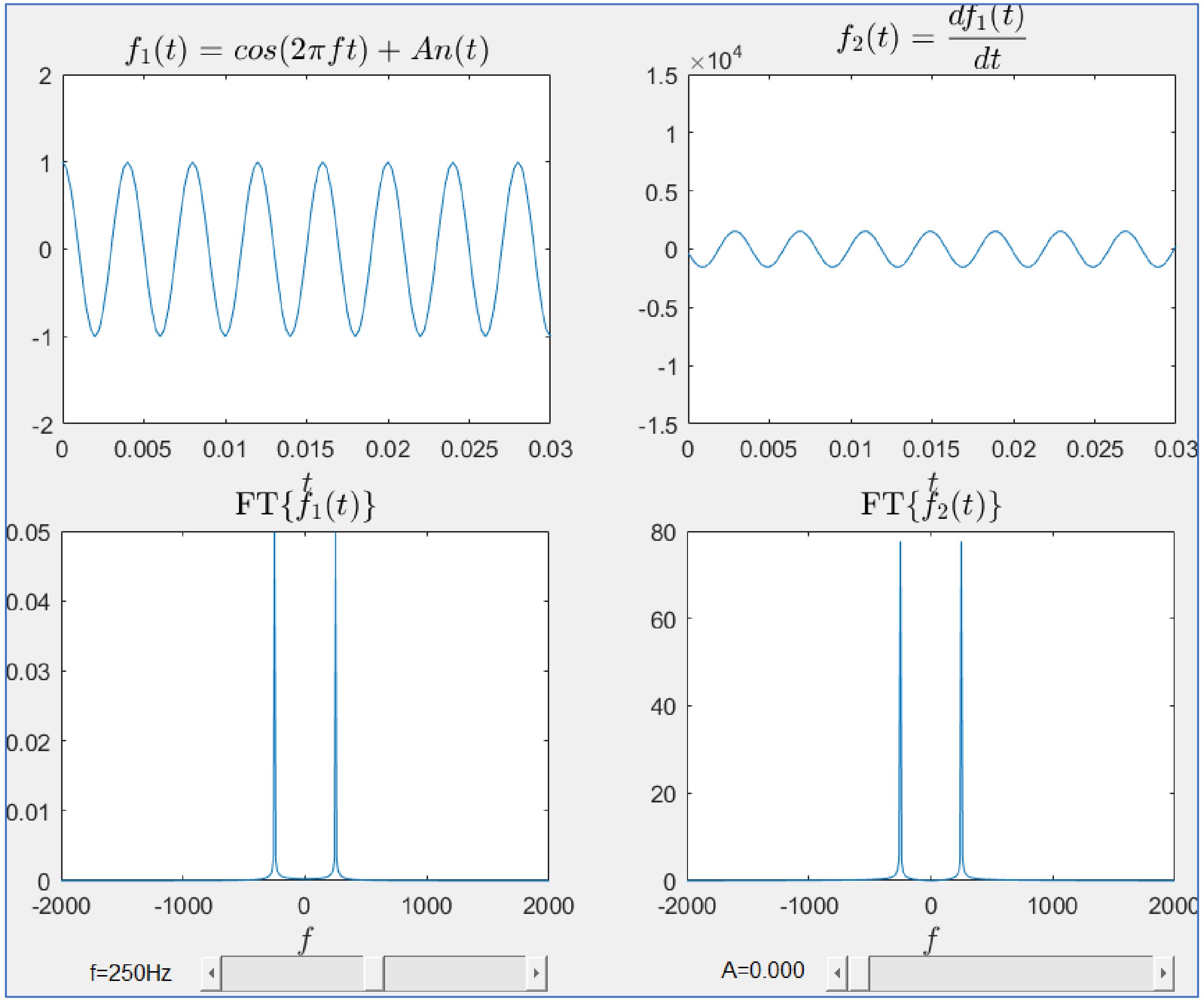
(1) Demo (Spectrum Obtained via FFT Processing)



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Example 7

Given $\mathscr{F}\left[\delta(t)\right]=1$, solve $\mathscr{F}\left[\delta'(t)\right]$

According to the time-domain differentiation property of the Fourier transform, obtain

$$\mathscr{F}\left[\delta'(t)\right]=\mathrm{j}\omega\cdot 1$$

8. Frequency-Domain Differentiation Property

If $\mathscr{F}\left[f(t)\right] = F(\omega)$, then $\mathscr{F}\left[tf(t)\right] = \mathrm{j}\dfrac{\mathrm{d}F(\omega)}{\mathrm{d}\omega}$

Or $\mathscr{F}\left[-\mathrm{j}tf(t)\right] = \dfrac{\mathrm{d}F(\omega)}{\mathrm{d}\omega}$

Generalized Form:

$$\mathscr{F}\left[(-\mathrm{j}t)^n f(t)\right] = \frac{\mathrm{d}^n F(\omega)}{\mathrm{d}\omega^n}$$

Or

$$\mathscr{F}\left[t^n f(t)\right] = (\mathrm{j})^n \frac{\mathrm{d}^n F(\omega)}{\mathrm{d}\omega^n}$$

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Example 9

Given $\mathscr{F}\left[f(t)\right]=F(\omega)$, **Solve** $\mathscr{F}\left[(t-2)f(t)\right]=?$

$$\mathscr{F}\left[(t-2)f(t)\right]=\mathscr{F}\left[tf(t)-2f(t)\right]$$

$$=\mathrm{j}\frac{\mathrm{d}F(\omega)}{\mathrm{d}(\omega)}-2F(\omega)$$

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9. Time-Domain Integration Property

If $f(t) \leftrightarrow F(\omega)$, **then**

$$\int_{-\infty}^{t} f(\tau)\,\mathrm{d}\tau \leftrightarrow F(\omega)\cdot\left[\frac{1}{\mathrm{j}\omega}+\pi\delta(\omega)\right]$$

When $F(0)=0$, $\quad \int_{-\infty}^{t} f(\tau)\,\mathrm{d}\tau \leftrightarrow \dfrac{F(\omega)}{\mathrm{j}\omega}$

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10. Frequency-Domain Integration Property (for reference)

Given $\mathscr{F}\left[f(t)\right]=F(\omega)$,

Thus $\mathscr{F}^{-1}\left[\int_{-\infty}^{\omega}F(\omega)\,\mathrm{d}\omega\right]=-\dfrac{f(t)}{\mathrm{j}t}+\pi f(0)\delta(t)$

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11. Time-Domain Convolution Theorem

$$\text{If } \mathscr{F}\left[f_1(t)\right] = F_1(\omega),\ \mathscr{F}\left[f_2(t)\right] = F_2(\omega)$$

$$\text{Then } \mathscr{F}\left[f_1(t) * f_2(t)\right] = F_1(\omega) \cdot F_2(\omega)$$

Time-domain convolution $\xleftrightarrow{\text{FT}}$ Frequency-domain multiplication

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12. Frequency-Domain Convolution Theorem

$$\text{If } \mathscr{F}\left[f_1(t)\right] = F_1(\omega),\ \mathscr{F}\left[f_2(t)\right] = F_2(\omega)$$

$$\text{Then } \mathscr{F}\left[f_1(t) \cdot f_2(t)\right] = \frac{1}{2\pi} F_1(\omega) * F_2(\omega)$$

Time-domain multiplication $\xleftrightarrow{\text{FT}} \dfrac{1}{2\pi}$ **Frequency-domain convolution**

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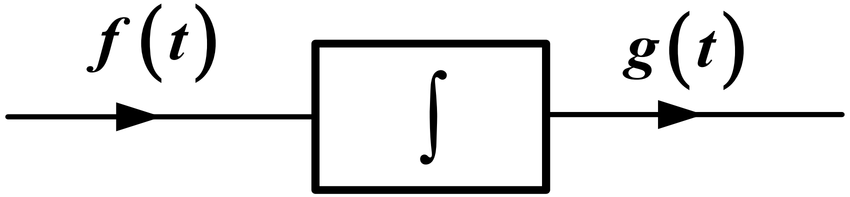
13. Applications of the Convolution Theorem

- Using Time-Domain Convolution to Find Spectral Density Functions
- Fourier Transform of the Integral $\int_{-\infty}^{t} f(\tau)\,\mathrm{d}\tau$
- Fourier Analysis for System (Zero-State) Response (Chapter 4)

13. Applications of the Convolution Theorem



$$g(t) = \int_{-\infty}^{t} f(\tau)\,\mathrm{d}\tau$$

Unit Impulse Response of an Integrator

$$h(t) = \int_{-\infty}^{t} \delta(\tau)\,\mathrm{d}\tau = u(t)$$

Response of a Signal Passing Through an Integrator

$$g(t) = f(t) * u(t) = \int_{-\infty}^{t} f(\tau)\,\mathrm{d}\tau$$

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13. Applications of the Convolution Theorem

$$\int_{-\infty}^{t} f(\tau)\,\mathrm{d}\tau = f(t) * u(t)$$

The time-domain integration property can be obtained through the convolution theorem

$$\mathscr{F}\left[\int_{-\infty}^{t} f(\tau)\,\mathrm{d}\tau\right] = \mathscr{F}[f(t)] \cdot \mathscr{F}[u(t)]$$

$$= F(\omega) \cdot \left[\frac{1}{\mathrm{j}\omega} + \pi\delta(\omega)\right]$$

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Summary

1. Linearity
2. Duality
3. Conjugation Symmetry
4. Time-Scaling Property
5. Time-Shifting Property
6. Frequency-Shifting Property
7. Time-Domain Differentiation Property
8. Frequency-Domain Differentiation Property
9. Time-Domain Integration Property
10. Frequency-Domain Integration Property
11. Time-Domain Convolution Theorem
12. Frequency-Domain Convolution Theorem
13. Applications of the Convolution Theorem

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Main Content

1. Fourier Transform of Cosine Signal
2. Fourier Transform of Sine Signal
3. Fourier Transform of Impulse Train
4. Fourier Transform of General Periodic Signals

Introduction

Periodic Signal:

The Fourier series F_k of a signal $f(t)$, discrete spectrum

Aperiodic Signal:

The Fourier transform $F(\omega)$ of a signal $f(t)$, continuous spectrum

How to compute the Fourier transform of a periodic signal?

What is the relationship with the Fourier series?

$f(t)$, **Periodic** / **Aperiodic** — **Unified analysis method :** **Fourier transform analysis method**

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1. Fourier Transform of Cosine Signal

Euler's formula

$$\cos\omega_0 t = \frac{1}{2}\left(e^{j\omega_0 t} + e^{-j\omega_0 t}\right)$$

Given

$$\mathscr{F}[1] = 2\pi\delta(\omega)$$

Frequency shifting property

$$\mathscr{F}\left[1\cdot e^{j\omega_0 t}\right] = 2\pi\delta(\omega - \omega_0)$$

$$\mathscr{F}\left[1\cdot e^{-j\omega_0 t}\right] = 2\pi\delta(\omega + \omega_0)$$

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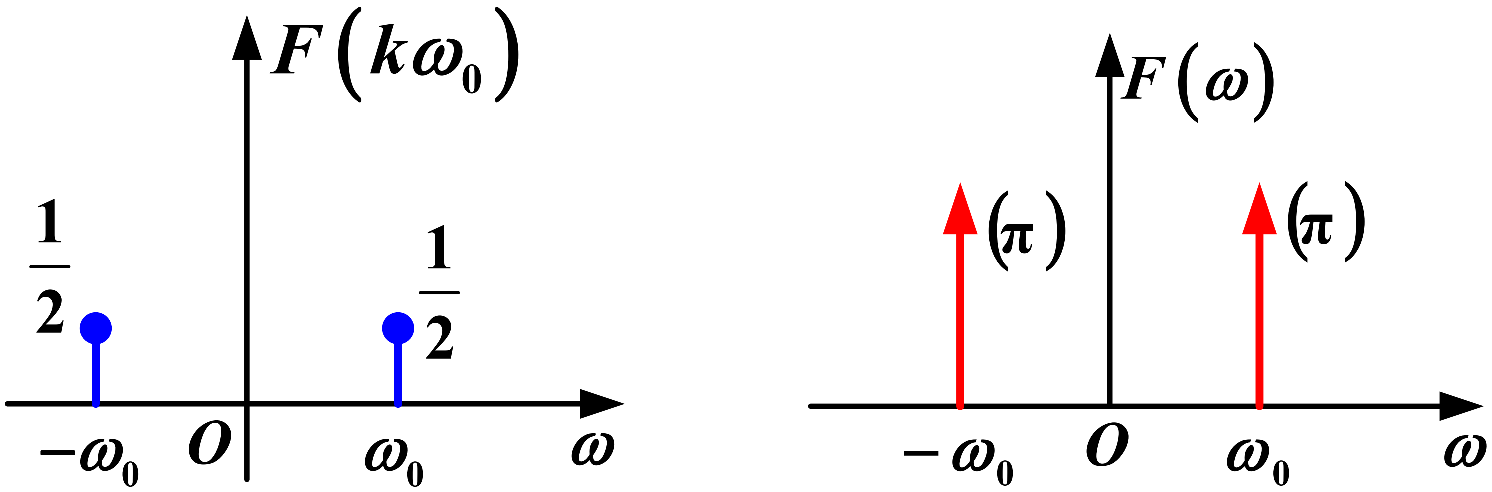
1. Fourier Transform of Cosine Signal

Fourier transform of cosine signal is

$$\cos\omega_0 t \xleftrightarrow{\text{FT}} \pi\left[\delta\left(\omega+\omega_0\right)+\delta\left(\omega-\omega_0\right)\right]$$

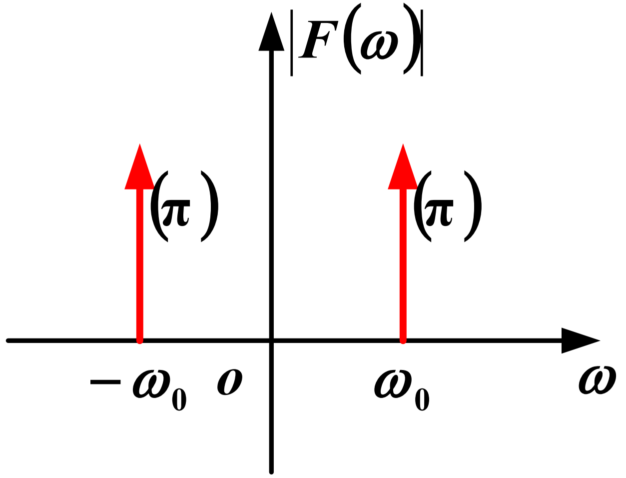


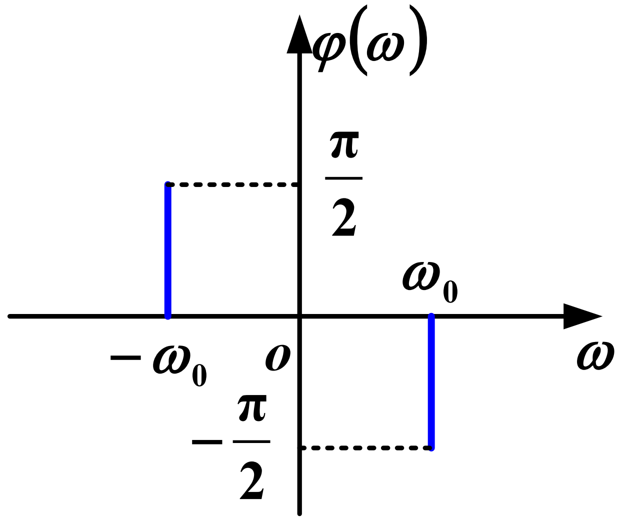
2. Fourier Transform of Sine Signal

$$\sin\omega_0 t = \frac{1}{2j}\left(e^{j\omega_0 t} - e^{-j\omega_0 t}\right)$$

$$\xleftrightarrow{FT} -j\pi\delta(\omega-\omega_0) + j\pi\delta(\omega+\omega_0)$$

$$= \pi e^{j\left(-\frac{\pi}{2}\right)}\delta(\omega-\omega_0) + \pi e^{j\frac{\pi}{2}}\delta(\omega+\omega_0)$$





- The amplitude spectrum of $\sin(\omega_0 t)$ is the same as $\cos(\omega_0 t)$
- The phase spectrum is different

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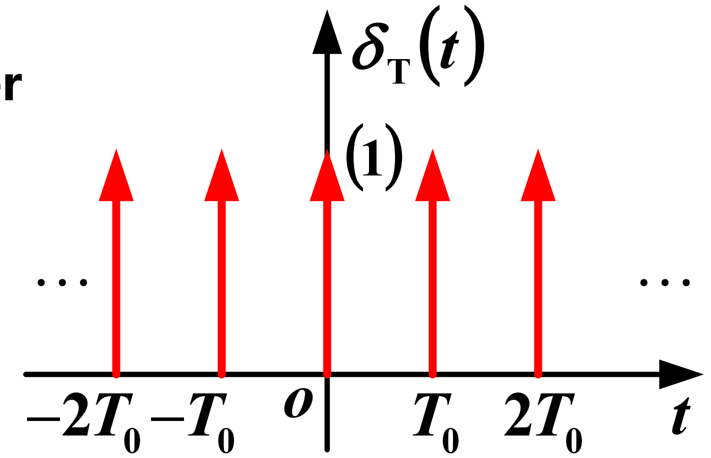
3. Fourier Transform of Impulse Train

$$\delta_T(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT_0)$$ k is an integer

$\delta_T(t)$



In the Fourier series, the coefficient is

$$F_k = \frac{1}{T_0}\int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} \delta(t)\, e^{-jk\omega_0 t}\, \mathrm{d}t = \frac{1}{T_0}$$

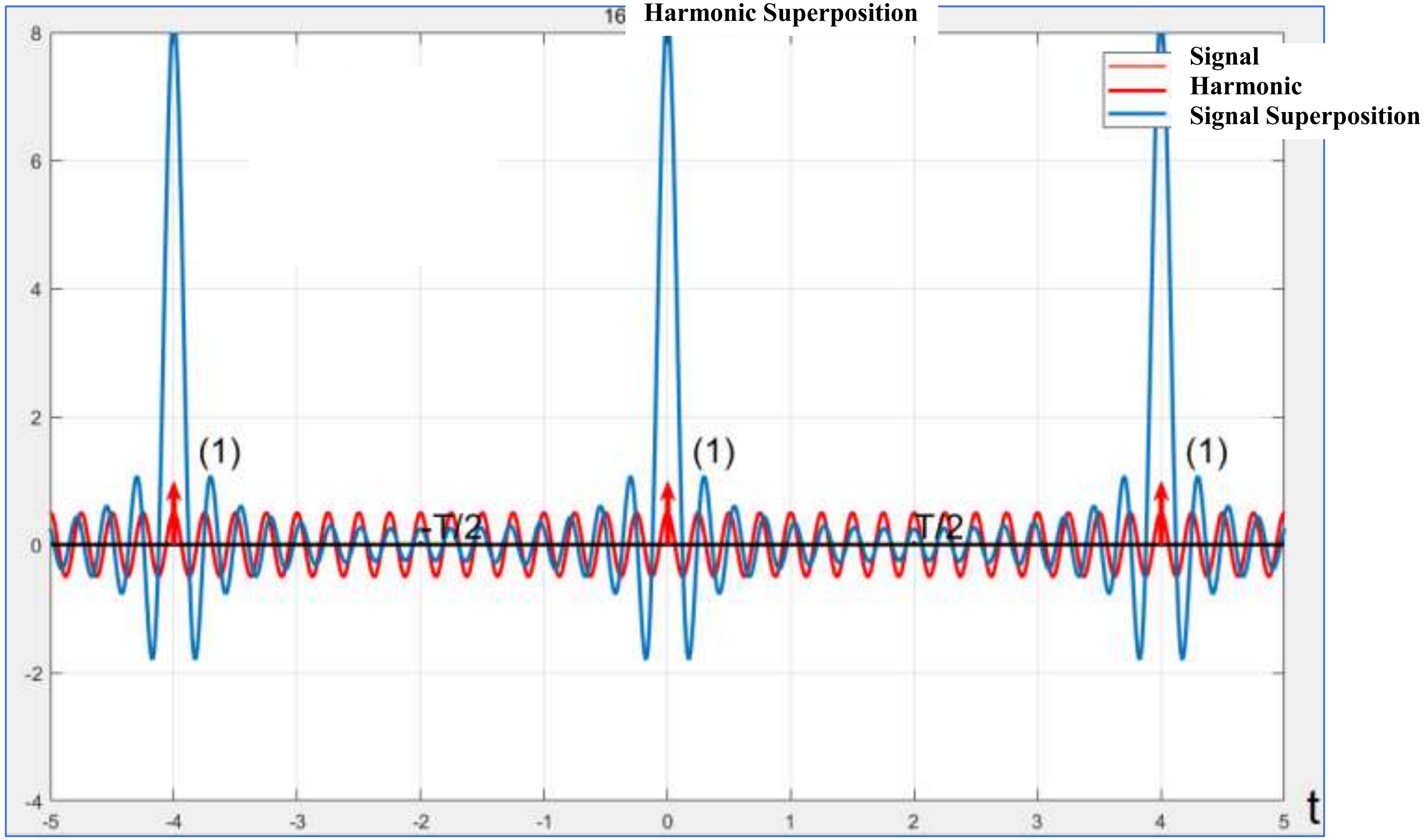
$$\left(\omega_0 = \frac{2\pi}{T_0}\right)$$

The Fourier series representation

$$e^{j\omega_o t} + e^{-j\omega_o t} = 2\cos(\omega_o t)$$

$$\delta_T(t) = \frac{1}{T_0}\sum_{k=-\infty}^{\infty} e^{jk\omega_0 t} = \frac{1}{T_0} + \frac{2}{T_0}\sum_{k=1}^{\infty}\cos(k\omega_0 t)$$

3. Fourier Transform of Impulse Train



$$\delta_T(t) = \frac{1}{T_0} + \frac{2}{T_0}\sum_{k=1}^{\infty}\cos\left(k\omega_0 t\right)$$

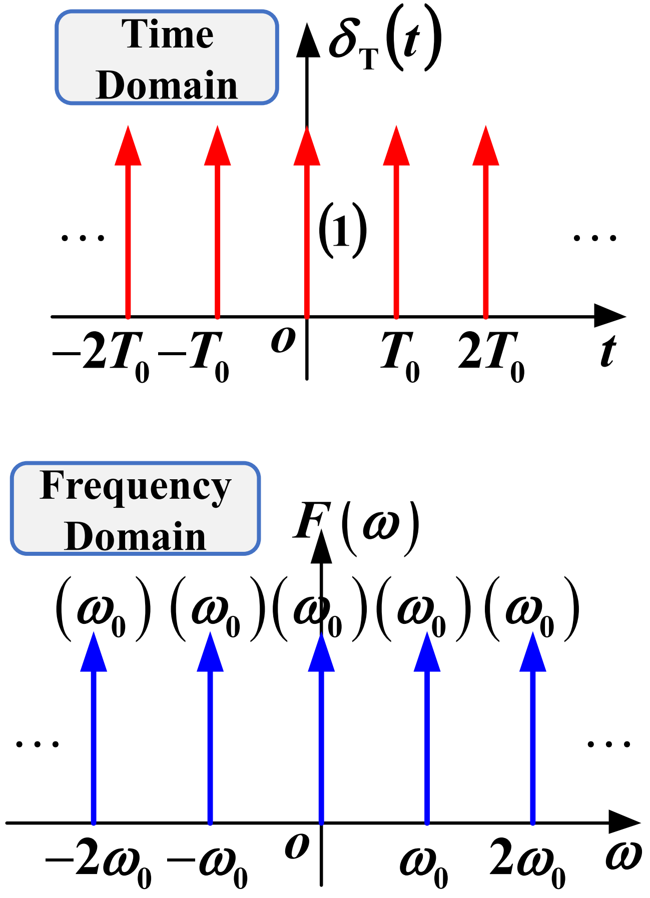
3. Fourier Transform of Impulse Train

$$\delta_T(t) = \frac{1}{T_0}\sum_{k=-\infty}^{\infty} e^{jk\omega_0 t}$$

Given $e^{jk\omega_0 t} \leftrightarrow 2\pi\delta(\omega - k\omega_0)$

$$\therefore F(\omega) = \frac{1}{T_0}\sum_{k=-\infty}^{\infty} 2\pi\delta(\omega - k\omega_0)$$

$$= \omega_0 \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_0)$$



The frequency spectrum of $\delta_T(t)$ remains an impulse train, with both strength and spacing of ω_0

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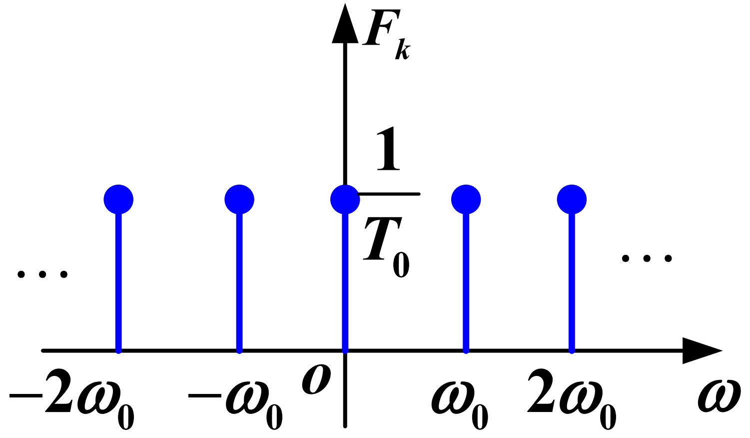
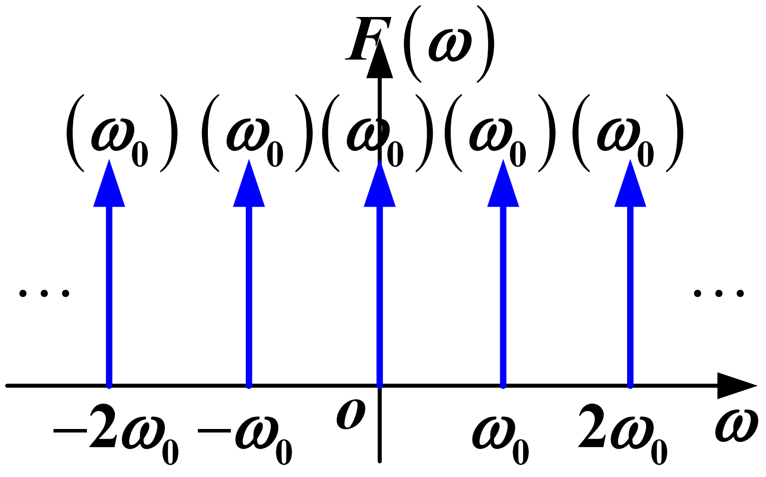
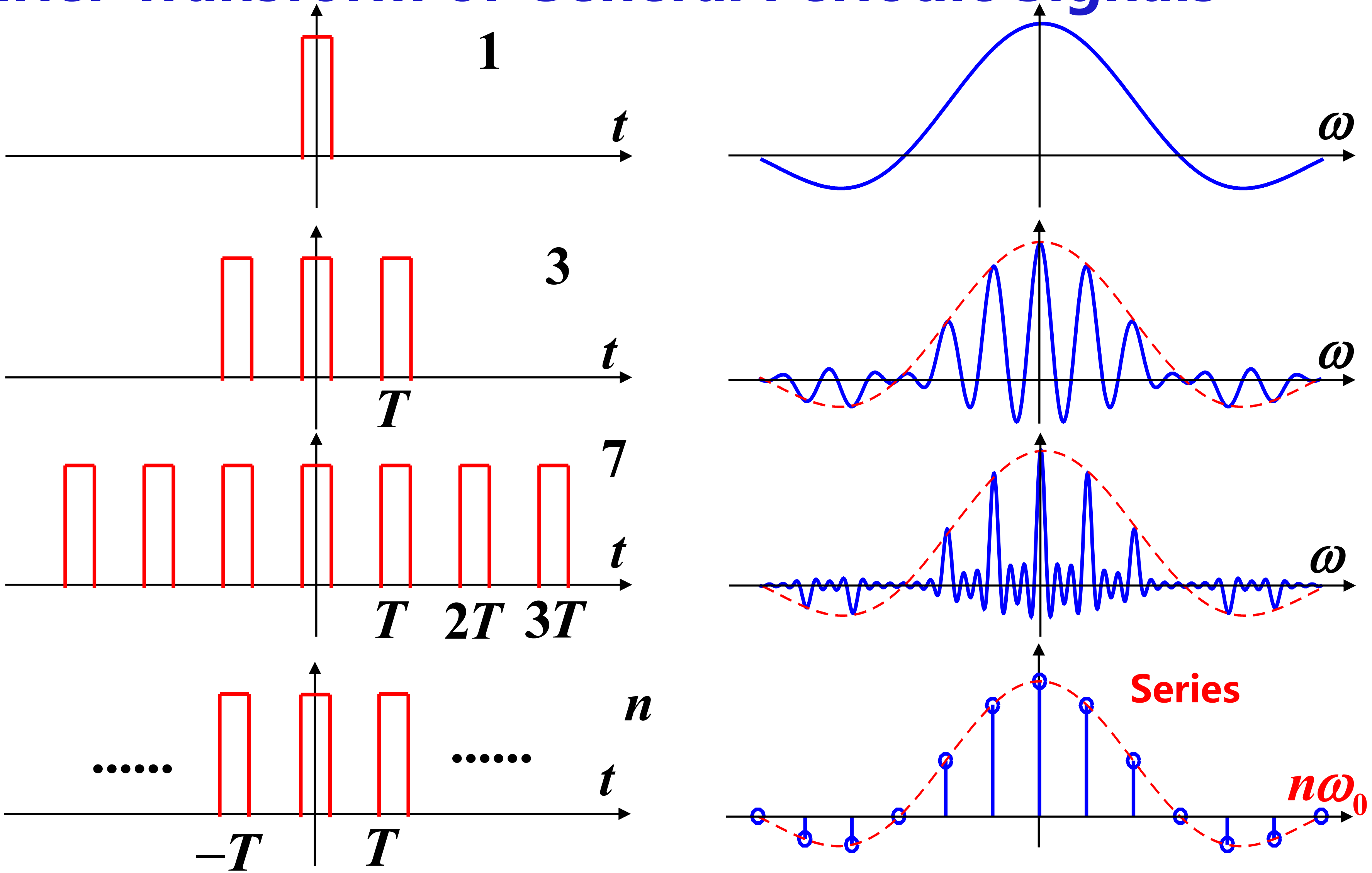
3. Fourier Transform of Impulse Train





Does not satisfy convergence
Bandwidth is infinitely wide

4. Fourier Transform of General Periodic Signals



4. Fourier Transform of General Periodic Signals

Periodic signals can be represented in the form of Fourier series

$$f_{\mathrm{T}}(t)=\sum_{k=-\infty}^{\infty}F_k e^{jk\omega_0 t} \qquad \left(T_0=\frac{2\pi}{\omega_0}\right)$$

The Fourier transform of a periodic signal (using the definition)

$$F_T(\omega)=\sum_{k-\infty}^{\infty}F_k\cdot 2\pi\delta(\omega-k\omega_0)$$

$$=2\pi\sum_{k-\infty}^{\infty}F_k\cdot\delta(\omega-k\omega_0)$$

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4. Fourier Transform of General Periodic Signals

$$F_T(\omega) = 2\pi \sum_{-\infty}^{\infty} F_k \cdot \delta(\omega - k\omega_0)$$

(1) The spectrum of $f_{\mathrm{T}}(t)$ consists of a series of impulses

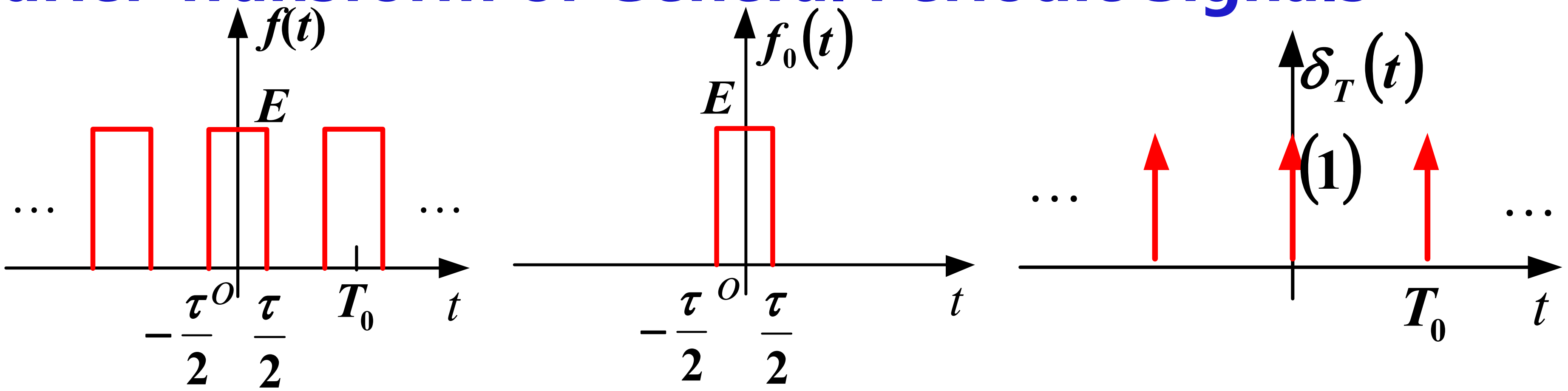
Positions: $\omega = k\omega_0$ (harmonic frequencies)

Intensities: $2\pi F_k$, proportional to F_k

(2) The amplitude of spectral lines is not finite because $F(\omega)$ represents spectral density

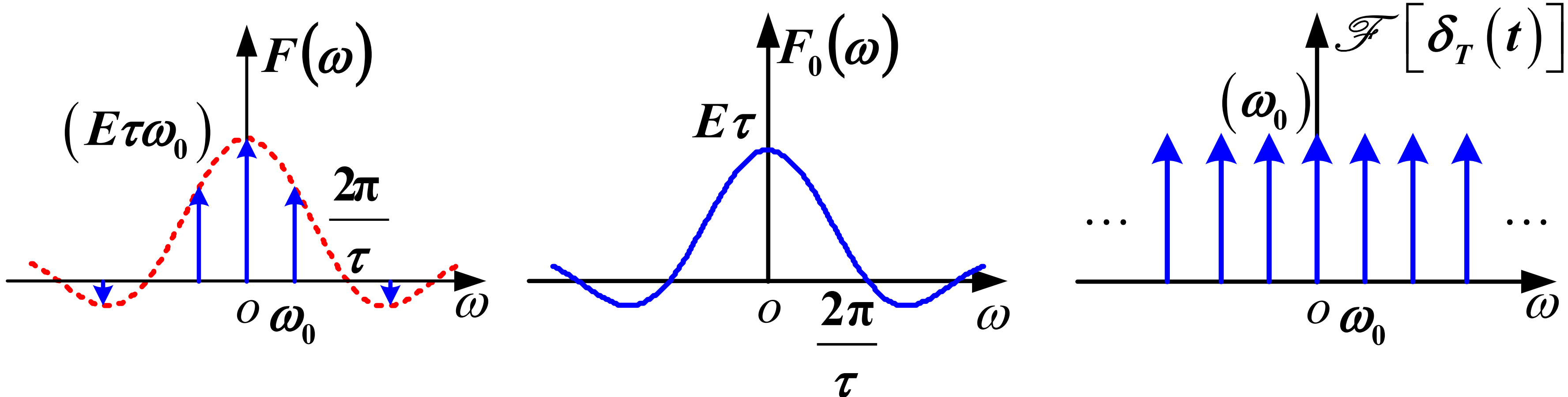
For periodic signals, $F(\omega)$ exists only at discrete points $\omega = k\omega_0$, with an infinitely small frequency range and infinite amplitude

4. Fourier Transform of General Periodic Signals

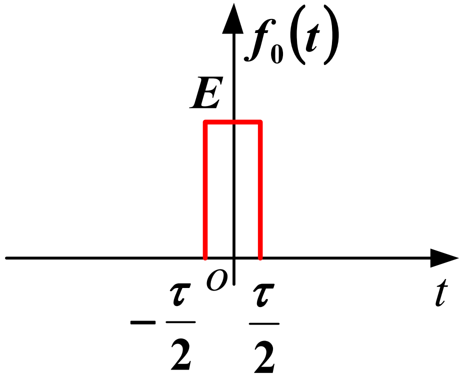


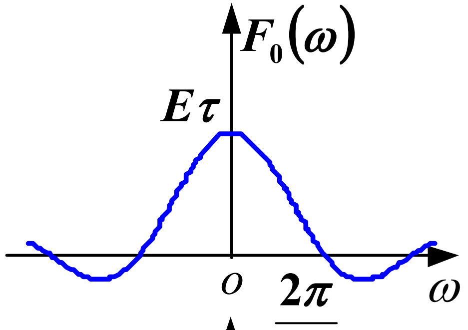
$$F(\omega) = F_0(\omega)\cdot \omega_0 \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_0)$$

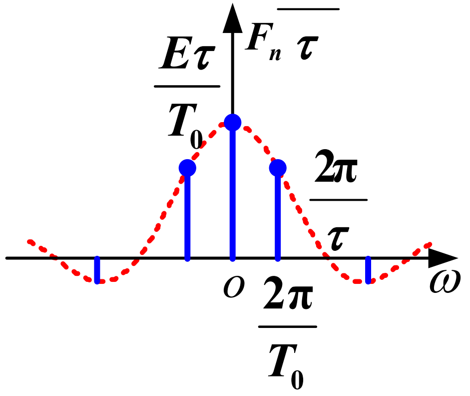
$$f(t) = f_0(t) * \delta_T(t)$$



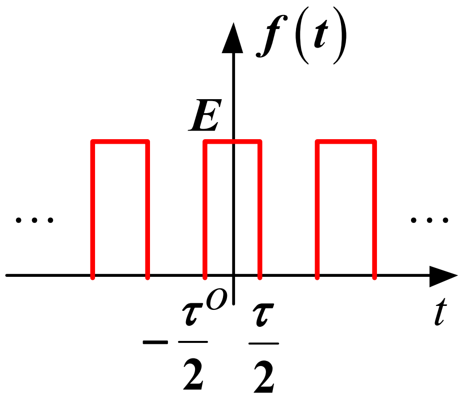
4. Fourier Transform of General Periodic Signals

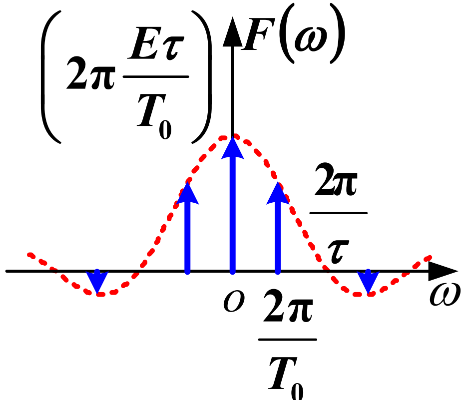






Fourier series spectrum





Fourier transform spectrum

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Summary

1. Fourier Transform of Cosine Signal
2. Fourier Transform of Sine Signal
3. Fourier Transform of Impulse Train
4. Fourier Transform of General Periodic Signals

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Thank you for listening !